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WILSON EDUARDO IKEDA

CULTURAL DIMENSIONS AND BEHAVIORAL FINANCE: CROSS-NATIONAL EVIDENCE OF INVESTOR BIASES

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WILSON EDUARDO IKEDA

CULTURAL DIMENSIONS AND BEHAVIORAL FINANCE: CROSS-NATIONAL EVIDENCE OF INVESTOR BIASES

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RESUMO

Esta tese investiga como a cultura nacional influencia a manifestação de vieses comportamentais — excesso de confiança, efeito disposição e comportamento de manada — entre investidores. Estruturada em cinco capítulos no formato de artigos, a pesquisa utiliza dados de 64 mercados ao longo de 15 anos e aplica métodos paramétricos e não paramétricos para identificar relações lineares e padrões não lineares subjacentes. Os resultados revelam que a cultura exerce papel determinante na formação dos vieses e que períodos de crise podem modificar essa influência, tanto em intensidade quanto em direção. Ao demonstrar que a cultura molda significativamente o comportamento do investidor, o estudo contribui para o avanço das finanças comportamentais e sugere que programas de educação financeira devem ser adaptados ao contexto cultural para maior eficácia. Além disso, oferece insights práticos para empresas ajustarem estratégias de negociação e posicionamento de mercado conforme o perfil cultural predominante. Ao analisar vieses sob a ótica cultural, esta pesquisa estabelece uma base sólida para estudos futuros e reforça a conclusão central: a cultura é um fator-chave na determinação do comportamento do investidor.

Classificação JEL: F01; G01; G41; Z10

Palavras-chave: Cultura Nacional, Vieses comportamentais, Excesso de confiança, Efeito Manada, Efeito Disposição, Dimensões Culturais de Hofstede

ABSTRACT

This dissertation investigates the role of national culture in shaping the manifestation of key behavioral biases - overconfidence, disposition effect, and herding - among investors. Structured as five article-based chapters, the study utilizes a comprehensive dataset covering 64 markets over a 15-year period (2010–2024) and applies both parametric and non-parametric techniques to capture linear relationships and underlying non-linear dynamics. The findings demonstrate that cultural factors significantly influence investor behavior and that financial crises can modify this influence in both magnitude and direction. By establishing culture as a critical determinant of bias formation, the research advances the field of behavioral finance and underscores the importance of culturally tailored investor education for enhanced effectiveness. Furthermore, the results provide actionable insights for firms seeking to align trading practices and market strategies with cultural profiles. By framing behavioral biases through the lens of national culture, this work lays a robust foundation for future research on cultural determinants of investment decisions and affirms the central conclusion: culture is a decisive factor in shaping investor behavior.

JEL codes: F01; G01; G41; Z10

Keywords: National Culture, Behavioral biases, Overconfidence, Herding, Disposition Effect, Hofstede's Cultural Dimensions

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1 INTRODUCTION

Culture is one of the most critical drivers of decision-making and consists of collective conclusions that have worked in the past and are transmitted to future generations (Goodell, Kumar, *et al.*, 2023). There are several concepts of culture. Hofstede (1980, 2001) explained culture as the shared mental programming that distinguishes social groups through common values expressed through symbolic elements and institutional systems. Kluckhohn and Strodtbeck (1961) elucidated culture as a set of established patterns that describe how societies resolve universal human problems by selecting preferred approaches to managing time and social bonds. Trompenaars and Hampden-Turner (1997) interpreted culture as a social system that enables people to handle their social bonds and showed that culture functions as an energetic system that transforms through dialogue. According to Inglehart (1997), cultural values develop from economic stability and family background experiences, which lead to social transformation through modernization processes. Schwartz (1992, 2012) established a motivational framework that explains culture as a collective value system that directs human actions through established patterns that vary in social value. The GLOBE Project (House *et al.*, 2004; Javidan *et al.*, 2006) defined culture as a society's collective values and practices that shape leadership and organizational behavior. Geertz (1993) joined other anthropologists in seeing culture as a set of historical traditions that make meaning through symbols. In summary, these basic definitions show culture as a multi-dimensional concept that includes group values, ways of thinking, social structures, and historical traditions. All these parts guide the forms of human action in many kinds of groups.

Social systems can only function because human behavior is not arbitrary but somewhat predictable. For every behavioral prediction, we consider both the individual and the situation. We presume that each person has a degree of mental programming that persists over time, leading to similar behavior in similar circumstances. Consequently, the more precisely we understand a person's mental programming and the situation, the more accurate our prediction will be (Hofstede, 2001).

The investor's behavioral reaction to a piece of information could have a relation with his/her national culture, which refers to the behavioral norms and conventional beliefs of most people in a particular country, as the collective set of everyday experiences that people of the same culture share would influence their cognitive and emotional approach. Therefore, differences in national culture may shed light on examining investors' behaviors in stock markets (Chang & Lin, 2015).

Behavioral finance analyzes decision-making processes, including the influence of emotional factors on investors' reasoning (Ricciardi & Simon, 2000). Decision-making is a complex mental

activity influenced by the decision-maker's psychological factors, in which the decision-maker chooses an alternative after collecting information (Sattar *et al.*, 2020). In this way, Behavioral Finance is an interdisciplinary area of finance that combines the principles of psychology, sociology, economics, and finance and tries to explain the impact of different emotional, social, and psychological behaviors on the financial decisions of investors that are not always rational, since decision-makers face various behavioral anomalies and biases (Zahera & Bansal, 2019). As behavioral finance suggests, influenced by various biases, investors make irrational investment decisions, and they understand that this can help them avoid losses through emotion-driven speculation while maintaining a balance between rationality and personal preferences.

1.1 Research Question

An understanding of how the values of thought, feeling, and action that are typical of national cultures come to play in financial markets' cognition, emotion, and behavior will perhaps shed much light on the origins of their anomalies, inefficiencies, and crises.

In light of growing academic interest in behavioral biases that contribute to market anomalies across global economies, this study explores whether such biases manifest uniformly among investors or vary in intensity across different cultural contexts. Specifically, it seeks to answer: Do investors worldwide exhibit the same set of behavioral biases, or are these biases shaped and amplified by national cultural values within diverse capital markets?

To address this research question, we focus on three prominent behavioral biases observed in capital markets - overconfidence, disposition effect, and herding- and examine whether these biases manifest similarly or differ in intensity across countries with distinct cultural values. Using a comprehensive 15-year dataset (2010–2024) covering 64 nations, this study employs regression-based estimation techniques to uncover how cultural frameworks influence the prevalence and strength of these investor biases in global equity markets.

As a result, we expect to find differences in investor behavior biases across different cultural environments.

1.2 General and Specific Objectives

The primary objective of this research is to understand how national cultural values influence investor behavior in financial markets. Specifically, we aim to investigate whether investors' behavioral biases, such as herding, overconfidence, and the disposition effect, manifest similarly or with the same intensity across different capital markets worldwide.

Furthermore, this research seeks to (i) further enhance the interface between behavioral finance and cultural finance by imputing cultural dimensions in the analysis of investor decision-making, (ii) specify what particular aspects of national culture relate to both the development and strength of behavioral biases, and (iii) give an organized picture of a comparison among different capital markets regarding investor biases through the frame of national cultural values.

1.3 Motivation

The motivation for this study comes from the growing recognition that financial markets are not based on calculations in a rational vacuum but are simultaneously and profoundly influenced by psychological and cultural factors. Though biases such as overconfidence, herding, and the disposition effect have been extensively documented in behavioral finance, relatively little work has addressed cross-cultural variations in these biases. Since investors operate within national cultures that mold their values and decision-making tendencies, it is important to know whether behavioral biases vary uniformly across countries or are influenced by cultural differences. Thus, this research attempts to bring together the fields of behavioral finance and cultural finance.

1.4 Relevance

This study has the potential to stimulate academic discussion and enhance practical applications. Academically, it advances interdisciplinary communication among finance, psychology, and cultural studies by empirically testing investor biases in relation to national cultural values. It provides evidence on how specific cultural dimensions exacerbate or mitigate anomalies in financial decision-making behavior. On a practical level, for policymakers, regulators, and increasingly active institutional investors operating in a more integrated global market, these results are highly relevant because the revealed cultural patterns in investor behavior will help lead to better risk management strategies, an optimal regulatory framework, and more effective international financial governance collaboration.

1.5 Contributions

This study makes some major contributions.

1 Theoretical Contribution: It enhances the area of behavioral finance through an infusion of cultural considerations into a review of a set of empirically generated investor biases, thereby broadening the explanatory framework to existing models.

2 Empirical contribution: By using regression-based estimation techniques and daily data from public companies from 64 countries for the period between 2010 and 2024, totaling more than

0.5 billion transactions, this study provides strong cross-country evidence on how national cultural values relate to the prevalence of behavioral biases.

3 Comparative Contribution: By comparing investor behavior across these different cultural settings, this research explains how biases vary around the world's capital markets and pinpoints specific cultural elements that contribute to these disparities.

4 Practical Contribution: It provides useful hints to investors, regulators, and policymakers about the way cultural contexts bias market anomalies and inefficiencies for them to design culturally responsive investment and regulatory strategies.

This work advances the new field of cultural finance by showing that without understanding the culture in which decisions are made, one cannot fully grasp investor behavior. It also keeps a running discussion between behavioral finance and cultural studies, blending theory with practical implications for the world's financial system.

1.6 Research in Chapters

The research is divided into five chapters:

1.6.1 How Does National Culture Shape Investor Behavior? Mapping a Research Agenda

A systematic literature review, complemented by a bibliometric analysis, presents the state-of-the-art from the literature to determine if and how national culture influences investor behavior. Through an analysis of 60 studies published between 2009 and 2024 in journals classified as ABS 2 to 4*, we identified three dominant themes in the literature: investor decision-making; the market and investor cultural preferences; and culture and behavioral biases. We also identify unexplored research areas and directions for future work in this promising field as the financial world becomes increasingly interconnected and cultural diversity among investors grows.

1.6.2 The Cultural Impact on Business Strategy: Faith, Law, and Economy

This study examines the interrelationship between a society's national culture, economy, religion, and legal system. The research reveals that developed countries exhibit greater individualism and masculinity and less uncertainty avoidance. The study also shows that Muslim-majority societies are more collectivist, Orthodox countries are less indulgent, and Protestant-majority cultures exhibit less power distance. Legal origin also reveals differences: countries with Germanic and French legal origins exhibit greater individualism and uncertainty avoidance than societies with Common Law origins. Scandinavian societies have a dominant feminine dimension, while countries with a French legal origin exhibit greater power distance, and societies with a Germanic legal origin have less

indulgence. The results demonstrate that economic status, dominant religion, and the origin of the legal system have strong implications for a country's culture.

1.6.3 Cultural Dimensions and Investor Overconfidence: Lessons from Global Market Crises

The empirical investigation into the relationship between national culture and investor overconfidence bias, using Hofstede's dimensions and daily trading volume of publicly traded companies across 64 stock exchanges during the pre-pandemic (2015-2020), pandemic (2020-2021), and post-pandemic (2022-2024) periods. Regardless of the period analyzed, societies with lower Power Distance and higher levels of Indulgence tend to exhibit greater overconfidence, as do, unexpectedly, countries with higher Long-Term Orientation. Individualism, Uncertainty Avoidance, and Masculinity, however, show a reversal: a positive association before the pandemic becomes negative during and after the pandemic. These dynamic findings offer timely insights into culture, psychology, and financial behavior.

1.6.4 Investor Bias Across Borders: Cultural Influences on the Disposition Effect

This chapter examines the influence of national culture, as measured by Hofstede's dimensions, on investors' disposition effect bias. This bias is defined as investors' tendency to quickly sell gains on rising stocks and hold long-term losses on falling stocks. Through the analysis of daily stock market data from 64 countries over a 15-year period (2010 to 2024) and the use of techniques for detecting linear and non-linear patterns, we show that societies with greater masculinity, long-term orientation, and indulgence are associated with a lower disposition effect. More complex patterns were detected in the relationships among the disposition effect, individualism, uncertainty avoidance, and power distance. The study also shows that economic development reduces bias. Thus, the research demonstrates that financial education programs and policy interventions should draw on cultural profiles to understand investor psychology.

1.6.5 Behavioral Shifts in Times of Crisis: Herding and Culture in Pandemic-Era Markets

The final empirical study investigates how cultural values and economic development shaped herd behavior among investors across 64 countries before, during, and after the COVID-19 pandemic. Our results show that higher GNI per capita is a consistently strong protective factor against herd behavior. Cultural influences are heterogeneous and vary over time: power distance became a substantially stronger predictor of increased herd behavior during the recovery period, long-term orientation persistently mitigated herd behavior and strengthened in the post-pandemic period, and uncertainty avoidance went from a negligible influence before the pandemic to a clear deterrent

during and after the crisis. Individualism exerts a small positive effect over the period, while indulgence lost its pre-pandemic protective role during the pandemic, and masculinity reversed its sign, reducing herd behavior in the post-pandemic period. This study contributes methodologically by integrating parametric and nonparametric approaches and, more substantively, by mapping how societal culture can change investor behavior during times of crisis. The findings underpin targeted regulatory oversight strategies, culturally sensitive crisis communication, and investor education, aiming to reduce correlated trading and increase market resilience in a world of recurring shocks.

2 THEORETICAL FRAMEWORK

2.1 From Values Orientation to Cultural Dimensions: Foundations and Critiques

Studies of human interactions among various societies depend on cultural analysis using many frameworks from international business, sociology, and education (Kirkman *et al.*, 2006; House *et al.*, 2004; Inglehart & Baker, 2000).

Kluckhohn and Strodtbeck (1961) developed a very influential framework about basic human values. Their Values Orientation Theory argues that, across cultures, significant human questions must be addressed regarding social relations, time orientation, nature orientation, and interpersonal relationships. The model is attractive for its generalizability, but it is simplistic and has been empirically tested only sparingly (Hills, 2002). It did, however, provide a platform for later efforts that were more structured in analyzing cultural variation and comparing cultures.

Hofstede's Cultural Dimensions Theory, developed from surveys IBM conducted in over 70 countries during the late 1960s and early 1970s, is perhaps most remembered for its focus on employees. Hofstede (2011) came up with six dimensions: power distance, individualism vs. collectivism, masculinity vs. femininity, uncertainty avoidance, long- or short-term orientation, indulgence versus restraint, providing a breakthrough for relating national culture to values at work and significantly enhancing cross-cultural analysis within international business research (Tung & Verbeke, 2010). Hofstede's work has also been adopted elsewhere because of its perceived ease of application in academia to bridge the gap between lessons taught and practice for managers and policymakers who expect cultural differentiation in negotiation, leadership, and organizational design (Kirkman *et al.*, 2006). Hofstede's research has become the most frequently cited contribution in the social sciences. By 2017, it ranked about 10,000 times in international business literature (Beugelsdijk *et al.*, 2017); "Culture's Consequences" has nearly 80,000 citations worldwide, underscoring its truly lasting impact. Most significantly, it was by examining the applicability of American management theories across different national contexts that their assumed universality was first brought into question (Jackson, 2020).

Despite this widespread use, Hofstede's model has been heavily criticized. To use nations as if they were cultures is to ignore internal diversity and variation between subcultures (Baskerville, 2003). Anthropologists and sociologists elsewhere protested the reduction of culture to numerical indices; they sought recognition of its complexity, dynamism, and contextual dependency (Baskerville, 2003; Signorini *et al.*, 2009). There were doubts about the data's representativeness, since it relied on responses from IBM employees who might have been influenced by their corporate

environment when answering the questions (McSweeney, 2002). It is accused of oversimplification, reducing culture to overlapping dimensions that are, in fact, difficult to replicate (Smith *et al.*, 1996). Another major criticism of Hofstede is that, by applying positivist principles, he treats culture as a static phenomenon when, in reality, it is dynamic and ever-evolving (Jackson, 2020). This has also been criticized as an ecological fallacy, since national averages are often taken to apply to individuals within the nation, thereby masking intra-country variation (Kirkman *et al.*, 2006). More than that, decades-old data underpin Hofstede's dimensions. The contemporary relevance of these has to be tested in the modern globalized world (Beugelsdijk *et al.*, 2017). Critics of applications in education have noted the misdirection caused by applying workplace-derived dimensions to student-teacher relationships since this helps reduce dynamics informed not just by culture but also by socio-economic and institutional factors (Signorini *et al.*, 2009).

Trompenaars identified seven dimensions based on relational dilemmas, including universalism vs. particularism and achievement vs. ascription (Hampden-Turner & Trompenaars, 1997). Although his method has been acclaimed for its practical relevance to managers compared with Hofstede's, Hofstede has later reanalyzed and indicated that only two of Trompenaars' dimensions were statistically significant; in effect, both fell within his own construct of individualism (Hofstede, 1996). Trompenaars and Hampden Turner defended their circular, dilemma-based approach, rejecting Hofstede's linear categories and emphasizing reconciliation over opposition (Hampden Turner & Trompenaars, 1997). Cultural change was articulated by Inglehart's Postmaterialist Values Theory as a generational shift from materialist to postmaterialist values and provided a framework linking value orientation to modernization and democratization processes, albeit criticized as overly focused on economic development (Inglehart & Baker, 2000) and Schwartz's Theory of Basic Human Values conceptualized ten universally motivating values arranged into bipolar dimensions with psychometric rigor across more than eighty countries, even if potentially too abstract for ready diffusion among managers (Schwartz, 2012). The GLOBE Project differentiated between values ("should be") and practices ("as is") and related these to leadership effectiveness, conducting one of the most extensive empirical studies across 62 societies (House *et al.*, 2004). Despite such comprehensiveness, however, criticism has been leveled against the GLOBE Project regarding both its complexity and debates over negative correlations between values and practices (Javidan *et al.*, 2006; Tung & Verbeke, 2010).

Together, these theories highlight the dual nature of cultural measurement, revealing its potential insights and inherent limitations. Kluckhohn and Strodtbeck laid a basic, somewhat

simplistic foundation. The Hofstede model is explicit, influential, and practical- the model has driven decades of scholarship and practice in real terms outside slight methodological inadequacies and oversimplification- it has been criticized for. Trompenaars discussed more dilemma-oriented and reconciliatory emphases but less static robustness. Inglehart explained cultural change by reducing it to the paths of economic development. Schwartz submitted psychometric rigor as usability. GLOBE tied culture to leadership, but did so in a way that made things complicated and contentious.

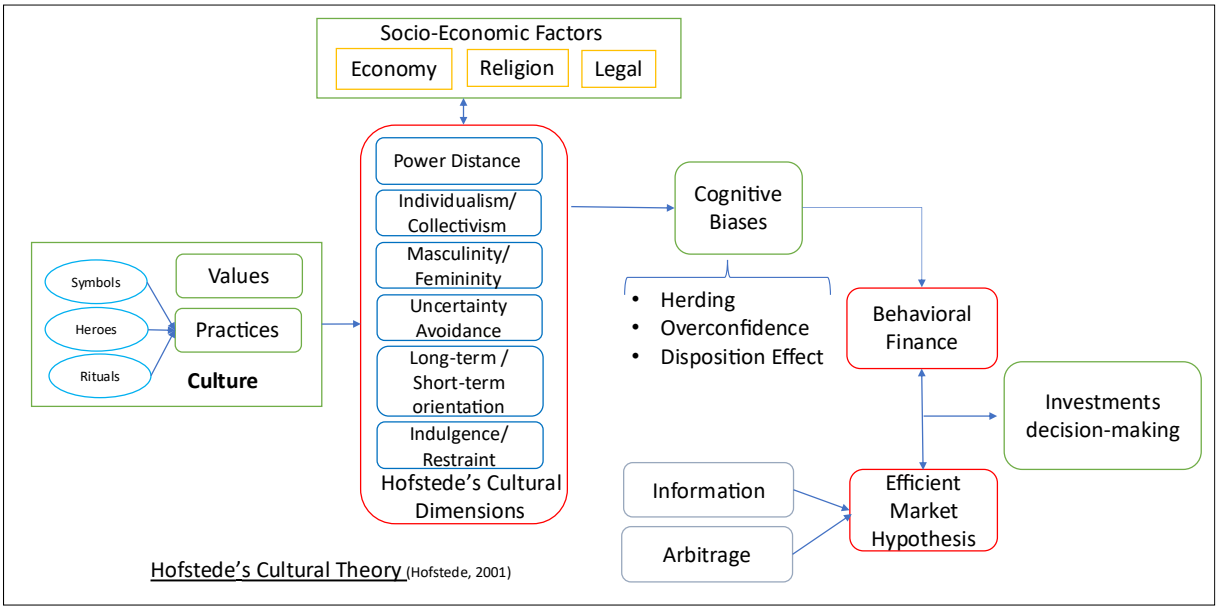
Though Hofstede has of late been the target of increasing criticism, his work is particularly apt for any study requiring a relatively macroscopic comparison across nations over time. In an analysis of 64 countries and their investor behavioral biases from 2010 to 2024, it is much easier to relate to and compare across different contexts when there is a recognizable, consistent baseline, such as Hofstede's cultural dimensions. Clarity, international reach, and long-lasting impact make it appropriate for assessing how cultural values may affect issues such as investor behavior. For these reasons, along with its strengths and wide acceptance, ours is based on the Hofstede framework.

Figure 1 - Theoretical Framework depicts the theoretical framework underpinning this research comprises several interrelated concepts and theories that collectively provide a lens for examining and understanding the relationship between national culture and investors' behavioral biases.

2.2 Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) is a fundamental concept in modern financial theory that has been the leading investment theory for over five decades. EMH is represented by Fama (1970) in three forms: (i) Strong-form efficiency, in practice, says that stock prices reflect all information-public, private, and even secret information such that investors cannot gain a competitive advantage; (ii) Semi-strong form: Stock prices respond to publicly available financial information including announcements, balance sheets of companies and their asset values; and (iii) Weak efficiency: Current stock prices fully reflect all past price data so that this information cannot be used to forecast future price movements.

Figure 1 - Theoretical Framework



Source: Elaborated by the author

The theory holds that investors cannot beat the market by relying on information as a primary tool for investment decision-making. They cannot outperform the market as the information is not proprietary and accessible to all. Therefore, no individual can be considered an investment expert or market specialist, as these traits apply to all investors. Any new information cannot generate abnormal profit as it is immediately available to the market and quickly reflected in stock prices. Even active managers cannot achieve high returns by exploiting available confidential information. The market objectively and informatively anticipates future movements, integrating information into market prices more effectively than insiders. Maximum returns from continuous trading are prevented when all information is reflected in stock prices. Market efficiency also depends on the investment method used by individual investors. EMH postulates that stock market participants act rationally. Rational investors focus on expected utility, leading to high returns and the formation of rational expectations. In markets where rational expectations are universal, prices fully reflect all available information; thus, equilibrium is achieved. EMH asserts that rational investor behavior is the fundamental assumption underlying all investment activities. If investors act irrationally and make random investment choices, equilibrium prices can deviate, but these deviations are short-lived because the irrational actions of different investors offset each other. Since irrational investors tend to buy more when stocks are either overpriced or underpriced, they generally earn lower returns than rational investors. So, these investors may be at a loss; they may witness depreciation in the value of their assets and a decline in the general ratings of the stock market. On the other hand, when rational investors engage in arbitrage to restore prices to equilibrium, this also indicates that markets are

efficient and thus achieve the goal of maximum profitability. The EMH maintains that, while not all investors act rationally, market dynamics are assumed to prevail under the rationality and efficiency assumptions (Konstantinidis *et al.*, 2012).

2.3 Behavioral Finance

EMH has contributed to the misreading of stock market bubbles and has since been demoted by academicians and financial practitioners with the advent of Behavioral Finance (BF) in the early 1990s. BF is not merely an extension of EMH but rather an opposite and substitutive framework. This discipline attempts to describe how human psychology (in particular, behavior) affects investment decisions. It assumes that some financial results may be caused by less rational investor behavior influenced by biases, psychological factors, and heuristics (Shiller, 2003).

Heuristics are simple, experience-based problem-solving methods or shortcuts suggested to explain decision-making. Heuristics work through the unconscious process of attribute substitution. The most common heuristics used by investors include (i) Representativeness, whereby an individual relates some unfamiliar new event with a familiar one and finds some characteristics in common between two different happenings; (ii) Anchoring – a mental shortcut used in decisions that relies on an initial piece of information referred to as an anchor; (iii) Herding – the tendency for people to follow groups (or herds), then group behavior in decision-making situations; (iv) Overconfidence – the tendency for individuals to overestimate their skill or ability. More specifically, this type of overconfidence involves excessive confidence in one's knowledge and information, which may result in mistaken investment decisions (Konstantinidis *et al.*, 2012).

Kahneman and Tversky (1979) developed the initial outline of prospect theory, stating that investor decisions are also based on several cognitive illusions, or choices in which people weigh gains and losses differently, mainly dependent not on what is lost but on what has been gained. Examples of most common biases are: (i) Loss Aversion, which means people show more sensitivity to possible losses than to equivalent gains, (ii) Mental Accounting, the establishment of separate mental accounts determined by individual-specific characteristics through which past experiences and financial events are recorded, (iii) Regret Aversion, an investor's motivation not to feel the emotional pain that will be caused by making a bad investment decision; and (iv) Disposition Effect, the tendency observed in investors whereby appreciated assets are sold off quickly while depreciating ones are retained. This results in holding onto stocks rather than realizing a loss, thereby delaying the loss and reducing anticipated regret when acknowledging a failed investment decision.

Behavioral Finance is the body of finance that challenged the Efficient Market Hypothesis by incorporating psychological and behavioral factors into the investment decision-making process, thereby becoming the dominant paradigm in finance. It argues that investors do not make optimal investment decisions because their choices are heavily influenced by psychological processes.

2.3.1 Overconfidence

Overconfidence is a behavioral bias characterized by the tendency to overestimate one's knowledge, judgment, and ability to control outcomes, placing more weight on personal information than on public information (Bouteska & Regaieg, 2018; Trejos *et al.*, 2019). Overconfidence consistently increases trading volumes, portfolio concentration, and risk-bearing (Odean, 1998). An investor who overestimates his abilities may misconstrue market signals and engage in excessive trading, leading to market mispricing and anomalies that further exacerbate the destabilizing effects of this bias on financial markets (Trejos *et al.*, 2019; Bouteska *et al.*, 2023).

2.3.2 Disposition Effect

Though initially identified by Shefrin and Statman (1985) in the context of the Disposition Effect (DE), or the tendency for investors to sell winning stocks while holding onto losing ones, subsequent studies noted its prevalence across myriad demographics and markets. Odean (1998) was the first to empirically validate this tendency, demonstrating that people liquidate winning investments too soon but retain losing investments for long periods. Later work confirmed that gender, age, and the strength of the disposition effect; more developed markets and education reduce the strength of the DE (Frino *et al.*, 2015; Breitmayer *et al.*, 2019). Recent research has indicated even more strongly how significant this bias can be under certain conditions: low financial knowledge, use of passive strategies, large portfolio stakes, home bias, and conditions that increase information asymmetry all create reluctance to realize losses. This makes clear both the global scope and multifaceted nature of DE (Ahn, 2022; Hwang *et al.*, 2024; Fan & Neupane, 2024; Han & Preda, 2025; Sui & Wang, 2025).

2.3.3 Herding

Herding occurs when investors change their choices in response to others' choices, driven by factors such as information gaps, pay incentives, or a tendency to follow the crowd (Bikhchandani & Sharma, 2000). Even though herding is often associated with crazy market phenomena such as bubbles or mispriced assets, it can also occur in rational situations through mechanisms such as informational cascades and social thresholds (Hirshleifer & Teoh, 2003). Facts from studies show

that herding narrows the spread of returns during significant market moves because everyone goes with the flow rather than making their own judgments about each asset (Christie & Huang, 1995). Cross-country studies further reveal variation in herding intensity: while limited in the United States and Hong Kong, it is more pronounced in Japan and especially in emerging markets such as South Korea and Taiwan, where macroeconomic factors play a stronger role than firm-specific information (Chang *et al.*, 2000).

2.4 Religion, Economy, and Legal

Culture, religion, economy, and law are the four key elements that mold societies and their global interactions. Hofstede (2011) described national culture as the collective beliefs, values, customs, behaviors, and artifacts that society members utilize to navigate their world. He views "culture" as the collective mental programming that distinguishes one group from others, suggesting that culture is always a collective phenomenon. Shared mental coding differentiates groups and is often associated with tribes, ethnic groups, nations, and organizations. Cultural aspects related to society, nation, and gender, learned from early childhood, are deeply ingrained in our minds, more so than the cultures of occupation or organization we learn later in life. Societal cultures are primarily value-based, since values represent broad preferences of one situation over another.

The relationship between culture and religion has long been a topic of discussion. It is imperative at this point to note the mutual influence between the two spheres: just as culture informs religion, so does religion prescriptively inform cultural expression (Beyers, 2017). Besides, culture embodies the ways and means by which humans apprehend and appreciate reality; Religion is an instrumental means for interpretation and understanding. Thus, the relation arises from an axiom of inseparability (Abdulla, 2018).

Cultural background and the values attached to it are closely related to economic considerations (Kostis, 2021). In a capitalist society, culture has increasingly become commodified. What we wear, eat, how we house ourselves, entertain, or travel, and other cultural elements are deeply influenced by, and in some cases dictated by, the economic situation in which these choices are made in all societies. Thus, culture is very much shaped by its relations with the economy (Clammer, 2016).

Regulations and rules often reflect the culture they come from (Silbey, 2018). The interaction between law and culture has long been a subject of academic interest, with the two seen as social entities constantly arguing, negotiating, and reconciling. In the view of socio-legal theory, they are connected; continuous disputes and conflicts determine their shapes. Law and culture do not exist as

separate ideas but rather continuously inform each other through constant interaction and conflict. In the end, while being influenced by society's cultural norms, religious beliefs, or economic structures, it, in turn, influences them (Yazdiha, 2017).

2.5 Hofstede's Cultural Theory

2.5.1 Culture

According to Hofstede (2001), the extent to which behavior can be predicted (determined by individuals' mental programming and other contextual factors) is necessary for the existence and functioning of social systems or structures of relationships among individuals, groups, or institutions. Mental programming instills a strong determination to behave similarly in similar situations, creating consistent behavioral patterns. This programming varies from individual to individual, with some components distinctive and others common. Mental programming runs at three dimensions: (i) the universal level, this implies the core commonalities of all humans and represent biological function and expressive behaviors like smile or weep; (ii) collective level, common to particular group or category of people and carry cultural elements (e.g. language, elder respect, personal space, perception of activity); and (iii) individual level which detail distinctive features of mental programming that make up personal identity and allow different behavior within same culture. Mental programs can be both inherited (genetic) and learned from experience. The universal level is likely entirely inherited, while the individual level is partly inherited, explaining differences in capabilities and temperament among siblings raised in similar environments. The collective level is mainly learned, as evidenced by shared mental programming among people with different genetic makeup but similar learning experiences. Societies, organizations, and groups have methods for preserving and transmitting mental programs across generations.

Values and culture are critical elements in understanding mental programming. Values are non-rational feelings that have intensity (relevance to us) and direction (indicating whether outcomes are good or bad). They are programmed early in our lives and become visible through behavior. On the other hand, culture is the collective mental programming that differentiates one group from another. It manifests in visible elements such as symbols (words, gestures, pictures, or objects with complex meanings understood only by those within the culture), heroes (individuals, real or imaginary, who embody valued characteristics and serve as behavioral models), and rituals (socially essential collective activities that bind individuals to the norms of the group), collectively referred to as practices. These practices are interpreted differently by insiders and outsiders. Culture to a group

is what personality is to an individual. The common characteristics influence a group's response to its environment. Societies, being the most complete human groups, are significant in cultural studies. They are self-sufficient social systems with interdependent categories. The homogeneity of national culture varies across societies and can be especially low in newer nations. However, different cultural groups within a society usually share certain traits that make them recognizable to outsiders (Hofstede, 2001).

Hofstede (2001) proposes that societal mechanisms maintain cultural patterns across generations. At the core of this is a system of societal norms, or shared value systems, among significant groups in the population. These norms originate from various ecological factors that affect the physical environment (e.g., geography, demography) and the social environment (e.g., urbanization, economy, technology). Societal norms shape the creation and maintenance of societal institutions, such as family structures, education systems, political systems, and legislation. Once established, these institutions reinforce the societal norms and the ecological conditions that led to their formation. While institutions may change, this does not necessarily affect societal norms. If these norms remain unchanged, the persistent influence of a majority value system gradually aligns the new institutions with societal norms.

2.5.2 National Culture Dimensions

Hofstede (2011) describes the six cultural dimensions. A dimension is a cultural feature that can be quantified and compared with those of other cultures. Every country has been ranked relative to others based on its scores across these dimensions. These dimensions are statistically unique and can appear in any combination, though certain combinations are more prevalent than others.

2.5.2.1 Power Distance (PDI)

Power Distance is a concept that describes how people with less power in organizations and societal structures, like families, perceive and accept the unequal distribution of power. Those in power do not define this concept, but those governed by it do. It implies that the degree of inequality in a society is as much accepted by those who are led as by those who lead. Power and inequality are fundamental aspects of every society, though the extent of inequality varies. The scores are generally higher in Eastern European, Latin, Asian, and African countries and lower in Western countries where Germanic and English are spoken. These scores suggest that power distance is more accepted in the former group of countries than in the latter (Hofstede, 1983).

2.5.2.2 Individualism and Collectivism (IDV)

Individualism and Collectivism are societal characteristics that measure the degree of group integration in a society. Individualistic societies are characterized by loose ties among individuals, in which each person is expected to care for themselves and their immediate family. On the other hand, collectivist societies are characterized by intense, cohesive in-groups, often extended families, that provide protection in return for unwavering loyalty and often oppose other in-groups. This dimension addresses a fundamental aspect of all societies. Developed and Western countries are individualistic, while less developed and Eastern countries are collectivist. Japan, interestingly, holds a middle ground in this regard (Hofstede, 2011; Brilei *et al*., 2000).

2.5.2.3 Masculinity and Femininity (MAS)

Masculinity and Femininity, as societal characteristics, refer to the distribution of values between genders, a fundamental aspect of any society. Hofstede (2001) found that women's values vary less across societies than men's. Men's values range from being very assertive and competitive, which are significantly different from women's, to being modest and caring, which are similar to women's values. The assertive end of the spectrum is termed masculine, while the modest, caring end is termed feminine. In 'feminine' countries, women share the same modest, caring values as men. In 'masculine' countries, women are somewhat assertive and competitive, but not as much as men, leading to a gap in values between the genders. Masculinity is high in countries such as Japan, German-speaking countries, and some Latin countries, such as Italy and Mexico. It is moderately high in English-speaking Western countries, low in Nordic countries and the Netherlands, and moderately low in some Latin and Asian countries, such as France, Spain, Portugal, Chile, Korea, and Thailand (Hofstede, 2011).

2.5.2.4 Uncertainty Avoidance (UAV)

Uncertainty Avoidance, distinct from risk avoidance, refers to a society's level of comfort with ambiguity and its ability to handle unstructured situations that are new, unknown, or different from the norm. Societies that avoid uncertainty strive to reduce the occurrence of such situations by implementing strict codes of conduct, laws, and rules, discouraging dissent, and adhering to an absolute truth. These societies tend to be more emotional and driven by internal nervous energy. On the other hand, societies that accept uncertainty are more open to diverse opinions, have fewer rules, and are more empiricist and relativist at the philosophical and religious levels, allowing for various perspectives. People in these societies are typically more composed and contemplative and are not expected to express their emotions outwardly. The Uncertainty Avoidance Index is generally higher

in Eastern and Central European, Latin American, Japanese, and German-speaking countries and lower in English-speaking, Nordic, and Chinese-speaking countries (Hofstede, 2011; Rieger & Wang, 2012).

2.5.2.5 Long-term versus Short-term Orientation (LTO)

This dimension, initially identified in a survey among students from 23 countries using a questionnaire created by Chinese scholars, was associated with Confucianism and hard work (Minkov & Hofstede, 2011). It was first named Confucian Work Dynamism by the study's lead author, Michael Harris Bond, due to its strong correlation with recent economic growth. However, as the study involved students from various countries, many were unfamiliar with Confucius, so it was renamed Long-Term versus Short-Term Orientation. The long-term orientation, corresponding to Confucian Work Dynamism, values perseverance, thrift, status-based relationships, and a sense of shame. The short-term orientation values reciprocating social obligations, traditional respect, face protection, and personal steadiness and stability. These values, already present in Confucius' teachings around 500 BC, are not solely Confucianism but are still found in countries with a Confucian heritage (Minkov & Hofstede, 2011). East Asian and Eastern and Central European countries tend to have a long-term orientation. South and North European and South Asian countries have a medium-term orientation. Short-term orientation is expected in the United States, Australia, Latin America, Africa, and Muslim countries (Hofstede, 2011).

2.5.2.6 Indulgence versus Restraint (IDG)

This dimension focuses on aspects not covered by the other five dimensions and is known from happiness research. Indulgence represents a society that allows for the relatively free fulfillment of primary, natural human desires for enjoyment and fun. On the other hand, restraint represents a society that controls the fulfillment of needs and regulates them through strict social norms (Breitmayer *et al.*, 2019). This sixth dimension is somewhat complementary to the Long-versus Short-Term Orientation dimension, with a weak negative correlation (Minkov & Hofstede, 2011). Indulgence is usual in South and North America, Western Europe, and Sub-Saharan Africa. Restraint is prevalent in Eastern Europe, Asia, and the Muslim world. Mediterranean Europe sits in the middle on this dimension (Hofstede, 2011).

3 HOW DOES NATIONAL CULTURE SHAPE INVESTOR BEHAVIOR? MAPPING A RESEARCH AGENDA

3.1 Introduction

Hofstede (2011) defines "culture" as a collective mindset that distinguishes one group of people from others and represents a shared program of values and beliefs embraced by each group. Culture is often associated with tribes, ethnic groups, nations, and organizations, and these associations reflect how societies see themselves (Triandis, 2002) and their social environment (Triandis, 1972), thereby linking culture to societal memory (Kluckhohn, 1954). Cultural aspects learned from early childhood are deeply embedded, offering subconscious ideas about how to behave (Kluckhohn, 1954). Furthermore, cultures are rooted in (often unconscious) values that reflect successful past experiences, and that promise valuable insights for future generations (Goodell, Kumar, *et al.*, 2023). In turn, a national culture manifests as dominant customs and traditional beliefs embraced by most people within a nation, along with institutions that contribute to shaping the national community's identity (Beugelsdijk & Maseland, 2011).

Because each national culture is shaped by norms and institutions, it also influences economic, political, religious, educational, and philosophical systems (Goodell, 2019; Inglehart, 1988), as well as individual behaviors. In this sense, culture is likely to be an essential driver of financial decision-making, risk-taking (Adhikari & Agrawal, 2016; Baele *et al.*, 2014; Williamson, 2010), investors' reactions to unexpected information, and investment choices (Chang & Lin, 2015). However, predicting such effects of cultural diversity challenges traditional assumptions about how people act. That is, rational decision-makers would always act to maximize their utility or investment returns and minimize their risk (Fama, 1970) by making prudent, logical decisions (Nadler & Breuer, 2019). According to the efficient market hypothesis, market prices accurately reflect all publicly available information because investors are rational in their information processing and market choices. But if culture, along with other non-pecuniary or emotional motives, defines what constitutes utility maximization, then such assumptions of rational decision-making are insufficient. Accordingly, finance researchers recognize both rational and irrational drivers of human judgment that depend on context (Cox Jr., 2023; Ganuthula, 2024). As Shiller (2016) puts it, there might be a single way to behave rationally, but there are countless ways to behave irrationally. Even goal-oriented investors are bound in their rationality (Simon, 1947), due to both human cognitive limits and the nature and complexity of the environments. Although some behavioral finance research predicts impacts of cultural factors on investors' behavior and risk preferences (e.g., Aren & Hamamci, 2020;

Mourouzidou-Damtsa *et al.*, 2019; Srivastava *et al.*, 2020), the question of how national culture actually shapes investor behavior and decision-making remains unanswered.

To address them, we turn to Hofstede's contributions. In addition to offering an influential definition of culture, Hofstede (2011) provides quantifiable measures that span six dimensions: power distance, individualism, masculinity, uncertainty avoidance, long-term orientation, and indulgence. Despite some reasonable criticisms of the dimensions (e.g., Beugelsdijk *et al.*, 2015) and the external validity and internal consistency of the scale (e.g., Gerlach & Eriksson, 2021), Hofstede's cultural dimensions continue to enjoy broad acceptance in cross-cultural, empirical research spanning various academic fields (Orr & Hauser, 2008).

For example, Xu (2022) shows that cultural tendencies toward risk aversion and ambiguity avoidance play a crucial role in explaining these differences. Moreover, Boahen and Mamatzakis (2025) pointed out that some cultural characteristics (e.g., high power distance, masculinity, and uncertainty avoidance) might encourage unethical practices, suggesting that strengthening legal environments can help counteract the negative effects of these cultural traits. Furthermore, Goodell (2019) shows that national culture influences financial systems and uses Hofstede's framework to establish that differentiating between intended rationality and institutional theory is critical for cultural finance studies. Culture in finance pertains to the way national culture influences financial systems. This can occur by shaping behavioral norms and the logic of appropriateness or by creating environments that affect incentives for individuals to maximize their utilities (Karolyi, 2016). In their bibliometric analysis of culture and finance, Goodell, Kumar, *et al.* (2023) also provide a broad overview of the role of culture in finance and identify key themes, including governance, financial decision-making, investing, and social trust. Building on this foundation, we focus explicitly on the intersection of national culture and investor behavior, using Hofstede's framework. In support of this effort, we undertake bibliometric and qualitative reviews to determine (1) the most prolific contributors (journals, countries, institutions, affiliations, authors) to the field, (2) the most cited articles, (3) the most important themes pertaining to culture and investors' behavior in capital markets, and (4) which trends and gaps indicate areas for further research.

Bibliometric analysis provides an effective means to map the evolution of a research domain (Ellegaard & Wallin, 2015) and establish a theoretical framework for continued, empirical research (Lim *et al.*, 2022). We complement this quantitative bibliometric analysis with a qualitative content analysis of articles published in the cultural finance field between 2009 and 2024 in ABS2+ journals. Briefly, the quantitative analysis identifies *Pacific-Basin Finance Journal* (8), *International Review*

of Financial Analysis (5), *Journal of International Finance Markets Institutions and Money* (5), and *Journal of Banking and Finance* (4) as the journals that have published the most relevant articles; furthermore, *International Review of Financial Analysis* features the top ten most cited articles. We determine that Schmeling (2009) is the most influential author.

Turning to the content analysis, we also identify three thematic clusters. First, studies that address investors' decision-making detail how information scarcity or overload exerts distinct impacts on investors' behavior across cultures. For example, cultures that prioritize short-term information indicate greater investment myopia than those with collective mindsets. Second, the market and investors' cultural preferences cluster involves research into how cultural practices (e.g., Hajj pilgrimage in Saudi Arabia, Chinese lunar calendar) and superstitions (e.g., lucky numbers) affect financial market dynamics. Third, research in the culture and behavioral biases cluster links investor behaviors and biases (e.g., herding, disposition effect) to cultural factors. By combining these insights, we can address the intricate, varied patterns of investor behavior across culture-specific countries and economic regions.

To establish these contributions, we present Hofstede's framework in Section 3.2, then describe the methodology in Section 3.3. Section 3.4 contains the results of the bibliometric analysis, which we use in Sections 3.5 and 3.6 to analyze the content of some selected, relevant studies. After we highlight some current research trends in Section 3.7, we outline critical directions for further research, in an attempt to establish relevant insights for academic, professional, and societal stakeholders. Finally, we conclude in Section 3.8.

3.2 National culture and Hofstede's framework

Culture can be defined in many ways. Hofstede's cultural dimensions theory refers to cross-cultural differences in organizational and national culture, defined as "the collective programming of the mind that distinguishes the members of one group or category of people from others" (Hofstede, 2011, p. 3). Hofstede (2011) reflecting his argument that there are three levels of human mental programming: universal (shared by all humans, biological, likely inherited), collective (common to people belonging to certain groups or categories, learned), and individual (unique, at the individual personality level, which produces a range of behaviors within the same collective, likely both learned and inherited). Underscore its innovative introduction of an "objective" measurement of culture that uses specific dimensions to understand and explain differences across national cultures,

this framework has been applied frequently in various research domains[1] (Goodell, Kumar *et al*., 2023). In particular, it underlies studies that define national culture as the subset of institutions related to collective identity (e.g., Baele *et al*., 2014), such that it exerts strong influence on the behavior of individual members of the culture across different contexts (Adhikari & Agrawal, 2016). In turn, culture has been shown to influence financial decision-making through different mechanisms, including individual decision-making at the micro level (Mihet, 2013) and cultural norms created by formal institutions (Baele *et al*., 2014).

Although it offers an effective measure of cross-country differences, Hofstede's framework has also come in for criticism, including allegations that it relies on flawed assumptions (e.g., McSweeney, 2002) and outdated input data (e.g., Taras *et al*., 2023). Scholars question its generalizability (Schwartz, 1994), reliability, stable factor structure, and measurement invariance (Gerlach & Eriksson, 2021). Seemingly in response to such concerns, Hofstede has refined and revised the model of culture. Specifically, the original version included four dimensions: power distance, individualism–collectivism, masculinity–femininity, and uncertainty avoidance. These metrics constituted a single framework that claimed to reflect international differences in corporate risk-taking and performance (Hofstede *et al*., 2010). Later revisions added the long-term orientation and indulgence–restraint dimensions. A long-term orientation, also referred to as Confucian dynamism, seeks to distinguish the types of thinking demonstrated by Eastern (e.g., Chinese) and Western cultures (Hofstede & Bond, 1988). Hofstede *et al*. (2010) added the indulgence–restraint dimension to address happiness.

3.2.1 Power Distance

Power distance relates to "the extent to which the less powerful persons in a society [or organization] accept inequality in power and consider it as normal" (Hofstede, 1986, p. 307). In societies that score higher on power distance, people accept a hierarchical order in which supervisors and senior family members have different rights, obligations, and privileges, without requiring any further justification. Members of societies that score low on power distance instead seek to equalize power, do not exhibit reverence for senior members of the organization or family, and demand justifications for power inequalities.

[1] Some alternative competing frameworks include Schwartz's Value Dimensions (Schwartz, 1992, 1994), the World Values Survey (WVS, www.worldvaluessurvey.org), and the Global Leadership and Organizational Behavior Effectiveness (GLOBE) framework (Lubetsky, 2008).

3.2.2 Individualism–Collectivism

This dimension refers to the relationship between the individual and the group. In societies with high individualism values, each person puts their interests above those of the group, with the expectation that individuals should look after themselves and their immediate family (Hofstede, 1980, 2001). Collectivist societies instead prioritize group interests and promote the belief that, from birth onward, everyone is integrated into a strong, cohesive group that looks after them with unquestioning loyalty. Individualism–collectivism is a frequently used dimension in cross-cultural research.

3.2.3 Masculinity versus Femininity

Masculinity and femininity are societal constructs that shape the distribution of gender-based values in society. Hofstede (2001, p. 281) refers to "the degree to which such masculine values as advancement, earnings, training, up-to-datedness [versus] such feminine values as the friendly atmosphere, position security, physical conditions, and cooperation are valued." Masculine societies embrace assertiveness, competitiveness, heroes, and material rewards for success, such that they tend to be more competitive. Feminine societies instead display preferences for cooperation, humility, and caring for the weak, which produces a more consensus-oriented worldview. This dimension also implies a strong emphasis on the gender roles established in society (Basabe *et al*., 2002).

3.2.4 Uncertainty Avoidance

This dimension reflects "the extent to which people within a culture are made nervous by situations which they perceive as unstructured, unclear, or unpredictable, situations which they, therefore, try to avoid by maintaining strict codes of behavior and a belief in absolute truth" (Hofstede, 1986, p. 308). If a culture scores low on uncertainty avoidance, its members are likely to take a more relaxed behavior toward uncertainty and ambiguity. High uncertainty avoidance instead suggests that the culture emphasizes the importance of formal rules and social control to ensure certainty and conformity (Hofstede, 2001). Members of the latter cultures likely share a common understanding of what is true and perceive issues in black-and-white terms, which, in turn, can determine how trust can be established.

3.2.5 Indulgence versus Restraint

Initially introduced in happiness research, the indulgence versus restraint dimension was integrated later by Hofstede *et al*. (2010). Indulgence suggests that societies allow for free gratification of basic human desires, enjoying life, and having fun; restraint refers to societies that

suppress gratification and regulate it with strict norms. Indulgence, versus restraint, is associated with a greater inclination for free speech (Hofstede *et al*., 2010). Relatively few studies have evaluated this dimension (e.g., Breitmayer *et al*., 2019; Cakici & Zaremba, 2023; Drobetz *et al*., 2023; Persakis & Koutoupis, 2024; Zhan, 2019).

3.2.6 Long- versus Short-Term Orientation

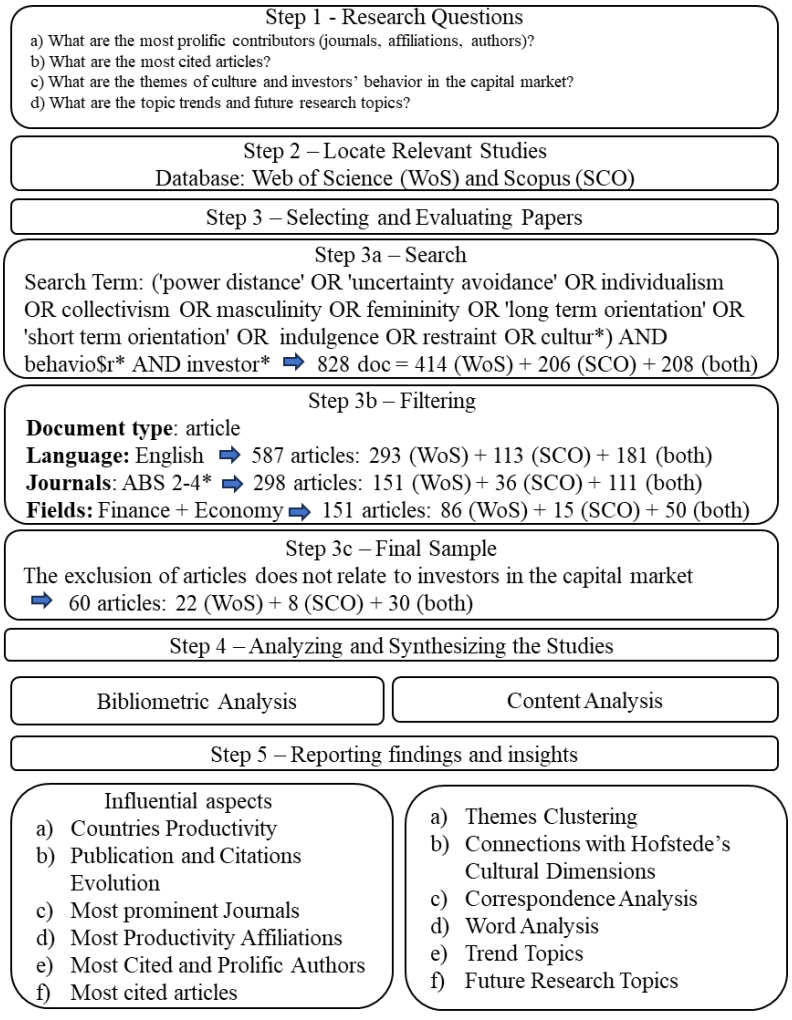
Finally, this dimension relates to how forward-looking a society is; as a newly introduced dimension, it offers relatively less clarity (Minkov *et al*., 2018). When originally introduced, with the label "Confucian dynamics" (Bond & Hofstede, 1989; Hofstede & Bond, 1988), it claimed to assess cultural orientations toward Confucianism. By revising this conceptualization to refer to a long-term orientation, Minkov and Hofstede (2012) were able to replicate the dimension in a wider sample of countries, using WVS items that seemed to capture the concept. It is based on values such as endurance, saving, hierarchical relationships, and honor, which reflect the Confucian principles of bringing the past into the present, national pride, respect for tradition, preservation of "face," and the need to fulfill social obligations, all while prioritizing future rewards and delayed gratification. In contrast, a short-term orientation focuses on immediate outcomes, such as meeting social duties, safeguarding reputations, and ensuring personal consistency, without the same emphasis on long-term planning or sacrifice. In this sense, a long-term orientation shares common points with the indulgence–restraint dimension when it comes to restraining oneself from indulgence to ensure future rewards. For example, educational attainment and long-term orientation display a strong correlation (Figlio *et al*., 2019).

3.3 Research Methodology

This research combines a bibliometric analysis with a systematic literature review (SLR). As we noted previously, bibliometric analyses are well-established and increasingly popular, likely due to their quantitative nature, such that they support assessments of vast data and thereby reduce the risk of interpretation bias (Donthu *et al*., 2021). Such analyses can reveal how scientific fields, social aspects, and conceptual aspects have evolved, such that they produce insightful maps of the structure of knowledge in a particular field of research (Ninkov *et al*., 2022). In contrast, SLR represents a structured methodology that provides a well-defined protocol for identifying, selecting, and critically evaluating extant literature. It adopts a comprehensive methodology to ensure rigor, objectivity, and transparency (Tranfield *et al*., 2003). For our SLR, we adopt the five-step protocol proposed by Khan *et al*. (2003): (1) formulate research questions by planning the search strategy and identifying

appropriate keywords; (2) locate relevant studies by identifying suitable repositories and target journals; (3) select and evaluate papers based on specific inclusion and exclusion criteria; (4) analyze and synthesize the studies; and (5) report the findings and insights. For the latter two steps, we use VOSviewer (Van Eck & Waltman, 2014), IRaMuTeQ software (Loubère & Ratinaud, 2014), and the bibliometrix package (Aria & Cuccurullo, 2017). Figure 2 illustrates our methodology.

Figure 2 – Research Methodology



Source: Elaborated by the authors

To address our research questions, we collected data in May 2024 from Clarivate Analytics' Web of Science (WoS) and Elsevier's Scopus databases, which represent primary, comprehensive citation indexes for numerous scientific areas and feature partial overlap (Kumpulainen & Seppänen, 2022). By integrating these data sources, we seek more comprehensive results and greater robustness for the bibliometric analysis. To these databases, we applied the following keywords in a compound query to select documents: investor and its variants AND behavior and its variants AND culture and its variants OR Hofstede cultural dimensions: power distance OR uncertainty avoidance OR

individualism OR collectivism OR masculinity OR femininity OR long-term orientation OR short-term orientation OR indulgence OR restraint. This initial search yielded 828 documents: 414 from WoS, 206 from Scopus, and 208 from both databases. Next, to restrict the scope, we included only papers written in English and excluded those published in journals not ranked in the 2021 Academic Journal Guide of the Chartered Association of Business Schools (ABS), as well as duplicates, resulting in 677 papers. Specifically, we restrict the sample to high-quality papers (ABS2+ rating) in the finance and economy fields (e.g., Goodell, Kumar, *et al*., 2023; Migliavacca *et al*., 2023).[2] Next, we read the abstracts as part of the screening process; if this step left any uncertainty about their relevance, we read the full papers for clarification. Through this screening process, we removed 91 articles because they related to corporate investment decisions (43), employed investments other than those in capital markets (17), adopted mathematical modeling (7), dealt with government policy (5), or covered other topics (19). The final database thus comprised 60 articles published between 2009 and May 2024 that represent the input for the bibliometric analysis.

For the SLR, we began with a computer-aided thematic analysis, which helps overcome challenges inherent to conventional thematic analyses (Niedbalski & Ślęzak, 2021). We use Reinert's (1990) technique to measure the frequency and organization of thematic classes statistically; it aims to increase the internal likeness of words within thematic categories and the distinctions across such categories. The stability of the outcomes is incorporated as a stopping criterion. This method then generates a tree diagram that illustrates a hierarchy of thematic classes; the closeness of these classes suggests a shared vocabulary. Furthermore, this method relies on lists of the most distinctive content words and most representative text segments, determined by chi-square (χ^2) values (Reinert, 1990). For this effort, we relied on the IRaMuTeQ software, which is based on R and Python (Loubère & Ratinaud, 2014). This approach only accounts for content words (e.g., nouns, verbs, adjectives, adverbs); it excludes grammatical endings to form "lexemes," and it also converts words to lowercase and eliminates accents. All these features are conveniently accessible in the IRaMuTeQ software and occur parallel to the identification of word syntaxes (Ratinaud, 2018; Sarrica *et al*., 2016).

3.4 Assessments of Influence

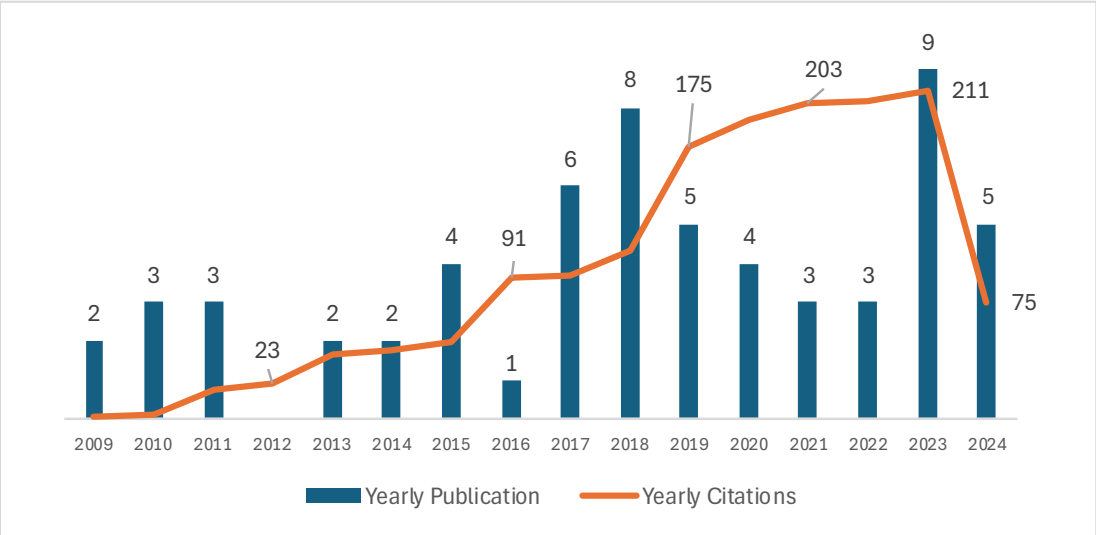
The bibliometric analysis pinpoints the most influential journals, as well as the most researched countries, institutions, affiliations, authors, and cited articles published in those journals.

[2] Bibliometric analysis relies on citations. The criteria for including articles should reflect the journal's citation count, not just journal rankings (Goodell, Oriani, *et al*., 2023). Noting the high citation rates of several ABS 2 journals in finance, we chose to include journals rated ABS2, along with those with rated 3, 4, or 4*.

3.4.1 Publication patterns

Figure 3 displays the publication and citation patterns for research on national culture and investors' behavior, revealing a significant increase in scientific production in recent years. Since 2017, the number of articles published has more than doubled compared with the previous period. Regarding citations, articles pertaining to culture and investors' behavior have grown consistently, from fewer than 100 citations prior to 2017 to more than 200 as of 2023.

Figure 3 - Evolution of Publications and Citations



Source: Elaborated by the authors

3.4.2 Influential journals and contributors

Table 1 depicts the most influential journals, in terms of both the total number of published papers and the number of incoming citations, through VOSviewer software.

Table 1 - Most prolific journals (papers and citations)

Rank	Journal	Number of Papers	Rank	Journal	Number of Citations
1	Pacific-Basin Finance Journal	8	1	Journal of Empirical Finance	440
2	International Review of Financial Analysis	5	2	Journal of Banking and Finance	299
3	Journal of International Financial Markets, Institutions, and Money	5	3	Pacific-Basin Finance Journal	155
4	Journal of Banking and Finance	4	4	Financial Review d	153
5	Journal of Behavioral Finance	3	5	International Review of Economics and Finance	78
6	European Journal of Finance	3	6	Financial Management	70
7	International Review of Economics and Finance	2	7	International Review of Financial Analysis	55

8	Financial Management	2	8	Journal of Financial Economics	50
9	Quarterly Review of Economics and Finance	2	9	Journal of International Financial Markets, Institutions, and Money	49
10	Economics Letters	2	10	Review of Financial Studies	42
11	Journal of Economic Studies	2	11	Quarterly Review of Economics and Finance	30
12	Journal of Economic Behavior and Organization	2	12	Economics Letters	16
13	Research in International Business and Finance	2	13	Journal of Economic Studies	16
14	Review of Quantitative Finance and Accounting	2	14	Journal of Financial and Quantitative Analysis	15
15	European Financial Management	2	15	Journal of Behavioral Finance	12
16	Finance Research Letters	2	16	Journal of Economic Behavior and Organization	12

Source: Elaborated by the authors
Note: The lead journals are determined by a) the total count of published papers and b) the number of citations in articles published in journals rated ABS2+ in the *Academic Journal Guide.*

As it shows, the *Pacific-Basin Finance Journal* ranks first, with eight articles, followed by *International Review of Financial Analysis* and *Journal of International Financial Markets, Institutions, and Money*, both of which feature five publications. These journals thus account for 30% of our sample. However, this ordering of journals does not reflect the classification of the most cited articles. In Table 1, the top journal in terms of the number of citations is *Journal of Empirical Finance* (440), followed by *Journal of Banking and Finance* (299) and *Pacific-Basin Finance Journal* (155).

Turning to the authors, the United States is home to the most significant number in our sample (13 authors), though 41% of authors are from the Asia-Pacific region (China 10, Australia 8, Japan 5, Taiwan 5, and New Zealand 4). Authors from Europe (the United Kingdom, 10; France, 4; and Germany, 4) also represent a significant portion (29%) of the sample (for further details, see Appendix A). Table 2 lists the most prolific institutions, in terms of both the number of papers published and citations received. The Chinese Culture University in Taiwan, Newcastle University in the United Kingdom, and Concordia University in Canada are the most productive institutions in our sample. The most cited institution is Leibniz University Hannover in Germany, with 440 citations, followed by the University of Wisconsin in the United States, with 163 citations. Auckland University of Technology in New Zealand and the University of Groningen in the Netherlands each received 157 citations.

Table 2 - Most prolific institutions (papers and citations)

Rank	**Affiliation**	**Country**	**Number of Papers**	**Rank**	**Affiliation**	**Country**	**Number of Citations**

1	Chinese Culture University	Taiwan	3	1	Leibniz University Hannover	Germany	440
2	Newcastle University	United Kingdom	3	2	University of Wisconsin	United States	163
3	Concordia University	Canada	3	3	Auckland University of Technology	New Zealand	157
4	University of Wisconsin	United States	2	4	University of Groningen	The Netherlands	157
5	Curtin University	Australia	2	5	University of Franche-Comte	France	153
6	Monash University	Australia	2	6	University of Grenoble	France	153
7	National Cheng Kung University	Taiwan	2	7	California State University of San Bernardino	United States	153
8	University of Dubai	U Arab Emirates	2	8	University of Wyoming,	United States	126
9	Poznan University of Economics and Business	Poland	2	9	University of Kansas	United States	126
10	Konan University	Japan	2	10	National University of Kaohsiung	Taiwan	75
11	Fuzhou University of International Studies and Trade	China	2	11	University of Washington	United States	50

Source: Elaborated by the authors

Next, Table 3 contains the list of the 18 most influential authors, according to the number of papers they published and the number of citations received. Darren Duxbury from Newcastle University and Seungho Lee from the University of Aberdeen, both in the United Kingdom, are the most productive authors. Each has published three papers in our sample. Maik Schmeling from Leibniz Universität Hannover in Germany is the author who has received the most citations (440) to date. As the findings from Table 3 also indicate, these impactful authors have affiliations with different universities, often in the United States (5).

Table 3 - Most productive authors (papers and citations)

Rank	Author	Institution	Country	Number of Papers	Number of Citations
1	Duxbury, Darren	Newcastle University	United Kingdom	3	22
2	Lee, Seungho	University of Aberdeen	United Kingdom	3	15
3	Switzer, Lorne N.	Concordia University	Canada	2	15
4	Wang, Jun	University of Western Ontario	Canada	2	15
5	Zaremba, Adam	Montpellier Business School	France	2	8

6	Chang, Matthew C.	Chinese Culture University	Taiwan	2	3
7	Schmeling, Maik	Leibniz Universität Hannover	Germany	1	440
8	Beugelsdijk, Sjoerd	University of Groningen	The Netherlands	1	157
9	Frijns, Bart	Auckland University of Technology	New Zealand	1	157
10	Zouaoui, Mohamed	University of Franche-Comte	France	1	153
11	Nouyrigat, Genevieve	University of Grenoble	France	1	153
12	Beer, Francisca	California State University of San Bernardino	United States	1	153
13	Anderson, Christopher	University of Kansas	United States	1	126
14	Fedenia, Mark	University of Wisconsin-Madison	United States	1	126
15	Hirschey, Mark	University of Kansas	United States	1	126
16	Skiba, Hilla	University of Wyoming	United States	1	126
17	Chang, Chih-Hsiang	National University of Kaohsiung	Taiwan	1	75
18	Lin, Shih-Jia	National University of Kaohsiung	Taiwan	1	75

Source: Elaborated by the authors
Notes: The authors are ranked according to the number of papers published and then the number of citations received from other articles published in journals rated ABS2+ in the *Academic Journal Guide.*

Finally, Table 4 indicates the most cited articles. In detail, "Investor Sentiment and Stock Returns: Some International Evidence" (Schmeling, 2009) has received 440 citations; it explores the impact of investor confidence on anticipated stock returns in 18 developed countries. The results indicate a negative correlation between sentiment and overall stock market returns across these nations; when sentiment is positive (negative), future returns typically are lower (higher). The study also reveals that the effect of sentiment on stock returns is more significant in countries characterized by less transparency (defined as institutional quality in a stock market that enables efficient information flows) and cultural tendencies toward herd behavior and overreaction. It thus indicates a conditional impact of culture at the national level on investor behavior. Beugelsdijk and Frijns's (2010) article "A Cultural Explanation of the Foreign Bias in International Asset allocation" received the second most citations (157); it shows that national-level uncertainty avoidance, individualism, and cultural distance strongly influence foreign asset allocation decisions. Countries with high uncertainty avoidance tend to allocate fewer assets to foreign markets; individualistic societies instead exhibit more aggressive foreign asset allocations. Their paper thus reflects on the role of culture in

international market asset allocations. Finally, the third most cited paper, by Zouaoui *et al*. (2011) and titled "How Does Investor Sentiment Affect Stock Market Crises? Evidence from Panel Data," suggests that investor sentiment significantly influences the probability of stock market crises within a one-year horizon, which represents a challenge to standard finance theory. The authors also find that this relationship is more pronounced in countries with cultural tendencies toward herd behavior and weaker regulatory institutions.

Table 4 - Most cited articles

Rank	Author(s)	Title	Citations
1	Schmeling (2009)	Investor sentiment and stock returns: Some international evidence	440
2	Beugelsdijk & Frijns (2010)	A cultural explanation of the foreign bias in international asset allocation	157
3	Zouaoui *et al.* (2011)	How does investor sentiment affect stock market crises? Evidence from panel data	153
4	Anderson *et al.* (2011)	Cultural influences on home bias and international diversification by institutional investors	126
5	Chang & Lin (2015)	The effects of national culture and behavioral pitfalls on investors' decision-making: Herding behavior in international stock markets	75
6	Cronqvist *et al*. (2015)	Value versus growth investing: Why do different investors have different styles?	50
7	Huang (2015)	Thinking Outside the Borders: Investors' Underreaction to Foreign Operations Information	42
8	Ng & Wu (2010)	Peer Effects in the Trading Decisions of Individual Investors	37
9	Bley & Saad (2010)	Cross-cultural differences in seasonality	35
10	Adachi *et al.* (2017)	Google search intensity and its relationship to the returns and liquidity of Japanese startup stocks	34
11	Bergsma & Jiang (2016)	Cultural New Year Holidays and Stock Returns around the World	33
12	Cheema & Nartea (2014)	Momentum returns and information uncertainty: Evidence from China	32
13	Chourou *et al.* (2018)	How does national culture influence IPO underpricing?	30

Source: Elaborated by the authors
Notes: All these articles received at least 20 citations from other articles published in journals rated ABS2+ in the *Academic Journal Guide*.

3.5 Content Analysis

To uncover the structure of the research streams pertaining to culture and investors' behavior, we conducted a content analysis, using Reinert's (1990) method and the IRaMuTeQ software. We adopted the software's default parameters: 40 instances per text segment, 10 end clusters, and a maximum limit of 3,000 most common forms. The 60 abstracts from our sample comprised 219 text segments, and 201 of those segments (i.e., 91.78% of the total) could be categorized into three main thematic clusters: investors' decision-making, market and investors' cultural preferences, and culture

and behavioral biases, which account for 32.34%, 20.90%, and 46.77%, respectively, of extant research into how national culture sways investors' behavior. Appendixes B and C list these thematic cluster studies according to geographic coverage and Hofstede's cultural dimensions, respectively. The illustration of the clusters in Figure 4 also specifies the most significant words in each and the respective χ^2 values; values greater than 3.85 indicate statistical significance at a 95% level.

3.5.1 Investors' decision-making

The first cluster includes articles that seek to demonstrate how information and cultural contexts drive investors' decision-making. Information might come from company reports (Wijayana & Gray, 2018), the internet (Adachi *et al*., 2017; Lee *et al*., 2023; Qadan & Zoua'bi, 2019), news (Huang, 2015), ads (Aman *et al*., 2023), or word of mouth (Ng & Wu, 2010). As Aman *et al*. (2023) show, soft news obtained from entertaining television ads boosts the level of stock market activity in Japan. The ability to forecast returns is proportional to investors' exposure to international news (Huang, 2015). Qadan and Zoua'bi (2019) also find a positive correlation between investors' use of financial websites and uncertainty, and this influence seems to hold regardless of the accuracy or distortion of the information (Lee *et al*., 2023). Adachi *et al*. (2017) establish a positive association between the intensity of Google searches for stock returns and their liquidity. That is, investors' behavior is significantly shaped by news and information disseminated through online social media platforms.

The quantity and quality of information are also essential to investors' decision-making. Several studies address information scarcity (Azzi & Suchard, 2019; Speidell, 2009) or overload (Nguyen & Truong, 2013; Todea & Buglea, 2017), as well as the quality of the information (Cheema & Nartea, 2014). As Speidell (2009) notes, information scarcity and cultural nuances are critical challenges in frontier markets. Azzi and Suchard (2019) discuss information gaps in Chinese financial markets specifically; they assert that these still developing financial markets face challenges related to regulatory unpredictability and cultural and legal complexities. Other studies detail how the information transmission channels can affect investors' behavior (Mbarki *et al*., 2022). In addition, when investors focus on long-term outcomes, it significantly enhances post-merger performance, whereas a short-term focus tends to affect the success of takeovers (Breuer *et al*., 2020). Docherty and Hurst (2018) explore the influences of the cultural context on market dynamics and particularly the tendency to prioritize short-term information. They propose that investors neglect the long-term value of assets when they operate in cultures with a strong focus on immediate market information; such momentum effects are especially frequent in cultures that exhibit greater investment myopia. In

line with this perspective, Todea and Buglea (2017) show that in countries with a collective cultural mindset, stock prices are strongly influenced by broad market information. Applying psychological insights into cultural differences to understand the information conveyed by global stock markets, Nguyen and Truong (2013) propose that stock markets in individualistic and low uncertainty avoidance countries tend to contain more information content. Other papers in this cluster address related aspects, like choices (Cronqvist *et al*., 2015) and focus (Breuer *et al*., 2020).

Figure 4 - Relevant Themes



Source: Elaborated by the authors
Notes: The figure displays the most significant words of each cluster and their chi-square (χ^2) values; χ^2 greater than 3.85 is statistically significant at the 95% level.

Investor behavior, particularly their risk-taking, is thus shaped by cultural factors (Lee *et al*., 2019). For example, cultural masculinity is linked to higher risk appetites (e.g., Meier-Pesti & Penz, 2008; Zheng *et al*., 2013). In contrast, femininity appeared to have no significant impact on risk-taking (e.g., Meier-Pesti & Penz, 2008)

Risk and investment activity also correlate positively in societies that value individualism (Nguyen & Truong, 2013). Beyond risky behaviors, though, investors tend to act more unpredictably in uncertain policy environments (Persakis & Koutoupis, 2024). In such contexts, it suggests that cultures with low uncertainty avoidance—being more comfortable with ambiguity and unpredictability—may exhibit even more erratic or unpredictable investor behavior compared to cultures with high uncertainty avoidance, which typically prefer stability and clear guidelines. As Wijayana and Gray (2018) show, markets' reactions to poor earnings reports by multinational companies depend on the culturally informed accounting principles of their home countries. Therefore, cultures with high uncertainty avoidance may react more negatively to poor earnings reports due to a preference for stability and predictability. Similarly, long-term-oriented cultures may focus more on future potential than short-term setbacks. Companies operating in countries that lean toward conservatism and opaqueness face more adverse market reactions, whereas those favoring optimism and transparency evoke more positive reactions. Accordingly, cultures with high uncertainty avoidance and feminine traits (e.g., valuing transparency and care) may react more positively to optimism and transparency, while conservative and opaque practices may align with cultures with masculine traits (e.g., competitiveness, result-driven behavior) that can lead to adverse reactions due to a lack of trust. Investors' optimism explains the underperformance of Japanese firms following their initial public offering (IPO), for example (Ikeda, 2023). Japanese culture, which is typically long-term-oriented and high in uncertainty avoidance, may lead to cautious investor behavior and lower optimism about short-term gains, resulting in IPO underperformance. Investors also tend to maintain their initial optimism about cross-border acquisitions in the long term, as well as when the target firm is from a country with a large economy and that prefers to avoid uncertainty (Zeng *et al*., 2023). Hence, investors may maintain optimism in cross-border acquisitions when the target country has low uncertainty avoidance and aligns with long-term goals, as these factors reduce perceived risk. Cultural biases, such as a preference for stability (high uncertainty avoidance) or a focus on individual gains (individualism), might cause investors to overreact to uncertainty. In Duxbury and Yao's (2017) study, investors follow a momentum strategy when buying stocks and a contrarian strategy when selling, and this asymmetrical behavior appears influenced by cultural beliefs. It suggests that momentum strategies align with high uncertainty avoidance (a preference for predictability) and individualistic traits (a focus on personal gain), while contrarian strategies align with low uncertainty avoidance (comfort with risk and ambiguity) and collectivist cultures (a focus on group stability).

This cluster includes research that identifies emotions as determinants of investors' behaviors as well. Zouaoui *et al*. (2011) found that investor sentiment - a feeling about potential gains without evidence - influences stock markets more strongly in countries where cultural tendencies favor herd-like behavior. Furthermore, Zaremba *et al*. (2020) concluded that high-sentiment markets (i.e., markets where investors may overlook potential risks and focus more on potential gains) perform better than low-sentiment ones, and this effect appears more impactful in countries typified by a collectivist culture.

3.5.2 Market and investors' cultural preferences

This cluster encompasses studies that survey global data (Bergsma & Jiang, 2016; Chourou *et al*., 2018; Zhang *et al*., 2023) or geographic regions, such as the Asia-Pacific (Durand *et al*., 2013), Arabian Gulf (Bley & Saad, 2010), or frontier economic regions (Berk *et al*., 2017). But it also includes investigations of single countries: Chen *et al*. (2020) and Huang, Chiu, *et al*. (2022) examine China; Mahendra *et al*. (2021) focus on India; Huang, Rhee, *et al*. (2022) study Japan and Pakistan; Halari *et al*. (2015) and Wasiuzzaman (2018) both investigate Saudi Arabia; and Chen and Chien (2011) survey Taiwan. Regardless of the geographic span of their analyses, these articles focus on culturally relevant events (Bergsma & Jiang, 2016; Bley & Saad, 2010; Chen & Chien, 2011; Halari *et al*., 2015; Huang, Chiu, *et al*., 2022; Wasiuzzaman, 2018) and themes, such as superstition (Chourou *et al*., 2018; Durand *et al*., 2013), culture and traditions (Chen *et al*., 2020; Mahendra *et al*., 2021), values (Berk *et al*., 2017; Huang, Rhee, *et al*., 2022), and language (Zhang *et al*., 2023).

In relation to cultural celebrations, we find studies that show that stock markets in the Arabian Gulf exhibit day-of-the-week anomalies influenced by Islamic cultural and religious factors (Bley & Saad, 2010). Investigating the effects of the Hajj pilgrimage on Saudi Arabia's stock market, Wasiuzzaman (2018) identifies a small negative impact on average returns but a significant increase in market volatility and reduced trading volumes, due to investors' prioritization of religious devotion during this period. In Pakistan, calendar-related anomalies occur during Ramadan and other Islamic events (Halari *et al*., 2015). Bergsma and Jiang (2016) cite a boost in stock market performance during Lunar New Year cultural celebrations, which they attribute to positive holiday sentiment. Bonuses issued during Lunar New Year, typically distributed in January, encourage greater risk-taking among Taiwanese investors, especially following years that exhibit market growth (Chen & Chien, 2011). Similarly, Huang, Chiu, *et al*. (2022) note that auspicious days on the Chinese lunar calendar drive trading volumes higher, albeit with lower market returns, particularly in cities with substantial elderly populations and a stronger cultural inclination toward gambling. In Australia and New Zealand, social

norms shape investor behavior by leading them to avoid sin stocks (i.e., industries considered unethical or immoral, like tobacco, gambling, weapons, and adult entertainment). In contrast, investors in Japan and South Korea exhibit preferences for such stocks, whose collective adoption appears to alleviate the psychological discomfort typically associated with this type of stock ownership (Durand *et al*., 2013). Cultural values also inform financial practices, such as the extent to which investors underprice IPO (Chourou *et al*., 2018).

Research on the effects of superstition indicates that the culturally auspicious number 8 influences investor behavior in China, leading investors to prefer stocks priced with values ending in 8 (Chen *et al*., 2020). In India, the astrological phenomenon of Mercury's retrograde motion exhibits a consistent correlation with market returns (Mahendra *et al*., 2021). In Japan, shareholder perks attain particular value because investors demonstrate a preference for physical gifts over monetary equivalents, reflecting their psychological inclination (Huang, Rhee, *et al*., 2022). In frontier market stocks, pricing thresholds create psychological barriers and market liquidity, reflecting cultural dimensions, such as individualism (Berk *et al*., 2017). Cultural and linguistic nuances also appear to shape investors' behaviors; as Zhang *et al*. (2023) highlight, when the native language includes weak future tenses, such that it does not distinguish future events clearly, there is a stronger tendency toward future-oriented behavior and long-term investments.

3.5.3 Culture and behavioral biases

The studies we assign to this thematic cluster rely on global data sets (Anderson *et al*., 2011; Beugelsdijk & Frijns, 2010; Bhatta *et al*., 2018; Breitmayer *et al*., 2019; Cakici & Zaremba, 2023; Chang & Lin, 2015; Drobetz *et al*., 2023; El Hajjar *et al*., 2024; Schmeling, 2009; Zhan, 2019), as well as data from the Organisation for Economic Co-operation and Development (Roque & Cortez, 2014). The country-specific studies tend to focus on China (Liu *et al*., 2023; Wang & Zhang, 2024), Japan (Kohsaka *et al*., 2017), Saudi Arabia (Gabbori *et al*., 2024), or the United States (Balkanska, 2018). With these diverse inputs, they deal with investors' behavioral biases, including herding (Chang & Lin, 2015; El Hajjar *et al*., 2024; Gabbori *et al*., 2024; Schmeling, 2009; Wang & Zhang, 2024; Zhan, 2019), disposition (Balkanska, 2018; Breitmayer *et al*., 2019; Kohsaka *et al*., 2017), foreignness (Beugelsdijk & Frijns, 2010; Bhatta *et al*., 2018; Drobetz *et al*., 2023), home (Anderson *et al*., 2011; Roque & Cortez, 2014), loss aversion (Liu *et al*., 2023), and recency (Cakici & Zaremba, 2023) biases.

Herding is the most recurrent behavioral bias. It refers to the phenomenon in which investors follow the same investment strategy, such as purchasing or selling similar shares in a specific period

(Chang & Lin, 2015). Herding behavior is more noticeable in countries categorized as collectivist (Schmeling, 2009), which also exhibit greater market volatility (Zhan, 2019). For example, herding behavior in Saudi Arabia, a slightly collectivist country, is leveraged during Islamic events, though not only during Ramadan, implying that greater adherence can encourage herding, arguably because religious events tend to influence collective moods and actions (Gabbori *et al*., 2024). According to El Hajjar *et al*. (2024), increased religiosity fosters social herding among stock market traders, in line with the notion that strong religious adherence encourages conformity. In contrast, individualistic societies tend to exhibit less herding behavior, which reduces market volatility during financial crises, particularly in emerging markets (Zhan, 2019). Excessive optimism, overconfidence, and a weaker disposition effect instead encourage investor herding (Chang & Lin, 2015).

The disposition effect refers to investors' tendency to sell overvalued shares and hold undervalued shares (Breitmayer *et al*., 2019), particularly when they have greater risk tolerance (Kohsaka *et al*., 2017). It is also common in stock markets that include broadly dispersed analyst forecasts (Balkansa, 2018), implying that investors increase their selling behaviors when information uncertainty is greater. Breitmayer *et al*. (2019) add that women and older investors are more susceptible to this effect, as are countries in which long-term orientation and indulgence prevail.

We consider foreign and home investor biases together. A foreign bias implies deviations from the optimal portfolio allocation in a particular foreign market, such that investors invest more in domestic shares and ignore the benefits of diversifying abroad (Drobetz *et al*., 2023). When the cultural distance between countries is greater, investors appear to grow even less disposed to investing in foreign markets due to their poor understanding of them (Beugelsdijk & Frijns, 2010; Bhatta *et al*., 2018). Environments that demand decision-making with incomplete information enhance investors' anxiety and risk taking, as well as their desire for greater uncertainty avoidance (Basabe *et al*., 2002). Nevertheless, non-institutional investors are generally more inclined to favor investments in geographically close countries that share borders or those that appear less corrupt (Roque & Cortez, 2014). Such trends might reflect the influences of cultural profiles on ethical decision-making (Vitell *et al*., 1993). That is, Hofstede (1980) relates economic inequality and power distance to corruption, which also can be linked to economic inequality (e.g., Husted, 2000). High power distance cultures tend to display more corruption and economic inequality (e.g., Seleim & Bontis, 2009). Furthermore, cultures marked by high uncertainty avoidance foster more internal investments and less foreign diversification, whereas cultures characterized as high in masculinity and overconfidence (Anderson

et al., 2011), along with a long-term orientation, encourage investors to invest more abroad and diversify their portfolios.

The loss aversion bias arises because investors do not equally weigh gains and losses and instead make decisions according to their potential gains, rather than expected losses (Bouteska & Regaieg, 2018). They make riskier investments to recover from losses but prefer investments in safe assets that provide gains. Furthermore, Liu *et al*. (2023) explain that lower-income investors are impacted by the fear of losses and high costs of participating in the stock market, while wealthier individuals, after achieving gains, become more cautious to avoid losing what they have gained.

Finally, Cakici and Zaremba (2023) examine recency biases, which reflect cognitive distortions that cause people to remember and prioritize the latest information or experiences over older ones, such that they downplay the relevance of older information. Thus, companies that show high recent stock returns, but low past returns, tend to outperform those with low recent returns but high past returns. This mispricing bias tends to be more pronounced after stock market downturns, during extreme volatility periods, and in countries that feature both high individualism and great shareholder rights protection.

3.6 Correspondence Analysis

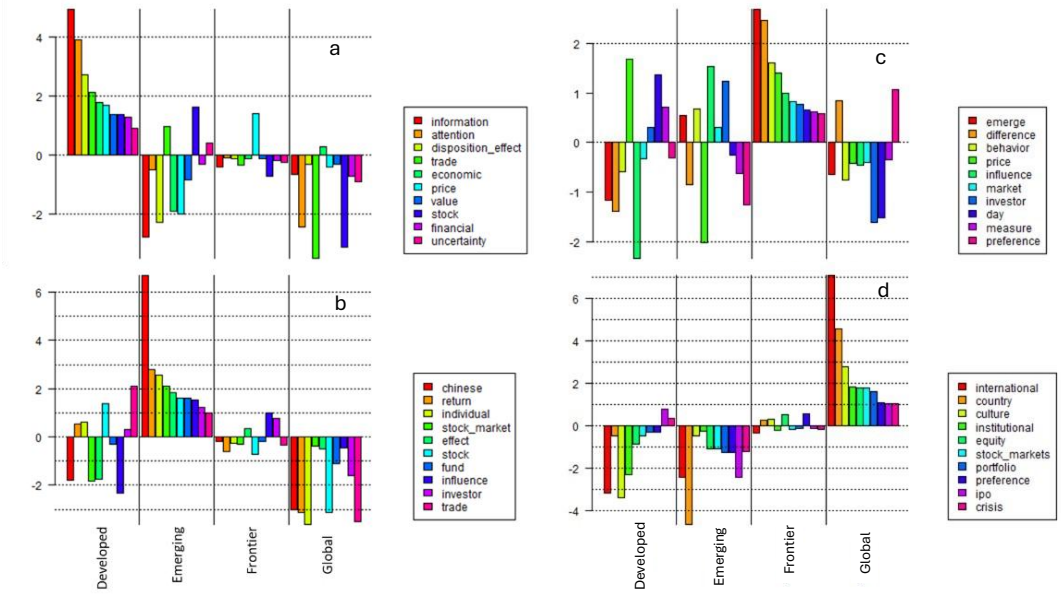
With a correspondence analysis, we sought to examine the most researched words, according to both national culture and economic classification. This method enables us to summarize available data through counts, compositions, or other ratio-scale measures. As part of a broader set of techniques that rely on singular value decomposition, correspondence analysis can be considered a counterpart of principal component analysis for categorical data. It assigns varying levels of importance to each row and column, according to their margins; that is, it applies the margins as weights to define the scales of the rows and columns, based on their interpoint chi-square (χ^2) distances (Greenacre, 2010). Appendix D contains the results of the correspondence analysis for each cluster's vocabulary and the positioning of words across each cluster. On the basis of this vocabulary distribution, we define the horizontal axis as cultural preferences and the vertical axis as investors' behavior or decision-making.

As noted, we use the correspondence analysis to identify the most researched words by country economic classification (Figure 5) and culture (Figure 6). In Figure 5, the most relevant words in the study sample relate to developed countries (quadrant a), emerging countries (quadrant b), frontier countries (quadrant c), and global regions (quadrant d), in line with Morgan Stanley Capital International's (MSCI, 2024) economic classification. The comparative analysis pertains to the most

common words for each country group, relative to the others (e.g., quadrant a depicts the most common words used in developed country studies, compared with studies of the other three groups of countries). It thus reveals that in studies of developed countries (quadrant a, column 1), particularly prevalent words refer to information (Aman *et al*., 2023; Huang, 2015; Qadan & Zoua'bi, 2019; Todea & Buglea, 2017), uncertainty (Balkanska, 2018; Persakis & Koutoupis, 2024; Qadan & Zoua'bi, 2019), the disposition effect bias (Balkanska, 2018; Kohsaka *et al*., 2017), decision-making information (Adachi *et al*., 2017; Aman *et al*., 2023; Huang, 2015; Qadan & Zoua'bi, 2019; Todea & Buglea, 2017), and investment strategies (Choy *et al*., 2023; Cronqvist *et al*., 2015; Ikeda, 2023; Lee *et al*., 2019; Persakis & Koutoupis, 2024; Switzer *et al*., 2017).

Studies in emerging countries (quadrant b, column 2) mostly refer to Chinese investors' decision-making (Azzi & Suchard, 2019; Cheema & Nartea, 2014; Lee *et al*., 2023; Zeng *et al*., 2023), Chinese cultural preferences (Chen *et al*., 2020; Huang, Chiu, *et al*., 2022), and investment returns (Duxbury & Yao, 2017; Halari *et al*., 2015). Moreover, these studies deal with cultural aspects, such as investors' superstitions (Chen *et al*., 2020; Mahendra *et al*., 2021), and calendar anomalies (Bley & Saad, 2010; Chen & Chien, 2011; Halari *et al*., 2015; Huang, Chiu, *et al*., 2022; Wasiuzzaman, 2018).

Figure 5 - Research Terms by National Economic Classification



Source: Elaborated by the authors

In studies from global regions (quadrant d, column 4), the most prominent words reflect international comparisons among countries and cultures (Anderson *et al*., 2011; Beugelsdijk & Frijns, 2010; Roque & Cortez, 2014; Zhan, 2019), investors' preferences (Breuer *et al*., 2020; Choy *et al*.,

2023; Roque & Cortez, 2014), IPO pricing behaviors (Chourou *et al*., 2018; Zhang *et al*., 2023), and economic crises (Mbarki *et al*., 2022; Zhan, 2019; Zouaoui *et al*., 2011). Global research also prioritizes terms related to foreign bias (Beugelsdijk & Frijns, 2010; Bhatta *et al*., 2018; Drobetz *et al*., 2023) and home bias (Anderson *et al*., 2011; Roque & Cortez, 2014). Finally, studies in frontier markets (quadrant c, column 3) cite cultural preferences (Berk *et al*., 2017) and investors' information (Speidell, 2009), through which they compare investors' culture and behavioral differences, in a global sense (Berk *et al*., 2017; Speidell, 2009).

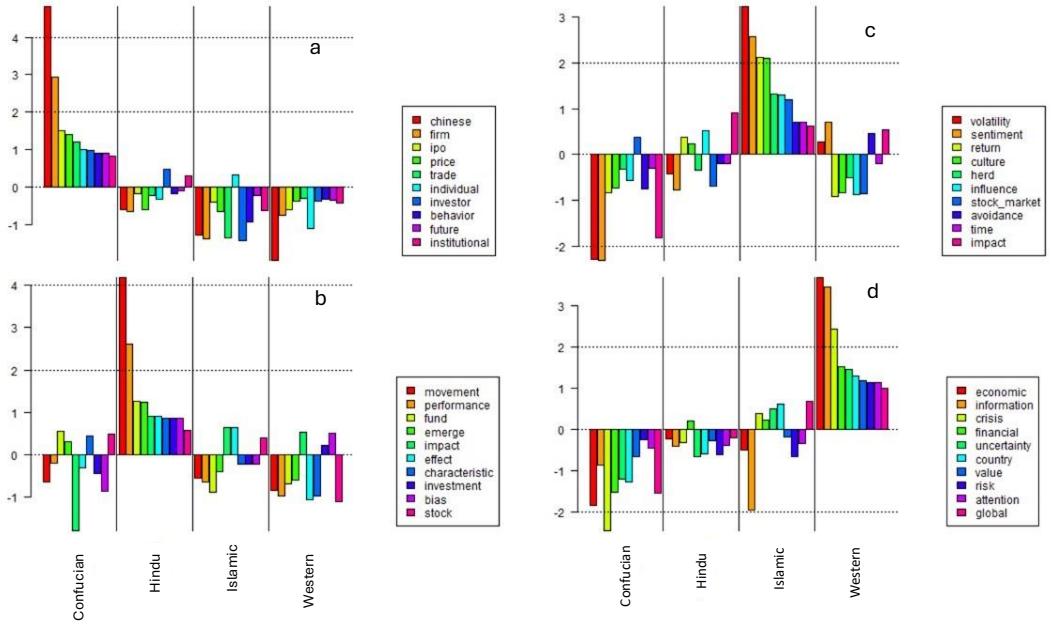
These analyses indicate that studies conducted in developed countries deal mainly with how investors process information for their decision-making and which investment strategies they pursue. In emerging countries, mainly China, the research highlights the impacts of cultural preferences, investor superstitions, and calendar anomalies. In frontier countries, scholars examine how pricing reflects cultural preferences. Studies that take a global perspective center on the influences of culture on economic crises, as well as foreign versus home biases.

In Figure 6, we depict the most relevant words in articles that address Confucian- (quadrant a), Hindu- (quadrant b), Islamic- (quadrant c), and Western- (quadrant d) oriented cultures. The pattern is similar to that in Figure 4. For countries that embrace Confucian philosophy (quadrant a; e.g., China, Japan, South Korea, Taiwan), the most common words are Chinese (Azzi & Suchard, 2019; Huang, Chiu, *et al*., 2022; Mbarki *et al*., 2022), firms (Cheema & Nartea, 2014; Lee *et al*., 2023; Zeng *et al*., 2023), investors' behavior (Chen & Chien, 2011; Duxbury & Yao, 2017; Lee *et al*., 2023; Ng & Wu, 2010), and future (Cheema & Nartea, 2014). This philosophy also appears linked to investigations of calendar anomalies (Chen & Chien, 2011; Huang, Chiu, *et al*., 2022) and superstitions (Chen *et al*., 2020).

The most used words in research conducted in Hindu countries (quadrant b; e.g., India) are movement (Mahendra *et al*., 2021) and performance (Nguyen *et al*., 2018). For Islamic countries (quadrant c; e.g., Indonesia, Iran, Malaysia, Pakistan, Saudi Arabia), the most frequent words are volatility (Gabbori *et al*., 2024; Wasiuzzaman, 2018), sentiment (Gabbori *et al*., 2024; Shah, 2024; Wasiuzzaman, 2018), return (Bley & Saad, 2010; Halari *et al*., 2015), and culture (Halari *et al*., 2015; Wasiuzzaman, 2018). Finally, studies conducted in Western settings (quadrant d) frequently refer to economics (Cronqvist *et al*., 2015; Persakis & Koutoupis, 2024), information (Huang, 2015; Qadan & Zoua'bi, 2019; Todea & Buglea, 2017), crisis (Lee *et al*., 2019; Zouaoui *et al*., 2011), uncertainty (Balkanska, 2018; Qadan & Zoua'bi, 2019), risk (Lee *et al*., 2019; Qadan & Zoua'bi, 2019), and

investment strategies (Cronqvist *et al*., 2015; Lee *et al*., 2019; Persakis & Koutoupis, 2024; Zouaoui *et al*., 2011).

Figure 6 - Research Terms by National Culture



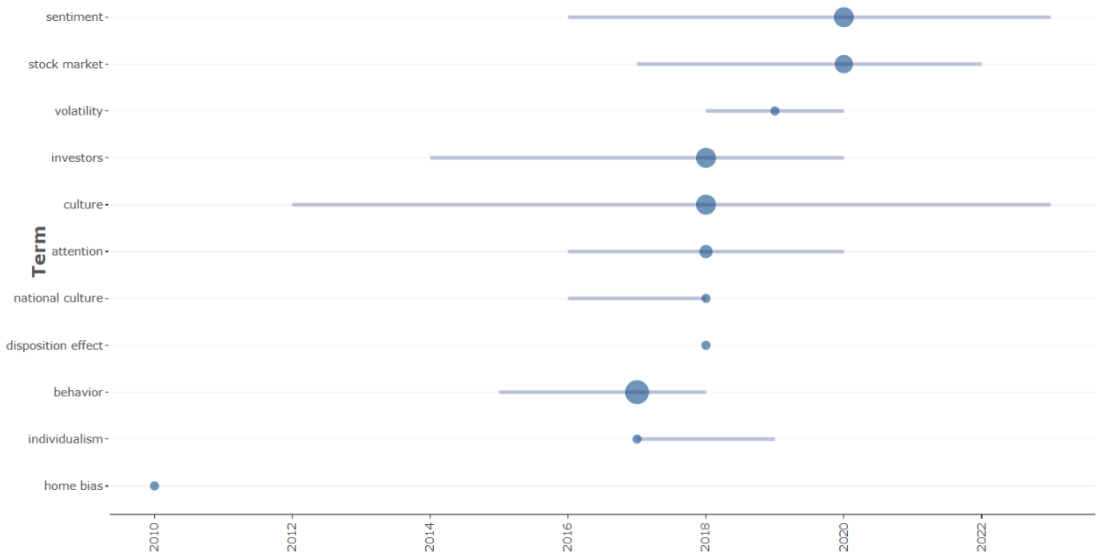
Source: Elaborated by the authors

The results of this analysis indicate that scholars studying Western cultures primarily focus on investors' decision-making process, but those studying Confucian cultures prioritize investors' information search (mainly in Japan), as well as how calendar anomalies (e.g., Lunar New Year bonuses) and superstitions (e.g., lucky numbers) affect investors' trading behavior. Research conducted in Hindu cultures often links cultural factors with stock market performance, and studies in Islamic cultures describe volatility in stock markets and how culture and religion (e.g., Muslim religious events) influence this effect.

3.7 Research Trends and Directions

To identify topic trends and research areas, we also conducted a topic trend analysis. With Figure 7, we illustrate the evolution of themes that it revealed, according to the keywords linked to each article in our sample. The lines in the figure represent the usage of each keyword over time (in years); circles on each line indicate the median years for the respective keywords, or the year each keyword was used most frequently. The size of these circles corresponds to the frequency of each keyword's use.

Figure 7 - Trend Topics for Culture and Investors' Behavior

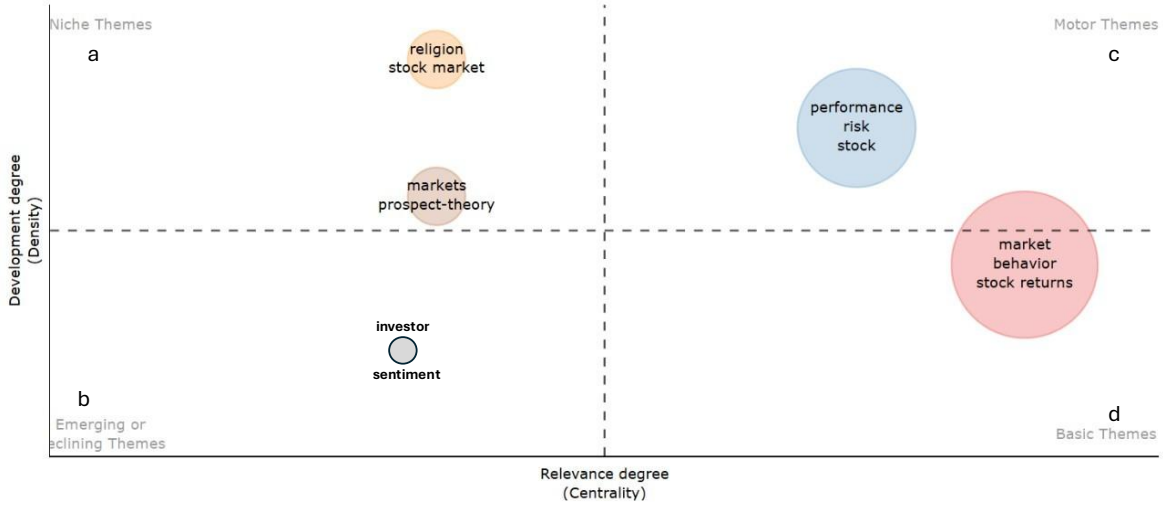


Source: Elaborated by the authors

For example, "culture" began to appear as a keyword topic in 2012; the related keyword "national culture" only emerged as a trend in 2016. "Individualism" has mainly appeared since 2017. When we consider the concept of investor behavior, the keyword "behavior" began being used with more frequency in 2017. The term "home bias" appeared in 2013; "disposition effect" started to appear in 2018. We also note that "sentiment" has been gaining popularity since 2016 and was often used in 2020. These findings suggest continued potential for exploring themes related to how culture influences investors' behavior.

Next, we sought to identify themes by examining clusters of author keywords and their interrelationships, defined according to two metrics: density (vertical axis) and centrality (horizontal axis). These metrics signal the importance of the topics and distinguish those that are significant from those that are not. A node's centrality and importance increase with the number of its connections to other nodes in the thematic network; a node's cohesiveness, which also depends on the density of a research field, indicates its potential for evolution and longevity (Clarke & Braun, 2017). A thematic map of culture and investor behavior, based on the articles in our sample that were published between 2021 and 2024, is illustrated in Figure 8.

Figure 8 - Thematic Map of Culture and Investor Behavior



Source: Elaborated by the authors

In the top-left quadrant, a, we find niche themes (e.g., religion, prospect theory), whereas quadrant b features emerging themes, such as investors' sentiment. Quadrant c reflects main trends, such as those linked to performance and risk in stock markets. Finally, the bottom-right quadrant d reveals predominant themes such as investors' behavior and stock returns. On the basis of these assessments, we propose several emerging frontiers that offer promising research opportunities, organized by the three thematic clusters.

3.7.1 Investors' decision-making

We still lack a sufficiently complete, clear picture of how culture shapes investors' decision-making. In Table 5, we list some relevant research questions in this domain, reflecting our sense that studies of cultural determinants of investor decision-making and their interactions with information processing promise to be fruitful research areas. Understanding cultural nuances can help researchers develop models that predict investor behaviors more accurately across different cultural contexts. Researching the impact of social media on investor behaviors that vary across cultures, such that it might affect how investors process information available online, offers another promising avenue. Scholars could examine the effects of investor myopia on market anomalies, too, as well as how investors react to natural disasters and global crises (e.g., COVID-19), and if and how these reactions differ across cultures. Similarly, moods might affect investor behavior variously in different cultural settings, in which case we need further research into emotional dynamics. Understanding these factors and their mechanisms could enhance financial theories and practices.

Table 5 – Research questions by cluster

A. Investors' decision-making cluster	
1	Do cultural differences influence investors' information processing and decision-making behavior?
2	How do investors react to news and information disseminated through social media platforms, as manifested in their investment strategies, across different cultural settings?
3	Does investor myopia affect other stock market anomalies across different cultures?
4	Are investors' responses to natural disasters and global crises shaped by national culture?
5	How do moods affect investor behavior in different cultural settings, particularly those influenced by religious beliefs?
B. Market and investors' cultural preferences cluster	
1	Do religious factors influence investors´ decision-making similarly across cultures (e.g., non-Muslim versus Muslim countries)?
2	How does investors' sentiment relate to their cultural preferences?
3	Do cultural preferences influence individual investors' risk-taking and time-horizon decisions?
4	Is gambling related to national culture?
5	How do saving habits relate to cultural preferences?
C. Culture and behavioral biases cluster	
1	How does the disposition effect interact with other behavioral biases, and what impacts do they have across dynamic international markets?
2	To what extent do institutional investors' strategies offset or reinforce herding behavior among individual investors?
3	How does social trust affect institutional investor herding behavior across different countries' religions, economic statuses, or legal systems?
4	Does recency bias affect other classes of assets (e.g., commodities, corporate bonds)?
5	How do cultural differences among investors influence home versus foreign biases in diversification strategies in portfolio allocations?

Source: Elaborated by the authors

3.7.2 Market and investors' cultural preferences

Continued research into how cultural preferences affect investor behavior might explore additional global events and religious influences; as several authors have pointed out, religion powerfully influences individual decision-making (e.g., Kumar *et al*., 2022). Scholars could also pursue a deeper understanding of how cultural norms and values shape investment choices. For example, examining the relationship between gambling and national culture might provide deeper insights into societal behaviors toward risk; studying the link between saving habits and cultural preferences could offer valuable perspectives on financial planning and consumption patterns across cultural settings. Determining how cultural values shape collective investor sentiment and stock market movements would also be useful to inform financial strategies and regulatory policies that are both culturally sensitive and effective in a global economic setting.

3.7.3 Culture and behavioral biases

Finally, we hope that continued research will attempt to clarify how the disposition effect interacts with other behavioral biases, such as overconfidence and loss aversion, to determine stock prices and market efficiency. The influence of the disposition effect on analyst disagreement has been established (Hur & Singh, 2019), but we still need theoretical explanations for this link. The European Union's unique combination of diverse cultures within an integrated financial market would provide a rich context for exploring this issue. Prior research establishes that herding behavior contributes to stock market volatility, but the extent to which institutional investors mitigate or amplify this effect is insufficiently understood, as are the implications of such behavior on international financial market dynamics. Another investor behavior that needs attention, especially with regard to the influences of cultural differences, is exhibitions of foreign and home bias, as manifested in international portfolio allocations. This question is particularly relevant as globalization trends continue to increase the availability of international investment opportunities. Understanding the cultural factors at play could suggest more effective diversification strategies. The effect of social trust on institutional investors' herding behavior in emerging markets lacks any comprehensive studies. Research related to the recency bias also should continue. Examining the impact of such biases on the prices and volatility of various asset classes might help reveal the broad effects of recent events on market behavior. For example, extended inquiries could explore how cultural backgrounds and bias affect other classes of assets (e.g., commodities, cryptocurrencies, corporate bonds), beyond stock markets.

3.8 Concluding Remarks

This study relies on both quantitative (bibliometric) and qualitative (content) analyses to determine how national culture, defined using Hofstede's cultural dimensions, affects investor behaviors. In our effort to determine (1) the most prolific contributors (journals, countries, institutions, affiliations, authors) to the field, (2) the most cited articles, (3) the most important themes pertaining to culture and investors' behavior in capital markets, and (4) which trends and gaps indicate areas for further research, we evaluated 60 articles, published between 2009 and 2024 in journals rated at least ABS2+. The analyses reveal that publications pertaining to national culture and investor behavior have been increasing since 2017, in both number and impact. Schmeling (2009) emerges as the most impactful contribution; the United States leads as the most prolific country, followed by China. The top three journals are *Pacific-Basin Finance Journal, International Review of Financial Analysis,* and *Journal of International Financial Markets, Institutions, and Money*.

The content analysis also identifies three major themes: investors' decision-making, such that studies investigate how information scarcity or overload can influence investors' behaviors across

various cultures; market and investors' cultural preferences, which consists of studies that outline how cultural practices and superstitions affect financial market dynamics; and culture and behavioral biases, a theme that details how investor behavioral biases (e.g., herding, disposition effect) relate to cultural factors. With a correspondence analysis, we derive the diverse patterns of investor behavior across countries with different cultural and religious backgrounds, as well as economic regions. Exploring how cultural factors shape investors' behaviors and convictions and then affect their investment decisions enlarges extant understanding, from a macro to a micro level. In this sense, our study extends beyond Goodell, Kumar, *et al*.'s (2023) bibliometric analysis of cultural finance. We also identify some unexplored research areas and directions for further research that can advance the field. Our findings are relevant to a broad spectrum of stakeholders within the academic sphere and across wider professional and societal contexts and can offer meaningful recommendations for financial markets as they grow increasingly interconnected and as cultural diversity among investors becomes even more pronounced.

4 THE CULTURAL IMPACT ON BUSINESS STRATEGY: FAITH, LAW, AND ECONOMY

4.1 Introduction

The four fundamental elements that determine how societies function and relate to each other include culture and religion, together with economy and legal systems. The mentioned elements show growing importance for financial operations, especially when making investment choices. Hofstede (2011) states that national culture, consisting of collective beliefs and values, traditions, actions, and objects, influences how people in society interact with their environment and their financial behavior.

Historians have studied in detail how cultural traditions have connected with religious beliefs throughout history. The two elements create a mutual relationship that affects how humans understand their financial world and the world in general (Abdulla, 2018; Beyers, 2017). People make investment choices based on their cultural background and religious beliefs, which leads to major impacts on their investment decisions.

The economy is deeply connected to cultural elements. The economic choices people make are influenced by the cultural values they hold (Kostis, 2021). The capitalist economic system drives cultural items toward commercialization, which shapes our decisions about different cultural elements (Clammer, 2016). The relationship between cultural elements and investment decisions shows how they influence economic behavior.

The cultural environment of a society determines what laws and regulations will be created (Silbey, 2018). According to the socio-legal perspective, law and culture exist as interconnected systems that transform each other through ongoing social conflicts and interactions (Yazdiha, 2017). The legal system of society develops through a process of mutual influence between cultural norms, religious beliefs, and economic frameworks that govern financial activities.

Research has not yet fully examined the connection between national culture and socioeconomic elements that affect Finance. This paper fills this gap, investigating how different national cultural aspects affect various socioeconomic metrics across nations. We leverage a dataset that includes measures of national culture (Hofstede dimensions), economic indicators, religious adherence, and legal systems.

Our study reveals that cultural dimensions can significantly influence economic behavior, including investment decisions. For instance, developed countries typically exhibit higher levels of individualism and the Freedom Index, which may influence their investment strategies. Emerging

and frontier economies tend to have lower individualism and a lower Freedom Index, but a higher uncertainty avoidance, which may be reflected in their risk-averse investment behavior. Frontier economies generally show lower masculinity, indicating different investment preferences. Furthermore, these findings have profound implications for firms making investment decisions, influencing market entry strategies, organizational structure, negotiation tactics, risk assessment, and legal considerations.

The remainder of this paper is organized as follows: Section 4.2 reviews the relevant literature, Section 4.3 describes the methodology, Section 4.4 presents the results, and Section 4.5 discusses the findings and concludes. This structure allows us to explore the intricate relationship between Culture and Finance, shedding light on how cultural factors can shape investment decisions.

4.2 Theoretical Framework

4.2.1 National Culture

During the 18th century, philosophers like David Hume and Immanuel Kant explored and compared the concept of 'national character'. In the following century, culture began to be considered in economic discussions, with theorists like Max Weber suggesting a connection between a society's ethical values and its economic prosperity, and Alexis de Tocqueville accurately foresaw the spread of cultural characteristics that originated in the former British Colonies of North America (De Jong, 2013).

While Hofstede is known for creating national indices based on variables that correlate at a national level and are theoretically tied to cultural phenomena, he was not the pioneer in this field. Over 30 years before Hofstede, Cattell (1949) initiated factor analysis of country data for the same objective. However, before Hofstede's study (1980), such research typically only revealed a significant economic development dimension and other factors that were either minor or hard to understand (Minkov & Hofstede, 2011).

After Hofstede, other researchers proposed different dimensions. For example, social scientist Shalom Schwartz surveyed individuals in 49 countries from 1988 to 1993, identifying seven values at the national level that closely resembled Hofstede's cultural dimensions (Schwartz, 1999).

Similarly, political scientists Ronald Inglehart and Christian Welzel identified two dimensions of cross-cultural variation using data from the World Values Survey (WVS) collected from nearly 100 countries between 1981 and 2014 (Inglehart & Baker, 2000).

Hofstede developed his dimensions to solve the core problems that society needs to handle

(Minkov & Hofstede, 2011). The initial four dimensions (Power Distance, Uncertainty Avoidance, Individualism versus Collectivism, and Masculinity versus Femininity) were the foundation for Hofstede's influential book, "Culture's Consequences".

Later, based on research by Canadian psychologist Michael Harris Bond in the Far East, a fifth dimension called "Long-Term versus Short-Term Orientation" was added. During the 2000s, Bulgarian academic Michael Minkov researched using data from the World Values Survey. This research led to a new interpretation of the fifth dimension and the introduction of a sixth dimension, "Indulgence versus Restraint" (Hofstede, 2011).

These six dimensions are defined as:

- Power Distance (PDI) describes how different cultures manage natural human social status differences between people.
- Uncertainty Avoidance (UAV) refers to how much tension a society endures when confronted with an unpredictable future.
- Individualism versus Collectivism (IDV) pertains to how individuals are assimilated into primary social groups.
- Masculinity versus Femininity (MAS) represents how emotional responsibilities are distributed between male and female roles in society.
- Long-Term versus Short-Term Orientation (LTO) relates to whether people's efforts are aimed toward the future or centered on the present and past.
- Indulgence versus Restraint (IDG) concerns the equilibrium between satisfying and regulating fundamental human desires associated with life's pleasures.

Each country has been evaluated and compared to others based on a score for each dimension. The dimensions exist independently of each other and appear in all possible combinations, but specific patterns emerge more frequently (Hofstede, 2011).

4.2.2 National Culture and Economy

As measured by Hofstede's dimensions, national culture has impacted various aspects of the economy. Research findings suggest that cultural elements are associated with various economic results. The relationship between individualistic values (IDV) and national GDP per capita shows a direct connection. The amount of development aid in a country's budget directly relates to its MAS score (Hofstede, 2011). Tang and Koveos (2008) demonstrate that GDP per capita shows a positive relationship with individualistic (IDV) values but a negative relationship with power distance (PDI)

values.

In the context of sustainability, countries with higher individualism (IDV), masculinity (MAS), and indulgence (IDG) have lower green bond issuance, whereas countries with high LTO and UAV have higher green bond issuance (Mertzanis & Tebourbi, 2023). Huang *et al.* (2022) show that power distance (PDI) creates negative effects on climate change performance, greenhouse gas emissions, renewable energy, and energy use. The study also finds that masculinity (MAS) has a negative relationship with climate change performance and greenhouse gas emissions.

A society that supports innovation includes five key elements: low power distance (PDI), high individualism (IDV), low uncertainty avoidance (UAV), long-term orientation (LTO), and high indulgence (IDV), and feminine (MAS) values (Espig *et al.*, 2021). Furthermore, by relating the Global Innovation Index with Hofstede dimensions, several scholars show that individualism (IDV) affects innovation significantly and positively (Rinne *et al.*, 2012; Khan & Cox, 2017; Prim *et al.*, 2017) as well as long-term orientation (LTO) (Khan & Cox, 2017; Prim *et al.*, 2017) and indulgence (IDG) (Khan & Cox, 2017; Prim *et al.*, 2017). In contrast, power distance (PDI) affects innovation significantly and negatively (Rinne *et al.*, 2012; Khan & Cox, 2017; Prim *et al.*, 2017). On top of that, Rinne *et al.* (2013) find that the link between creativity and individualism (IDV) is positive, suggesting that the principles of autonomy, independence, and freedom are essential for a country's creative potential.

Lee *et al.* (2017) find that countries with high intellectual capital tend to be individualistic (IDV), have low power distance (PDI), and have weak uncertainty avoidance (UAV). The research suggests that a nation should work to establish an environment that promotes equality, freedom, safety, and competitive activity to boost its intellectual capital. DeBode *et al.* (2020) suggest that the cultural aspect of femininity (MAS) was linked with a majority of the economic freedom indicators. Short-term orientation (LTO) leads to the highest level of business freedom in social systems. Less indulgent cultures (IDG) showed a positive connection between their values and higher levels of business and financial success. More individualism (IDV) led to more financial independence.

Furthermore, tax evasion is linked with high power distance (PDI) and low masculinity scores (MAS) countries (Tsakumis *et al.*,2007), collectivist (IDV) (Tsakumis *et al.*,2007; Richardson, 2008), and high uncertainty avoidance (UAV) (Tsakumis *et al.*,2007; Richardson, 2008). Achim *et al.* (2019) concluded that the growth of the shadow economy is connected with high collectivism (IDV), femininity (MAS), short-term orientation (LTO), and restrained (IDG) countries.

As the cultural dimensions of society significantly influence economic outcomes:

H1: Developed countries tend to present higher individualism (IDV), lower power distance (PDI), and lower uncertainty aversion (UAV) than emerging and frontier countries

4.2.3 National Culture and Religion

Religion functions as a cultural element of a nation, as religious practices and beliefs often reflect the political shifts, environmental changes, economic developments, and demographic shifts that occur in the national cultural framework. Hong *et al.* (2023) state that religion serves as an essential informal institution that maintains its vital position in social structures. Religious cultural communities that span across the globe (Catholicism, Protestant Christianity, and Islam) determine the cultural identity of different nations. The most substantial effect emerges from religious and cultural traditions that missionaries, conquerors, and colonizers introduce to their conquered lands (Johnson & Cohen, 2013).

Minkov and Hofstede (2011) discovered that national elements have a greater impact than worldwide religious beliefs. This research leads to the formation of nationally uniform and statistically identifiable groups of religious communities within the same country despite their nominal differences. The different local expressions of global religions fail to produce the same result because they do not merge into standardized religious groups that can be measured across international borders. The findings suggest that a shared national history is a significant cultural determinant regarding values, while a universally shared religion does not have the same influence.

Weber (1905) states that Protestants across all social classes demonstrate economic rationalism, which Catholics do not typically show. He theorizes that these disparities stem from the inherent nature of their respective religious doctrines rather than their political circumstances. He posits that the Protestant ethic influenced many people to work in the secular world, developing their enterprises, engaging in trade, and accumulating wealth for investment.

According to Blum and Dudley (2001), Protestant cities that had access to the Atlantic Ocean and printing facilities succeeded because they had superior transportation networks and fewer regulatory restrictions. The Protestant cities developed their own distinct specialization patterns, which depended on their geographical position relative to London. This characteristic was not observed in Catholic Europe. The economic network became closely-knit because Protestants developed trust in contracts with unknown individuals while maintaining their literacy skills. Therefore, they support Weber's theory in the context of information networks, not individual economic behavior. In addition, Schilpzand and De Jong (2021) found that the connection between Protestant adherence and work ethic is positive, though not as strong as that with economic

development. This paper posits that the capitalist spirit linked to Max Weber's Protestant Ethics is not tied to Protestantism but rather to the phase of economic modernization.

In other aspects, according to a study by La Porta *et al.* (1999), it was observed that nations predominantly populated by Catholics or Muslims tend to exhibit poorer governmental performance. Countries with a Catholic majority are often more interventionist, and their governments are typically less efficient. Compared to Protestant countries, Catholic nations generally fall short in providing public goods, have reduced transfers, lower government consumption, and fewer public sector employees. Similarly, Muslim-majority countries are usually more interventionist, less efficient, and display a lower degree of democracy.

Rosson and Fields (2008) examined the correlation between Hofstede's cultural dimensions and the expansion of evangelical Christianity from 1985 to 2000. It was discovered that a country's growth in evangelical Christianity was positively associated with greater power distance (PDI) and less individualism (IDV). Tang and Koveos (2008) found that Catholic countries are positively correlated with individualism (IDV), while Protestant countries are positively correlated with individualism (IDV) and negatively correlated with uncertainty aversion (UAV) and masculinity (MAS). Islamic countries presented a negative correlation with long-term orientation (LTO), in contrast with Confucian countries, which are significantly positively correlated with long-term orientation (LTO). Mättö and Niskanen (2019) found a correlation between trade credit, religion, and national culture variables. It is discerned that nations with a Catholic majority exhibit elevated levels of trade credit compared to those with a Protestant majority. Moreover, the degree of religiosity appears to influence trade credit exclusively within Catholic nations. The research further uncovers a positive association between trade credit and Hofstede's cultural dimensions: power distance (PDI) and uncertainty avoidance (UAV). More recently, Lehmann and Weiße (2025) state that the combination of cultural dimensions and religious beliefs influences entrepreneurial conduct as they establish core values that include autonomy, innovation, and risk-taking behavior.

Therefore, the cultural dimensions of a society are influenced by its predominant religion and national factors.

H2a: Protestant societies exhibit higher individualism (IDV) and lower uncertainty aversion (UAV) than Catholic societies

H2b: Islamic societies tend to have a lower long-term orientation (LTO) and lower Indulgence (IDG) than Catholic societies.

4.2.4 National Culture and Legal System

Civil Law, derived from Roman Law, comprises a set of comprehensive rules closely tied to justice and morality, while Common Law, originating in Britain, was primarily formed by judges resolving specific disputes based on case law and precedents. Modern commercial laws stem from three Civil Law traditions: French, German, and Scandinavian. Investors tend to have less robust legal rights under Civil Law than Common Law. The most notable contrast is between Common Law nations, which offer the most robust protection to shareholders and creditors, and French Civil Law nations, which offer the least protection to investors. German Civil Law and Scandinavian countries offer investor protection that falls somewhere between Common Law and French Civil Law countries. Regarding law enforcement quality, Scandinavian and German Civil Law countries lead the pack, followed by Common Law countries, with French Civil Law countries trailing behind. The evidence consistently points to the relative weakness of legal protections for investors in French Civil Law countries compared to those in Common Law countries, regardless of their income levels (La Porta *et al.*, 1998).

Countries under Common Law are negatively correlated with uncertainty avoidance (UAV). A higher score of uncertainty avoidance is generally associated with countries with precise laws and regulations (Tang & Koveos, 2008). Haggard and Haggard (2018) found that legal origin influences the number of steps, the duration required to establish a business, and the simplicity of obtaining credit. Furthermore, cultural aspects (such as power distance (PDI)) and religious beliefs play a significant role in accounting for the gender disparities in the ease of launching a business. Moreover, La Porta *et al.* (1999) investigated the correlation between a country's legal origin and the quality of its government. They discovered that countries with lower gross national product and those with French legal systems tend to have lower-quality governance. Wen *et al.* (2022) found that French Civil Law countries presented the worst innovative performance, while there is no robust evidence that Common Law countries surpassed German and Scandinavian Civil Law countries in technical innovation. More recently, Jung (2025) found that countries with higher individualism tend to have stronger property rights protections.

Then, a country's legal origin significantly interacts with its national culture, influencing various economic and societal outcomes.

H3: Countries with French Civil Law origins may have cultures with lower individualism (IDV), higher power distance (PDI), and higher uncertainty avoidance (UAV) than countries with origins in Common Law.

4.3 Methodology

4.3.1 Data Sources

Our source of Hofstede's six cultural dimensions is Hofstede Insights (2024), which offers data for 119 countries. Economic market classification is obtained from MSCI (2024). The Freedom Index is obtained from Worldwide Governance Indicators (World Bank, 2024) using the index named Voice and Accountability, which measures the degree to which citizens of a country can engage in the electoral process and enjoy rights such as freedom of speech, freedom to form associations, and the presence of a free press.

For religious demographics, the source is the Pew Research Center (2024), and the same source is used for Christian denominations (Pew Research Center, 2011). Religiosity is obtained from the seventh wave (2017-2021) of the World Values Survey (WVS) - a global research initiative committed to scholarly examining individuals' social, political, economic, religious, and cultural beliefs worldwide. The data from the WVS cover several essential aspects of people's religious orientation. Religiosity tracks how involved people are in religious services and how much importance they attach to their religious beliefs (World Values Survey, 2024).

Finally, the source of data on legal origins is La Porta *et al.* (2017), while the rule of law is obtained from Worldwide Governance Indicators (World Bank, 2024). The concept of the rule of law reflects how much trust individuals have in societal rules and their adherence to them. It specifically refers to the effectiveness of contract enforcement, the protection of property rights, the reliability of police services, and the judicial system. It also encompasses the potential for crime and violence (Word Bank, 2024).

4.3.2 Measurements

To assess the relation of socioeconomic factors with Hofstede dimensions, we adapt Chang and Lin (2015) by performing the multinomial logistic regression for the economy (Equation (1), religion (Equation (2), and Legal (Equation (3).

Economy

$$\text{Economy}_i = \alpha_0 + \alpha_1 PDI_i + \alpha_2 IDV_i + \alpha_3 MAS_i + \alpha_4 UAV_i + \alpha_5 LTO_i + \alpha_6 IDG_i + \alpha_7 Freedom_i + \varepsilon_i \quad (1)$$

where:

Economy_i is a categorical variable with the country's MSC classification (Developed, Emerging, or Frontier)

PDI_i, IDV_i, MAS_i, UAV_i, LTO_i, and IDG_i are Hofstede's Cultural Dimensions for each country *i*

Freedom_i is the Voice and Accountability index from Worldwide Governance Indicators for each country *i*

α_0, α_1, α_2, α_3, α_4, α_5, α_6, and α_7 are regression coefficients

ε_i refers to the error term

Religion

$$\text{Religion}_i = \beta_0 + \beta_1 PDI_i + \beta_2 IDV_i + \beta_3 MAS_i + \beta_4 UAV_i + \beta_5 LTO_i + \beta_6 IDG_i + \beta_7 Religiosity_i + \zeta_i \quad (2)$$

where:

Religion_i is a categorical variable of a country's majority religion (Catholic, Protestant, Orthodox, Muslim, Hindu, Jewish, or Confucianist country)

PDI_i, IDV_i, MAS_i, UAV_i, LTO_i, and IDG_i are Hofstede's Cultural Dimensions for each country *i*

Religiosity_i is obtained through WVS from Wave 7 – World Values Survey, and indicates how involved people are in religious services and how much importance they attach to their religious beliefs

β_0, β_1, β_2, β_3, β_4, β_5, β_6, and β_7 are regression coefficients

ζ_i refers to the error term

Note: Although Confucianism is not a religion, for this study, some Eastern Asian countries are considered Confucian nations (Sun, 2013).

Legal

$$\text{Legal}_i = \gamma_0 + \gamma_1 PDI_i + \gamma_2 IDV_i + \gamma_3 MAS_i + \gamma_4 UAV_i + \gamma_5 LTO_i + \gamma_6 IDG_i + \gamma_7 RuleLaw_i + \xi_i \quad (3)$$

Where

$Legal_i$ is a categorical variable of Legal Origin (Common Law, French Civil Law, German Civil Law, or Scandinavian Civil Law)

PDI_i, IDV_i, MAS_i, UAV_i, LTO_i, and IDG_i are Hofstede's Cultural Dimensions for each country *i*

RuleLaw is the Rule of Law index from Worldwide Governance Indicators

γ_0, γ_1, γ_2, γ_3, γ_4, γ_5, γ_6, and γ_7 are regression coefficients

ξ_i refers to the error term

Table 6 summarizes the variables' definitions, as well as shows their sources and their categories or range:

Table 6 – Variables Description – Economy, Religion, and Legal

Variables Dependent	**Definition**	**Category/Range**	**Source**
Economy	The country's MSC classification	Developed/ Emerging/ Frontier	MSCI (2024)
Religion	A country's majority religion	Catholic/ Confucianism/ Hinduism/Islamism/ Judaism/Orthodox/ Protestant	Pew Research Center (2024)
Legal	The country's legal origin	Common Law/ French Civil/ German Civil/ Scandinavian Civil	La Porta *et al.* (2017)
Independent			
PDI	Power of Distance	0 to 100	Hofstede Insights (2024)
IDV	Individualism	0 to 100	Hofstede Insights (2024)
MAS	Masculinity	0 to 100	Hofstede Insights (2024)
UAV	Uncertainty Avoidance	0 to 100	Hofstede Insights (2024)
LTO	Long Term Orientation	0 to 100	Hofstede Insights (2024)
IDG	Indulgence	0 to 100	Hofstede Insights (2024)
Freedom	Measures the degree to which citizens of a country can engage in the electoral process and enjoy rights such as freedom of speech, freedom to form associations, and the presence of a free press.	-2.5 to +2.5	The Voice and Accountability Index (World Bank, 2024)
Religiosity	Tracks how involved people are in religious services and how much importance they attach to their religious beliefs	0 to 100	World Values Survey (2024)

Rule of Law	Reflects how much trust individuals have in societal rules and their adherence to them.	-2.5 to +2.5	Worldwide Governance Indicators (World Bank, 2024)

Source: Elaborate by the authors

4.4 Results and Discussion

4.4.1 Data Description

Table 7 depicts the statistical description of the independent variables.

Table 7 – Statistical Descriptions – Economy, Religion, and Legal

Variable	Obs	Mean	Std. Dev.	Min	Max
IDV	119	40.277	22.622	0	100
UAV	119	66.017	21.412	8	100
PDI	119	66.723	20.562	11	100
MAS	119	46.538	17.573	5	100
LTO	109	40.385	20.71	1	100
IDG	98	45.378	23.186	0	100
Freedom	67	0.376	0.947	-1.61	1.775
Religiosity	80	0.619	0.224	0	0.951
Rule of Law	119	0.16	0.954	-2.197	1.958

Source: Elaborated by the authors

The cultural dimensions in the dataset show a clear preference for hierarchical systems and strict boundary definitions. Most countries maintain hierarchical systems, which result in high Power Distance and Uncertainty Avoidance scores according to research data. The dimensions of Individualism and Long-Term Orientation have lower average scores, which indicate that the society operates as a collective unit and prioritizes immediate objectives.

The three institutional indicators, Freedom, Rule of Law, and Religiosity, present significant differences in their measurements. The Freedom Index shows a broad range of scores with a low average value, which suggests that civil liberties vary substantially between different countries in the sample. The Rule of Law manifests differently across nations because their systems of legal enforcement and institutional authority differ. Religiosity shows moderate levels of influence through its average score, which remains stable while its values show limited variation between countries. Religious beliefs function as a stabilizing force that directs human actions by maintaining social norms.

The standard deviations of Individualism and Indulgence and Rule of Law variables demonstrate the wide range of values present in the dataset. The various data points in this variation allow researchers to perform complete international assessments and execute detailed evaluations.

The level of individualism and the strength of legal institutions in a country determine its entrepreneurial potential, yet power distance and uncertainty avoidance create barriers to reform implementation (Lehmann & Weiße, 2025).

4.4.2 Economy

The multinomial logistic regression analysis was conducted with 60 countries, categorized according to their economic status: Developed (reference category), Emerging, and Frontier. Table 8 shows the results of the multinomial regression.

Table 8 – Multinomial Regression Results - Economy

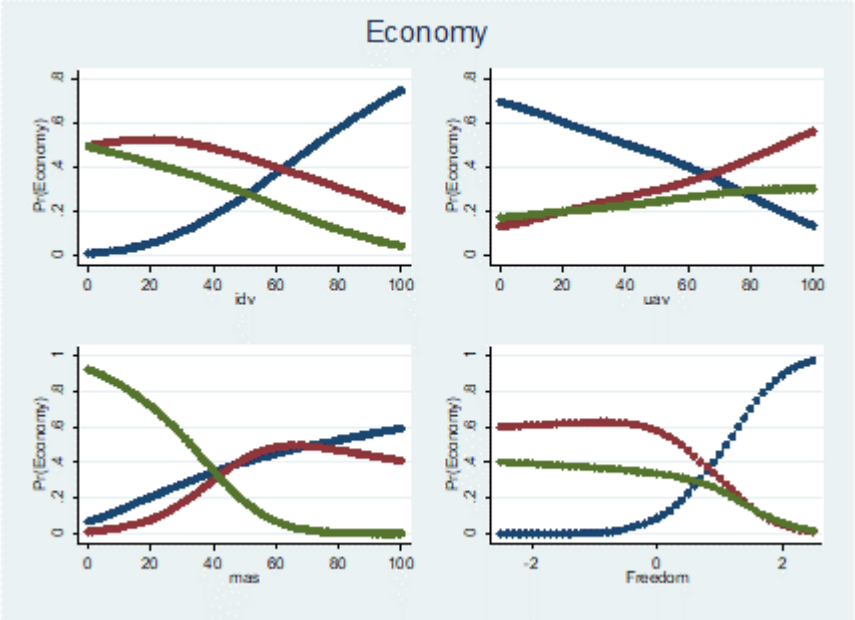
Variable	emerging	frontier
IDV	**-0.099****	**-0.013****
	(-2.00)	**(-2.03)**
UAV	**0.090****	**0.083***
	(2.23)	**(1.94)**
PDI	-0.073	-0.058
	(-1.22)	(-0.89)
MAS	-0.056	**-0.205*****
	(-1.44)	**(-3.03)**
LTO	-0.015	-0.020
	(-0.49)	(-0.57)
IDG	0.016	-0.027
	(0.39)	(-0.60)
Freedom	**-4.235***	**-4.408***
	(-1.79)	**(-1.82)**
Observations	60	
Log likelihood	-30.31	
LR $\chi^2(12)$	69.96***	
Pseudo R^2	0.536	

Source: Elaborated by the authors
Note: ***p<0.01 **p<0.05 *p<0.1

The results reveal significant cultural and Freedom Index differences across developed, emerging, and frontier economies. Emerging and frontier economies, compared to developed ones, have significantly lower individualism (IDV), higher uncertainty avoidance (UAV), and a lower Freedom Index. Beyond that, frontier economies also show significantly lower masculinity (MAS) compared to developed countries. Other cultural dimensions did not show significant differences.

Figure 9 illustrates the findings, showing that the higher the IDV and Freedom Index and the lower the UAV, the greater the likelihood of a society having a developed economy. Similarly, the figure shows that the higher the MAS, the greater the likelihood of a frontier economy culture.

Figure 9 - Cultural dimensions and Freedom Index in Economy



Source: Elaborated by the authors

Therefore, hypothesis H1, *"Developed countries tend to present higher individualism (IDV), lower power distance (PDI), and lower uncertainty aversion (UAV) than emerging and frontier countries"*, is partially confirmed. Developed countries show higher individualism (IDV) and lower uncertainty avoidance (UAV), as the results show. However, although power distance (PDI) is lower in developed countries, there is no statistically significant difference to confirm it.

Investment strategies in emerging economies should consider lower individualism and higher uncertainty avoidance, indicating a preference for stable, predictable investments. The lower Freedom Index in these regions suggests potential regulatory challenges. Lower masculinity in frontier economies implies that different societal values affect market trends. These cultural factors should be considered for informed and culturally sensitive investment decisions.

4.4.3 Religion

The multinomial logistic regression analysis was conducted using 96 countries, categorized according to their majority religion: Catholics (reference category), Buddhism, Confucianism, Hinduism, Islam, Orthodox, and Protestants. Table 9 depicts the results of modeling the cultural dimensions and religiosity index.

Table 9 – Multinomial Regression Result - Religion

Variable	Buddhism	Confucian	Hindu	Islam	Orthodox	Protestant

IDV	-1.787 (-0.01)	-1.072 (-0.00)	-0.225 (-0.01)	**-0.210***** **(-2.68)**	-0.106 (-1.45)	-0.113 (-1.25)
UAV	-0.482 (-0.01)	-1.676 (-0.01)	-2.269 (-0.01)	0.019 (0.42)	0.079 (1.61)	-0.235 (-1.29)
PDI	0.605 (0.00)	-0.159 (-0.00)	-0.720 (-0.00)	0.013 (0.31)	0.048 (1.19)	**-0.159*** **(-1.85)**
MAS	-1.466 (-0.01)	-0.126 (-0.00)	0.436 (0.01)	-0.014 (-0.32)	-0.031 (-0.72)	0.000 (0.00)
LTO	0.524 (0.00)	1.976 (0.00)	1.202 (0.00)	0.105 (1.46)	0.076 (1.09)	0.089 (0.79)
IDG	0.641 (0.00)	-0.539 (-0.00)	-1.472 (-0.00)	-0.048 (-1.55)	**-0.065*** **(-1.94)**	-0.055 (-0.78)
Religiosity	-61.096 (-0.00)	-40.923 (-0.00)	77.466 (0.00)	-4.763 (-1.24)	0.193 (0.05)	-0.854 (-0.20)
Observations		79				
Log likelihood		-38.47				
LR $\chi^2(12)$		182.79***				
Pseudo R^2		0.704				

Source: Elaborate by the authors
Note: ***p<0.01 **p<0.05 *p<0.1

The study reveals that cultural dimensions and religiosity vary significantly across different religious groups. While Buddhism, Confucianism, and Hinduism show no significant differences from Catholicism in all variables, Islam shows significant differences in individualism (IDV). Protestants differ significantly from Catholics regarding power distance (PDI), and Orthodox, regarding indulgence (IDG). Religiosity does not present statistically significant differences.

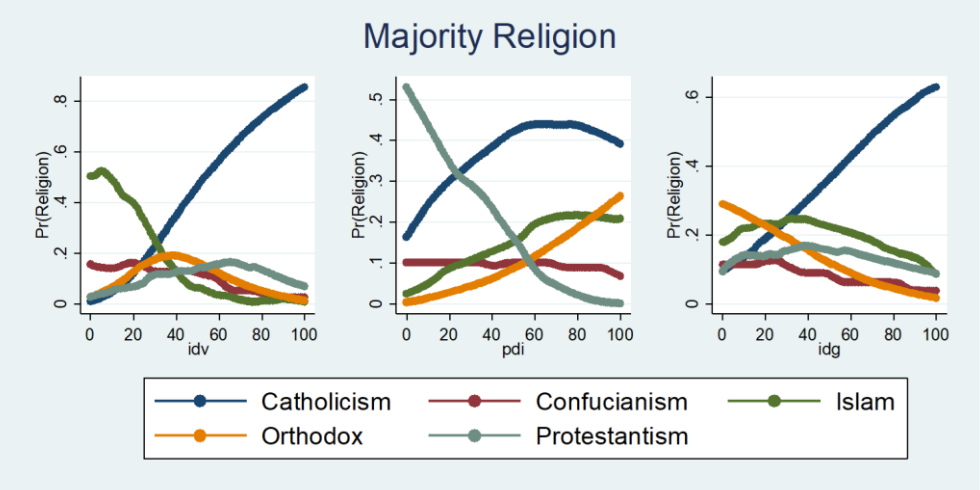
Figure 10 shows findings indicating that societies with a Protestant majority have lower PDI, while societies with a Catholic, Orthodox, or Islamic majority are more likely to have high PDI. Societies with a Catholic majority also have a greater tendency toward individualism (IDV) and greater indulgence (IDG). Conversely, societies with an Islamic majority tend to be more collective, presenting low IDV.

Although Protestant societies on average exhibit higher economic development, our results show that it is development itself, rather than Protestant affiliation per se, that drives higher individualism and lower uncertainty avoidance. Thus, Hypothesis 2a *"Protestant societies tend to exhibit higher individualism (IDV) and lower uncertainty aversion (UAV) than Catholic societies"* is not confirmed.

One plausible explanation is that the various denominations within Christianity, including Lutheran, Reformed, Anglican, and Pentecostal, each have distinct cultural influences that shape individualism and risk-taking behaviors. The second factor of modernization leads Protestant

communities with educated urban residents to adopt individualistic values and low uncertainty avoidance. However, Protestant communities in rural areas with lower education maintain Catholic-like beliefs. Finally, institutional quality serves as a moderating factor, as Protestant societies exhibit increased individualism and reduced uncertainty avoidance when rule-of-law institutions are strong; however, these differences become less pronounced when institutions are weak.

Figure 10 – Relationship between culture and religion



Source: Elaborated by the authors

For hypothesis H2b, *"Islamic societies tend to have a lower long-term orientation (LTO) and lower Indulgence (IDG) than Catholic societies"*. However, the results show Islamic societies have lower indulgence (IDG) than Catholic societies, but it is not statistically significant. Moreover, there is no information about long-term orientation (LTO) in Islamic societies.

Investors should adapt to religious and cultural differences by understanding the cultural dimensions of their target markets, tailoring their investment strategies, engaging local experts for insights, respecting local cultural norms, and continually learning and adapting to cultural changes. This approach can lead to culturally sensitive and successful investment decisions.

4.4.4 Legal Origin

The multinomial logistic regression analysis was conducted on a sample of 98 countries, categorized by legal origin: Common Law (reference category), French Civil Law, German Civil Law, and Scandinavian Civil Law.

Table 10 indicates the results of the multinomial logistic regression:

Table 10 – Multinomial Regression Results – Legal Origin

Variable	French	German	Scandinavian

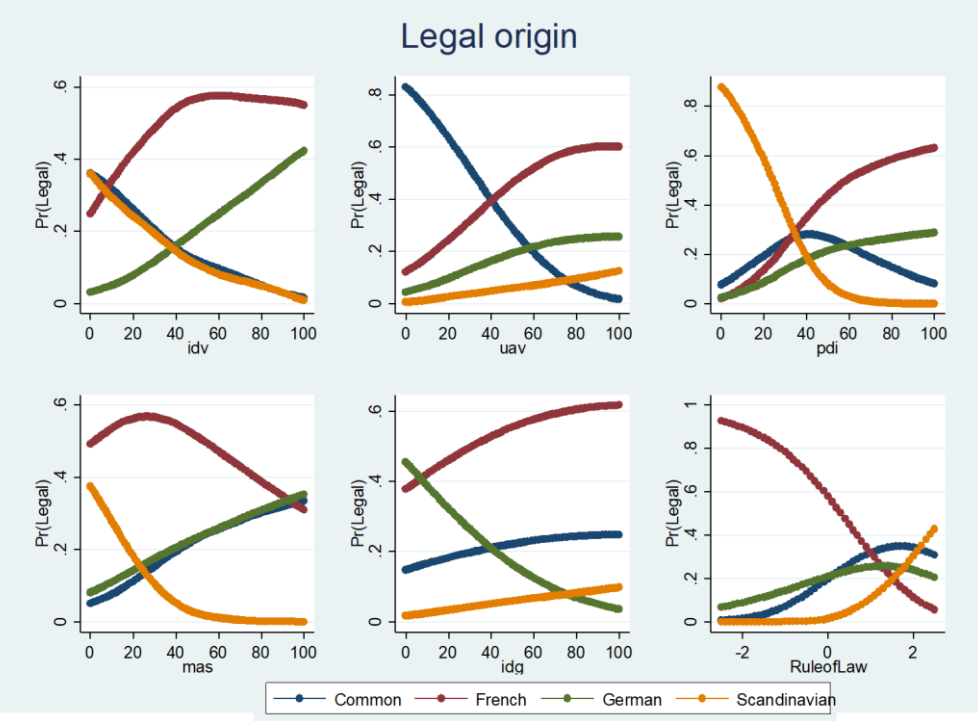
IDV	**0.098*** **	**0.117*** **	-0.115
	(2.81)	**(3.01)**	(-0.73)
UAV	**0.072*** **	**0.074*** **	0.115
	(2.82)	**(2.68)**	(1.39)
PDI	**0.052***	0.054	-0.185
	(1.78)	(1.56)	(-1.06)
MAS	-0.027	-0.000	**-0.168***
	(-1.11)	(-0.00)	**(-1.78)**
LTO	-0.017	0.031	-0.053
	(-0.73)	(1.17)	(-0.65)
IDG	-0.002	**-0.045***	0.024
	(-0.13)	**(-1.75)**	(0.41)
Rule of Law	**-1.865****	-1.069	2.692
	(-2.17)	(-1.06)	(0.84)
Observations		98	
Log likelihood		-62.78	
LR $\chi^2(12)$		104.64***	
Pseudo R^2		0.455	

Source: Elaborate by the authors
Note: ***p<0.01 **p<0.05 *p<0.1

The study found significant differences in cultural dimensions and the Rule of Law index across different legal categories. Specifically, French Civil Law countries have a higher individualism (IDV), power distance (PDI), uncertainty avoidance (UAV), and a lower Rule of Law index, compared to Common Law countries. German Civil Law countries show a higher individualism (IDV) and uncertainty avoidance (UAV), and lower indulgence (IDG) than Common Law countries. Lastly, Scandinavian Civil Law countries have a lower masculinity (MAS) than Common Law countries. The differences in other dimensions were not statistically significant.

Figure 11 displays results indicating that societies rooted in the French civil-law tradition exhibit high individualism (IDV), uncertainty avoidance (UAV), power distance (PDI), masculinity (MAS), and indulgence (IDG), alongside a comparatively low Rule of Law score. In direct contrast, cultures of Scandinavian legal origin show the opposite profile. German civil-law systems stand out for their high individualism and masculinity, moderate uncertainty avoidance and power distance, low indulgence, and robust Rule of Law. Finally, Common Law societies register low individualism, uncertainty avoidance, and power distance, coupled with high masculinity, moderate indulgence, and strong Rule of Law performance.

Figure 11 - Relationship between culture and legal origin



Source: Elaborated by the authors

Based on these findings, hypothesis H3 *"Countries with French Civil Law origins may have cultures with lower individualism (IDV), higher power distance (PDI), and higher uncertainty avoidance (UAV) than Common Law origin countries"*, is partially supported: French civil-law countries demonstrate significantly higher power distance and uncertainty avoidance than their Common Law counterparts. However, the individualism scores in both the Common Law and Scandinavian clusters remain unexpectedly low, probably because their legal systems have developed distinct paths, which negatively affect formal litigation when institutional strength is controlled for.

In our interaction model, the World Bank Rule of Law index, which captures enforcement speed, judicial independence, and bureaucratic efficiency, accounts for most of the positive variation in legal outcomes among high-institution, high-individualism nations such as Canada and Sweden. The remaining individualism effect reflects cultural preferences for private dispute resolution through arbitration, mediation, and other informal mechanisms, leading to a downward slope in both Common Law and Scandinavian origins.

The historical classification system of legal origin delivers valuable information, yet it does not explain how institutions function today. The legal environment consists of the actual Rule of Law and enforcement performance, which represent key operational elements. Research that combines legal origin with legal environment measurements produces better results for institutional risk

evaluation. A system reveals its basic origins in its legal origins, but its operational structure develops from the legal environment. The evaluation of risk, market-entry planning, and institutional cluster boundaries depends on the proper identification of these dimensions.

Furthermore, the outcomes suggest that legal origins significantly influence cultural dimensions, thereby affecting investment decisions. Specifically, French Civil Law countries' higher power distance, uncertainty avoidance, and lower Rule of Law index could indicate a more hierarchical structure and risk-averse behavior, affecting market trends and business practices. German Civil Law countries' higher individualism, moderate power distance, and uncertainty avoidance might imply different societal values that influence corporate governance. Scandinavian Civil Law countries' lower power distance and masculinity could suggest different market dynamics.

4.5 Conclusions

The study examines how national culture influences economic operations as well as religious traditions and legal frameworks. Using data from Hofstede Insights, MSCI, the World Bank's Worldwide Governance Indicators, the Pew Research Center, the World Values Survey, and La Porta *et al.* (2017), the results show that cultural dimensions together with religious affiliation and legal origin, drive variation in corporate market entry approaches, organizational structures and negotiation methods across developed, emerging and frontier economies.

The research found that developed countries generally exhibit higher individualism and higher Freedom Index scores. The results also indicate that people in these nations place greater importance on personal autonomy, and their societies focus on protecting individual rights. The Freedom Index and IDV scores of emerging and frontier economies remain low. However, their UAV scores remain high, suggesting that these societies tend to follow rules to seek security and avoid uncertainty. In addition, frontier economies demonstrate both lower masculinity (MAS) and lower Freedom Index scores, which indicate their populations follow distinct social values and cultural beliefs.

The majority religion also influences the cultural dimensions of a country. Muslims demonstrate lower individualism (IDV), which indicates their society follows a collectivist approach. Orthodox groups generally have lower indulgence, suggesting that they prioritize controlling and curbing the gratification of desires over the free enjoyment of life. Societies with a Protestant majority tend to have low power distance (PDI) and value equality and egalitarian relationships. People in these societies favor flat organizational structures and expect leaders to be approachable and receptive to input.

The research also discovered a correlation between a country's legal origins and cultural dimensions. French Civil Law countries have higher power distance (PDI) and uncertainty avoidance (UAV), suggesting a societal preference for clear hierarchies and rules. Scandinavian Civil Law countries show lower PDI and Masculinity (MAS), indicating a more egalitarian society. German Civil Law countries exhibit higher IDV, moderate PDI, and UAV, while Common Law countries exhibit higher Masculinity (MAS), lower PDI, and lower UAV.

The findings of this study have significant implications for firms' investment decisions:

1. Companies need to understand the cultural aspects of their target international market before they can execute market entry and product strategy. The Freedom Index and a nation's individualistic cultural values determine which products firms emphasize in their offerings. Organizations operating in countries with high uncertainty avoidance need to demonstrate that their products are safe and reliable to their customers.

2. The success of organizational structure and management effectiveness relies on cultural dimensions because they establish proper organizational design and effective management approaches. A hierarchical organizational structure works best in countries with high power distance. On the other hand, in countries with low power distance and high individualism, a flat organizational structure with a participative management style might be more suitable.

3. Businesses can develop negotiation approaches and business strategies through cultural dimension knowledge that suits their target market requirements. Businesses that operate in countries with high uncertainty avoidance must develop comprehensive contracts and well-defined business strategies. In countries with a high long-term orientation, firms might need to demonstrate commitment and perseverance.

4. The cultural aspects of a society establish the degree of risk that an investment will have. High uncertainty avoidance in countries leads to lower acceptance of new products and business models, which creates risks for innovative companies.

5. Businesses need to understand their domestic legal system because the legal origins of a nation affect cultural dimensions that determine investment choices. For instance, French Civil Law countries with higher power distance and uncertainty avoidance might have more bureaucratic procedures, which could affect the ease of business.

6. The influence of the majority religion on a country's cultural dimensions suggests that firms must be mindful of religious norms and values when operating in these countries. Business operations

across multiple areas would experience changes because of this influence, which extends from product development to marketing strategies.

In conclusion, understanding a country's cultural dimensions can provide valuable insights for firms making investment decisions. However, it is essential to remember that these are aggregate-level findings, and individual behaviors within each country can vary greatly. Therefore, firms should complement these insights with on-the-ground research to understand the specific context in which they will operate. Further research is needed to explore these variations and their implications.

5 CULTURAL DIMENSIONS AND INVESTOR OVERCONFIDENCE: LESSONS FROM GLOBAL MARKET CRISES

5.1 Introduction

Overconfidence refers to people's perception that they are more competent than others (De Bondt & Thaler, 1995), leading individuals to overestimate the accuracy of their knowledge, skills, and control over events (Bouteska & Regaieg, 2018). Overconfident investors tend to be risk-takers, are more active in the market, and have more assets in their portfolios than their rational counterparts (Gupta *et al*., 2018). This overconfidence results in mispricing and overvaluation in financial markets, as investors perceive themselves as having superior information-gathering skills and thus prefer their private information over public information (Trejos *et al*., 2019). This aggressive trading behavior, driven by overestimating their trading skills, leads to higher trading volumes and increased market volatility (Gervais & Odean, 2001). Consequently, overconfidence amplifies investor behavior bias in the market (Bouteska *et al*., 2023).

Scholars have shown that investors suffering from self-attribution biases tend to overweight private information signals (Daniel *et al*., 1998). Consequently, when public information arrives in the market, overconfident investors only partially adjust to the new information and take much longer to get closer to the whole information level (Dou *et al*., 2016). Such 'irrational' behavior, which Hofstede (2011) refers to as "collective mental programming," significantly impacts stock prices (Chang & Dong, 2006). Therefore, values and beliefs influence how people interpret information and respond to it (e.g., Brochet *et al*., 2019; Chui & Kwok, 2008; Nguyen *et al*., 2018)[3]. This social signaling bias, initially proposed by Bénabou and Tirole (2002), highlights how cultural differences account for variations in financial decision-making at the country, firm, and individual levels (e.g., De La Cruz *et al*., 2019). However, to date, research is limited regarding the role of national culture in explaining social signaling bias in variations on overconfident investor behavior, especially during disaster periods.

Previous research has demonstrated that investor sentiment and mood can be adversely influenced by various external events, including natural and human-caused disasters, which subsequently affect investment behavior (Dai *et al*., 2020; Dessaint & Matray, 2017; Wang & Young, 2020). Brounen and Derwall (2010) report that international equity markets tend to experience short-

[3] See De Bondt and Thaler (1995) and Barber and Odean (2001) for a review of behavioral finance.

term declines following terrorist attacks, with these effects being more pronounced than those observed after natural disasters such as earthquakes. During the COVID-19 pandemic, overconfidence bias diminished in developed markets, reflecting a broader decline in investor confidence. Conversely, investors continued to exhibit strong overconfidence in emerging and frontier markets, such as China, Taiwan, Turkey, Jordan, and Vietnam, engaging in speculative trading despite heightened market volatility (Shrothryia & Kalra, 2023) The heterogeneity in investor responses to adverse events appears to be partially shaped by cultural values (Fernandez-Perez *et al*., 2021), suggesting that culture functions as a socially transmissible construct that interacts with behavioral predispositions to influence financial decision-making in complex ways (Beckmann *et al*., 2008). As national culture encompasses typical behaviors and shared beliefs of most people within a country (Beckmann *et al*.,2008), we posit that national culture plays a significant role in shaping market sentiment, particularly regarding reaction to information shock and consequent investment decision-making processes.

To advance the field, this study examines whether variations in national culture influence overconfidence levels across sixty-four financial markets, employing Hofstede's (2011) national culture index over ten years. To account for temporal dynamics and periods of heightened uncertainty, the analysis is segmented into three distinct phases: the pre-pandemic period (January 2015 – March 2020), the pandemic period (March 2020 – December 2021), and the post-pandemic period (January 2022 – June 2024).

Hofstede's cultural index comprises six dimensions: (i) Individualism versus Collectivism (IDV) -how individuals relate to primary groups; (ii) Uncertainty Avoidance (UAV) - how much tension does a society experience when it faces an uncertain event; (iii) Power Distance (PDI) - how societies deal with inequalities; (iv) Masculinity versus Femininity (MAS) - how roles are distributed between men and women; (v) Long-term versus Short-term Orientation (LTO) - whether people's efforts are future- or past-/present-oriented; and (vi) Indulgence versus Restraint (IDG) - the balance between satisfying and regulating human desire. Every country has been rated and compared with others on each dimension (Hofstede, 2011).

We find that the cultural dimensions of Indulgence and Power Distance exert consistent and stable effects on overconfidence across the pre-pandemic, pandemic, and post-pandemic periods. Similarly, Long-Term Orientation maintains a stable influence; however, it produces an unexpected result, with higher levels of long-term orientation associated with increased overconfidence.

In contrast, the cultural dimensions of Individualism and Masculinity, which are typically positively associated with overconfidence, both reversed their associations during the pandemic and post-pandemic periods, indicating a behavioral shift under crisis conditions. Meanwhile, the Uncertainty Avoidance dimension reveals a striking behavioral shift across periods. The pre-pandemic phase unexpectedly shows a positive and statistically significant relationship with overconfidence. However, this association reverses during the pandemic and post-pandemic periods, becoming negative as individuals likely become more sensitive to ambiguity and disorder. This behavioral adaptation reflects a heightened tendency to seek structure and adhere to strict norms when confronted with unstructured and unpredictable conditions. These findings underscore the non-linear and dynamic nature of the relationship between cultural values and overconfidence, highlighting that culturally contingent factors shape investor responses to new information.

5.2 Data and method

5.2.1 Data

We collect daily trade volume data from 64 financial markets from January 2015 to June 2024 from the Capital IQ database, comprising 152,521 observations. To gather data related to Hofstede's six cultural dimensions, we use Hofstede Insights (2024), which offers data for 119 countries. Following the official declaration by the World Health Organization (WHO) on 11 March 2020, recognizing COVID-19 as a global pandemic (Padhan & Prabheesh, 2021), and to discern behavior patterns, the analysis is split into three distinct periods: pre-pandemic (January 2015 – 10 March 2020), pandemic (11 March 2020 – December 2021), and post-pandemic (January 2022 - June 2024).

5.2.2 Method

To measure overconfidence, we follow the approach of Chang and Lin (2015). First, and to avoid the influence of industrial differences, we calculate the average trade volume of all industries with a decrease in the industrial index in the stock market i for day t, as in the equation (4):

$$\bar{V}_{i,t}^{d} = \sum_{j=1}^{q_{i,t}} V_{i,j,t}^{d} / q_{i,t} \tag{4}$$

where:

$\bar{V}_{i,t}^{d}$ is the quotient of the trading volume of sector *j* with decreased sector indexes in stock market *i* for day *t,* divided by the average daily trading volume of sector *j* during the research interval.

$q_{i,t}$ is the number of sectors with decreased sector indexes in the stock market *i* for day *t*.

We, then, compute overconfidence for each country, considering the average trading volume of sector j during the research interval as a proxy for "normal" daily trading volume. Therefore, the excessive trading volume as a proxy for overconfidence (Odean, 1998) is calculated as in the equation (5):

$$OC_{i,t} = \bar{V}_{i,t}^{d} \quad (5)$$

where:

$OC_{i,t}$ is the overconfidence of the stock market i for day t

$\bar{V}_{i,t}^{d}$ is the quotient of the trading volume of sector *j* with decreased sector indexes in stock market *i* for day *t,* divided by the average daily trading volume of sector *j* during the research interval.

To conduct the multinomial logit analysis, we categorize the variable overconfidence into three distinct levels based on its quartile distribution. Observations with overconfidence values below the first quartile (Q1) are classified as exhibiting low overconfidence and are assigned the value -1. Conversely, observations with overconfidence values exceeding the third quartile (Q3) are considered to reflect high overconfidence and are assigned the value of 1. Values falling between the first and third quartiles (i.e., between Q1 and Q3) are categorized as neutral and are assigned the value of 0. Further, to examine how national culture relates to country variations in overconfidence, we then estimate the equation (6), (7), and (8):

$$\text{OCLvi} = \alpha_0 + \alpha_1 IDV_i + \alpha_2 UAV_i + \alpha_3 PDI_i + \alpha_4 MAS_i + \alpha_5 LTO_i + \alpha_6 IDG_i + \varepsilon_i \quad (6)$$

$$\sum_{h=1}^{3} \pi_h = 1 \quad (7)$$

$$\pi_1 = P(\text{OCLv}_\text{i} = -1);\ \pi_2 = P(\text{OCLv}_\text{i} = 0);\ \pi_3 = P(\text{OCLv}_\text{i} = 1) \quad (8)$$

where:

OCLv_i is the level of the OC.

For the first quartile of OC (p25), OCLv_i = -1; for the fourth quartile of OC (p75), OCLv_i = 1; otherwise, OCLv_i = 0 and

π_1 , π_2, π_3 represents response probabilities (OCLv = -1 for Low, OCLv = 0 for Neutral, and OCLv = 1 for High)

Culture entails the six dimensions of the Hofstede national culture index, which range from 0 to 100, and Ɛ is the error term.

Additionally, to mitigate the distorting impact of outliers, the final dataset has been winsorized at the lowest (p1) and highest percentiles (p99) of the overconfidence variable, yielding 140,836 daily observations. See Appendix E.

5.2.3 Robustness

For robustness, we employ the Generalized Additive Model (GAM). The GAM model relaxes the strict linearity assumption of linear models by modelling the explained variables as an additive combination of the explanatory variables. This approach offers considerable flexibility, particularly when the residual distribution of Hofstede's dimensions is unknown, while preserving additivity (Hastie & Tibshirani, 1986). The model specification is presented in Equation (**9**):

$$OC_{i,t} = \beta_0 + f_1(IDV_i) + f_2(UAV_i) + f_3(PDI_i) + f_4(MAS_i) + f_5(LTO_i) + f_6(IDG_i) + \varepsilon_{i,t} \quad (9)$$

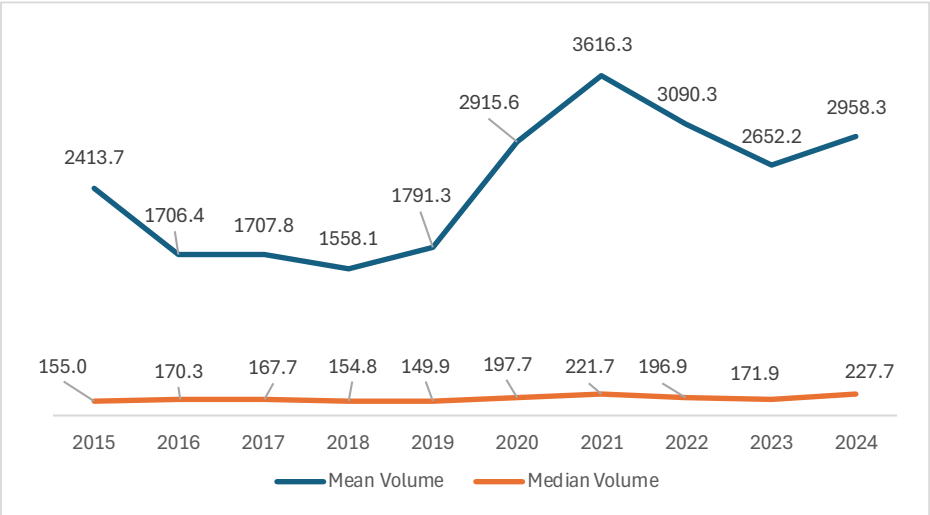
where:

f_i is a smooth function calculated to fit each of the six Hofstede's cultural dimensions for each country i, IDV_i, UAV_i, PDI_i, MAS_i, LTO_i, and IDG_i

5.3 Results

This paper investigates whether Hofstede's cultural dimensions determine overconfidence levels across 64 financial markets over 10 years, differentiating three different periods: pre-pandemic COVID period, COVID-19 period, and post-pandemic period. Figure 12 depicts the means and median yearly countries' trade volume (in millions).

Figure 12 - Countries' trade volume per year (in millions)



Source: Elaborated by the authors using Capital IQ database

Throughout the pandemic years (March 2020-2021), traded volumes experienced a significant surge. This period saw the highest recorded trading volumes (see, for example, Shrotryia & Kalra, 2021). As reported in Appendix E, the mean trading volume and the standard deviation vary significantly across the different countries. For example, China, Hong Kong, Indonesia, Russia, Thailand, and the United States exhibit the highest mean trading volume. These results align with those of Chiah and Zhong (2020), who found that trading volume increased significantly during the pandemic period, with an average rise of 54% compared to pre-COVID-19 levels, sustained by huge positive investor sentiment, heightened market volatility, and greater divergence in investor opinions during the crisis period.

Table 11 presents the descriptive statistics. The high standard deviation of Overconfidence indicates significant discrepancies in overconfidence levels across countries throughout the analyzed period. Further, Hofstede's six cultural dimensions also show significant variations, indicating different behavioral features at different levels of overconfidence. For example, Individualism achieves the highest value in the Netherlands (100) and the lowest in Nigeria (0); Uncertainty Avoidance displays a value of 100 and 99 in Greece and Portugal, respectively, and 8 in Singapore. The dimension Power Distance reports the lowest value in Austria (11), whereas Malaysia displays the highest value (100). The Masculinity dimension reports 95 in Japan and only 5 in Sweden. Japan and Venezuela report the highest values of Long-Term Orientation and Indulgence, respectively, scoring 100. The lowest values for Long Term Orientation are reported by Peru (5) and Colombia (6); for Indulgence, Egypt and Pakistan both report a low value of 0. See Appendix F.

Table 11 – Descriptive Statistics - Overconfidence

Variable	Obs.	Mean	S.D.	p25	Median	p75	Min	Max
Overconfidence	140,836	1.083	0.887	0.562	0.859	1.312	0.038	5.624
Individualism	152,521	48.244	23.720	28	47	70	0	100
Uncertainty avoidance	152,521	64.542	22.331	49	69	85	8	100
Power Distance	152,521	59.592	20.358	45	65	73	11	100
Masculinity	152,521	49.653	17.863	42	50	62	5	95
Long-term orientation	150,148	44.620	20.911	29	47	57	0	100
Indulgence	142,895	46.860	22.024	29	46	65	0	100

Source: Elaborated by the authors

Table 12 depicts the multinomial logistic regression results. Panel A and Panel B report results for the pre-pandemic and pandemic periods, respectively, whereas Panel C displays results for the post-pandemic period. Columns 1 to 6 span the partial models; the complete model appears in column 7. For brevity, we focus our analysis on the whole model. Further, to better capture the potential

nonlinearity relations between overconfidence levels and the cultural dimensions, we display in Figure 13 - Panel A (pre-pandemic), Panel B (pandemic), and Panel C (post-pandemic), the nonlinear prediction at 95% confidence interval of overconfidence levels based on assumed intervals of 0 to 100 for the six Hofstede cultural dimensions based on GAM regression results. See also Appendix G.

Table 12 – Multinomial logistic estimations. Dependent variable: Overconfidence

Panel A - Pre-Pandemic period: 2015-2020

Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Constant	-1.613***	-1.123***	-0.213***	-1.587***	-1.5165***	-1.264***	-2.350***
Individualism	0.0128**						0.0082***
Uncertainty Avoidance		0.0016***					0.0018**
Power Distance			-0.013***				-0.0049***
Masculinity				0.0116***			0.0138***
Long-term Orientation					0.01214***		0.0090***
Indulgence						0.0059***	0.0019***
Pseudo R^2	0.017	0.002	0.016	0.005	0.029	0.005	0.049
χ^2	2524.16***	328.20***	2283.71***	701.04***	4166.16***	726.52***	6621.47***

Panel B - Pandemic period: 03:2020-2022

Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Constant	0.3883***	2.8904***	2.3922***	1.384***	-0.6206***	1.175***	3.066***
Individualism	0.0157***						-0.0198***
Uncertainty Avoidance		-0.027***					-0.0249***
Power Distance			-0.0207***				-0.0171***
Masculinity				-0.0060***			-0.0110***
Long-term Orientation					0.0445***		0.0530***
Indulgence						-0.0007	0.0071***
Pseudo R^2	0.013	0.018	0.012	0.002	0.042	0.000	0.072
χ^2	693.49***	978.24***	663.56***	123.74***	2271.13***	1.19	3648.95***

Panel C - Post-Pandemic period 2022_06:2025

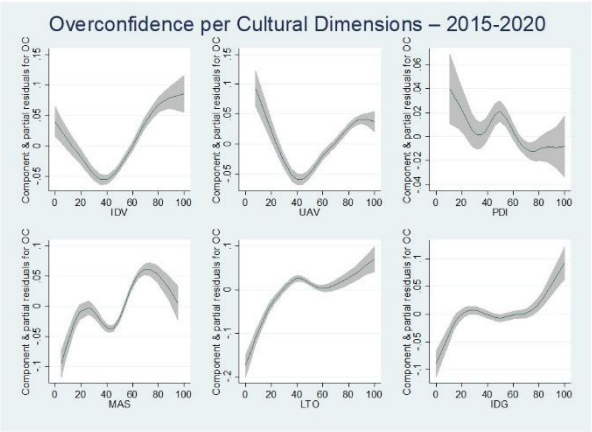
Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Constant	0.2685***	2.4502***	2.2266***	1.2398***	-0.3543***	1.0525***	3.8648***
Individualism	0.0133***						-0.0105***
Uncertainty Avoidance		-0.0238***					-0.0236***
Power Distance			-0.0219***				-0.0169***

Masculinity				-0.0076***			-0.0172***
Long-term Orientation					0.0313***		0.0282***
Indulgence						-0.0026***	0.00027
Pseudo R^2	0.011	0.014	0.013	0.003	0.032	0.001	0.056
χ^2	854.43***	1053.63***	973.70***	203.11***	2365.33***	35.20***	3913.08***

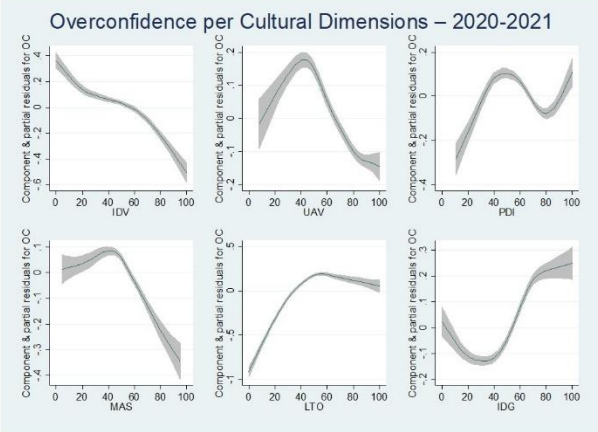
Note: Multinomial logistic estimations where the dependent variable is overconfidence. *p<0.1; **p<0.05; ***p<0.01
Source: Elaborated by the authors

Figure 13 - Predictive relationships of overconfidence levels by cultural dimensions

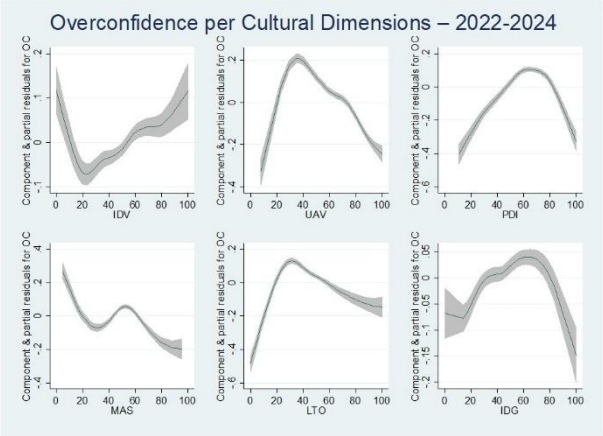
Panel A: Pre-Pandemic Period: 2015-2020



Panel B: Pandemic: 03:2020-2021



Panel C: Post-Pandemic: 2022-06:2024



Notes: The figure illustrates the nonlinear prediction of overconfidence at the 95% confidence interval, based on assumed intervals of 0 to 100 for the six Hofstede cultural dimensions. The nonlinear prediction is based on the results of the GAM regression, as specified in Equation (**9**9) and displayed in Appendix G.
Source: Elaborated by the authors

Results from multinomial logistic regression show that the dimensions Indulgence, Power Distance, and Long-Term Orientation show the same pattern across the three periods. However, Long-Term Orientation displays an unexpected positive relationship. The positive and statistically significant coefficient of Indulgence indicates that societies that allow for the gratification of basic human desires are prone to overconfidence. Graphical GAM non-linear results from Figure 13 indicate that overconfidence rises sharply for high levels of Indulgence (Panel A and Panel B). Still, in the post-pandemic period, this behavior seems restrained. In Figure 13, Panel C indicates a sharp decline in the post-pandemic period, suggesting that individuals seem to accept behavior according to more strict norms after a highly uncertain period. Power Distance behaves as expected, showing a consistently negative coefficient across all periods. It reflects that individuals who accept a hierarchical order without questioning it and seeking further justification restrain their overconfidence level (e.g., Hua and Wei, 2017). Graphical results underline a non-linear relationship, indicating that high levels of Power Distance imply a substantial decrease in overconfidence in the post-pandemic period. Long-term orientation displays unexpected behavior, reporting a positive and statistically significant coefficient over the three periods, suggesting that societies with strong future-oriented perspectives may exhibit overconfidence in long-term decision-making even during uncertain times. Though this behavior is typical in Short-Term Oriented cultures, where decisions are more often made in short-term periods that can lead to neglect of a deeper analysis and, therefore, amplify the individual overconfidence tendency (Barber & Odean, 2001; Souder *et al*, 2016). Although in the graphical representation of GAM coefficients (Figure 13, Panels B and C), we note a non-linear relationship showing that, since the pandemic period, high levels of Long-term Orientation have led to a decrease in Overconfidence levels.

Regarding the cultural dimensions of Individualism, Uncertainty Avoidance, and Masculinity, their influence on overconfidence reverses between the pre-pandemic and pandemic periods, with this inverse relationship persisting into the post-pandemic phase. This shift suggests a sustained behavioral adjustment in response to crisis conditions, reflecting the dynamic nature of culturally embedded investor responses to uncertainty. Individualism reports the expected positive relationship with Overconfidence in the pre-pandemic period, suggesting that in societies that value individualism, people put their interests above those of the group. This behavior changes during and after the pandemic, indicating that individuals value a strong, cohesive group sentiment in periods of high uncertainty. However, if we look at the non-linear results (Figure 13 – Panel C), the relationship between Individualism and Overconfidence returns positive, indicating that individuals seem to prioritize their interests above those of the group. The Uncertainty Avoidance dimension depicts a surprising result in the pre-pandemic period, showing a positive and statistically significant relationship with Overconfidence. Though in the following periods it reverses the signal, that is, positive to negative, it underlines that individuals display more uncertainty avoidance when they become more nervous in situations that they perceive as unstructured, unclear, and unpredictable, thus trying to follow strict codes of behavior. This pattern persists after the pandemic, reporting a strong non-linear relationship, underlining that as Uncertainty Avoidance increases, the overconfidence sharply decreases. The Masculinity dimension changes the pattern of behavior from the pre-pandemic period to the following periods. The positive relationship displayed in the pre-pandemic period suggests that societies more aligned with masculine behavior embrace assertiveness and material rewards and thus leverage overconfidence. During the pandemic and post-pandemic periods, the relationship becomes negative. This trend is clearly illustrated in Figure 13, Panels B and C, which emphasize the emergence of a negative association. These graphical results suggest that, following a period of profound global uncertainty, individuals have come to place greater value on cooperation and exhibit a more consensus-oriented worldview.

5.4 Discussion and conclusion

To address the gap in how national culture explains social signaling bias in variations on overconfidence at country levels, especially during disaster periods, this paper explores whether differences in national cultures determine overconfidence levels across 64 financial markets over 10 years. We found that the dimensions Power Distance and Indulgence maintain their expected influence across periods, that is, negative and positive relationships, respectively, with Overconfidence. The cultural Power Distance dimension remains negative across all periods,

indicating that “Less powerful person in a society (or organization) accepts inequality in power, considering it as normal” (Hofstede, 1986, p. 307); therefore, they refrain from individual overconfidence (e.g., Hua & Wei, 2017). Societies characterized by high levels of Indulgence tend to encourage individuals to prioritize enjoyment and personal gratification, which may lead to overestimating their capabilities. As a result, such individuals are more prone to overreact during heightened uncertainty (Cieślik *et al*., 2023). Further, we report that the Individualism and Masculinity dimensions reverse their influence from the pre-pandemic period to the following periods. In the pre-pandemic period, individualist societies leveraged their overconfidence (e.g., Chui *et al*, 2010; Hua & Wei, 2017; Nguyen & Truong, 2013; Yates & Oliveira, 2016), as well as Masculinity societies (e.g., Hua & Wei, 2017; Tan *et al*., 2019); Such societies tend to be more competitive, embrace assertiveness and prioritize their interests above the group. However, both dimensions exhibit a significant reversal during the pandemic and post-pandemic periods, suggesting that behaviors associated with assertiveness and risk-taking may undergo recalibration in response to heightened uncertainty. In such contexts, there appears to be a shift towards prioritizing collective interests, with individuals increasingly behaving as part of a cohesive group that seeks equitable outcomes for all its members. The dimensions of Long-Term Orientation and Uncertainty Avoidance yield unexpected results across the periods under examination. Specifically, overconfidence levels appear to increase with higher levels of Long-Term Orientation. This seemingly contradictory pattern may suggest that societies with a strong future-oriented outlook are more prone to exhibiting overconfidence in long-term decision-making, potentially due to an overreliance on strategic foresight and planning. As illustrated in Figure 13 (Panels B and C), the non-linear results reveal an inverted U-shaped trend during the pandemic and post-pandemic periods. Specifically, overconfidence increases at lower levels of Long-Term Orientation but declines sharply beyond a certain threshold. These findings suggest that societies with strong long-term orientation—characterized by an emphasis on perseverance, savings, respect for tradition, and fulfilment of social obligations—may exhibit more restrained and cautious decision-making, tempering overconfidence in uncertain contexts. This dimension aligns with restrained societies where individuals leverage future rewards. Uncertainty Avoidance dimension also reports a surprising result in pre-pandemic period, suggesting that contrary to low Uncertainty Avoidance societies tend to present more overconfidence due to a greater risk-taking tendency (Acker & Duck, 2008; Nguyen & Truong, 2013; Tan *et al*., 2019; Xu, 2022), the pre-pandemic contradictory result might indicate a form of defensive or compensatory overconfidence in stable times.

The findings of this study contribute to the existing literature by illustrating that investor responses to uncertainty and elevated risk are, to a significant extent, shaped by underlying cultural values. This highlights the dynamic nature of overconfidence behavior, which appears to respond to crises through culturally contingent shifts in investor behavior under heightened uncertainty. Despite its contributions, this study is subject to certain limitations. Most notably, the post-pandemic period remains relatively short, limiting the ability to assess whether the behavioral patterns observed during the pandemic will persist over the long term. Another important limitation is that there is no way to differentiate the trading volume generated by robots, which has grown substantially in the last decade. On the other hand, recent research (e.g., Oyeniyi *et al.*, 2024; Yang *et al.*, 2025) indicates that technology does not eliminate behavioral biases but rather systematizes them. The volume generated by robots acts as a mechanism for transmitting fundamental human impulses. Thus, total volume remains a valid proxy for the psychological inclinations of the culture in which capital is embedded. Nevertheless, using cross-country data spanning ten years offers a valuable contribution to the literature. The findings lay a foundation for future research to explore further and disentangle the complex relationship between cultural values and investor overconfidence.

6 INVESTOR BIAS ACROSS BORDERS: CULTURAL INFLUENCES ON THE DISPOSITION EFFECT

6.1 Introduction

Behavioral finance has changed how we view markets by incorporating psychological considerations into conventional economic theories (Shefrin, 2002). It has important implications for investment policies and market behavior (Barberis & Thaler, 2003; Odean, 1998; Shefrin & Statman, 1985). The disposition effect (DE) is the tendency for investors to sell appreciated assets quickly while holding on to depreciating ones (Shefrin & Statman, 1985). This behavior is inconsistent with classical finance theory, which assumes rational investor decision-making.

Culture emerges as a crucial theoretical determinant of the disposition effect (DE) by serving as the collective mental programming system that modulates investors' perceptions, reactions, and decision-making regarding gains and losses in the capital market (Hofstede, 2001; Goodell, Kumar, *et al*., 2023). The DE is fundamentally a behavioral bias rooted in Prospect Theory, driven by loss aversion (late selling of losers) and the desire to realize pride (early selling of winners).

The Behavioral Finance literature recognizes that investment decisions are influenced by emotional and psychological factors (Ricciardi & Simon, 2000; Zahera & Bansal, 2019), and national culture, by integrating values, ways of thinking, and social structures, provides the framework for these emotional responses (Geertz, 1993; Schwartz, 2012). As an aspect of human decision-making, investor behavior in selecting and reacting to different types of assets or information has been studied in various markets (e.g., Bharandev & Rao, 2019; Chen *et al*., 2007; Han & Preda, 2025) and among various types of investors (e.g., Ahn, 2022; Breitmayer *et al*., 2019; Fan & Neupane, 2024; Frino *et al*., 2015; Hwang *et al*., 2024; Shu *et al*., 2005); however, cultural preferences can affect investment behavior.

The comparative analysis using multinomial logistic regression and Generalized Additive Models (GAM) provides compelling evidence that cultural influences on the DE are neither uniform nor strictly linear. The divergence between the two modeling approaches underscores the complexity of behavioral responses across cultural contexts. While the logistic model suggests an inverse association between Masculinity (MAS) and Indulgence (IDG) and the prevalence of DE—contrary to prior theoretical expectations—GAM results reveal non-linear patterns that challenge conventional interpretations. For instance, Individualism (IDV), traditionally viewed as a mitigating factor against

behavioral biases, exhibits a positive relationship with DE in the GAM framework, suggesting threshold effects and interaction dynamics that linear models fail to capture.

These findings highlight the necessity of adopting flexible, non-parametric approaches when examining cultural determinants of investor behavior. The observed inconsistencies between models are not statistical anomalies but rather indicative of underlying behavioral heterogeneity. Cultural dimensions may exert influence only under specific conditions or at certain intensity levels, reinforcing the argument that investor psychology is context-dependent. Moreover, the role of economic development emerges as a stabilizing factor. Gross National Income per capita (GNIPC) consistently demonstrates a negative association with DE, implying that wealthier nations, likely characterized by more sophisticated financial systems and higher levels of investor education, exhibit reduced susceptibility to this bias. This result aligns with prior literature emphasizing the protective effect of financial literacy and institutional quality.

From a practical standpoint, these insights carry significant implications for policy design and investor education. Financial literacy programs and regulatory interventions should incorporate cultural profiling to address behavioral vulnerabilities effectively. A one-size-fits-all approach is inadequate; instead, strategies must be tailored to the cultural and economic realities of each market. By acknowledging the non-linear and context-specific nature of cultural influences, policymakers and practitioners can develop more nuanced frameworks to mitigate biases such as the Disposition Effect, thereby promoting market efficiency and investor welfare.

6.2 Literature Review

The Prospect Theory was proposed by Kahneman and Tversky in 1979 to explain how people make financial decisions based on gains and losses. Even though Shefrin and Statman (1985) have been credited with assisting in the identification of DE, a behavior bias that has to do with investors' tendency to sell appreciated assets and hold on to depreciated assets, this work has been expanded upon by other researchers who have sought to extend the investigation of DE to various demographic groups and market environments.

For example, Odean (1998) conducted the first empirical validation, demonstrating that individual investors tend to sell their winning stocks prematurely while holding their losing stocks for extended periods. Frino *et al*. (2015) and Breitmayer *et al*. (2019) show that gender and age act as demographic DE amplifiers, while many studies support the global nature of the bias (e.g., Bharandev & Rao, 2019; Chen *et al*., 2007). In addition, Breitmayer *et al*. (2019) show that countries

with better financial literacy and more developed markets and higher wealth levels tend to have lower DE, as education, institutional quality, and information access might reduce emotional biases.

The research by Ahn (2022) shows that investors who have limited financial knowledge, low investment experience, and who choose passive investment strategies will be most affected by DE bias. Hwang *et al*. (2024) quantify its magnitude by documenting that many investors refuse to dispose of holdings when they decline by more than 30 percent. Fan and Neupane (2024) demonstrate that short-term foreign institutional investors in India are most susceptible to bias under high information asymmetry, resulting in detrimental effects on their subsequent performance. Han and Preda (2025) demonstrate that home bias causes retail foreign exchange traders to maintain their losing positions, while Sui and Wang (2025) show that larger portfolio stakes intensify reluctance to realize losses due to cognitive dissonance and concerns about self-image.

Under Prospect Theory (Kahneman & Tversky, 1979), investors are naturally risk-averse when ahead and risk-seeking when down. High UAV cultures make investors more fearful of ambiguous situations, leading them to seek quick profits while holding on to unprofitable assets (Breitmayer *et al*., 2019). People tend to sell their successful investments early because of cultural influences from Prospect Theory, which shows that individuals value gains less than they fear losses. In other words, the higher a country's UAV score, the more reluctant investors in that society will be to risk giving up a sure gain by disproportionately selling winners, and to avoid the pain of a realized loss by holding on to losers (Shefrin & Statman, 1985). Moreover, Breitmayer *et al*. (2019) and Rieger *et al*. (2022) find a strong positive correlation between uncertainty avoidance and the disposition effect.

Furthermore, high power distance societies place the greatest value on showing respect to those in authority and on maintaining social standing (Hofstede, 2011). Investors in high power distance societies tend to follow established market trends and seek guidance from authority figures because admitting losses would result in loss of social standing (Hsee & Weber, 1999). The fear of admitting errors leads investors to maintain their losing positions, which strengthens the phenomenon of unequal gain and loss realization (Bontempo *et al*., 1997; Breitmayer *et al*., 2019; Frino *et al*., 2015; Hsee & Weber, 1999). The behavior follows cultural norms that prevent people from admitting mistakes while showing deference to those in authority and maintaining social harmony, thereby supporting the psychological aspects of DE (Rieger *et al*., 2022).

Nonetheless, people who think long-term tend to discount time less because they value future rewards as much as present rewards (Hofstede, 2011). LTO Cultures support patient-centered

strategic thinking, yet they prevent investors from making impulsive, short-term decisions that result in DE (Briley *et al*., 2000). The investors continue their financial backing despite current losses because they believe these temporary expenses will yield significant profits in future years (Rieger *et al*., 2022). High-LTO culture investors use long-term evaluation methods because these methods shield them from making impulsive decisions based on short-term market volatility (Rieger *et al*., 2022). The decision-making process of people from long-term-oriented cultures focuses on future results and social equilibrium, which acts as a defense mechanism against DE (Briley *et al*., 2000).

Across the literature, higher individualism tends to weaken DE, whereas lower individualism tends to amplify it. Rieger *et al*. (2022) found that investors from individualistic societies use rational thinking in their investment decisions, thereby reducing emotional bias. Briley *et al*. (2000) demonstrated that collectivist societies base their judgments on situational factors, yet individualist societies depend on dispositional factors. DE becomes stronger for investors from collectivist societies because they base their investment decisions on external events that confirm their poor investment choices. Reinforcing this, Breitmayer *et al*. (2019) demonstrated that people from individualistic cultures experience less DE than those from collectivistic societies because independent choice and personal responsibility protect against emotional decision errors.

Masculine culture promotes aggressive conduct and success because winning trades confer social status and business growth opportunities, yet investors stay in losing trades to hide their weakness from others (Breitmayer *et al*., 2019). Higher masculinity levels increase the risk of DE because they drive individuals to become more competitive and performance-oriented (Lawal & Sakariyaku, 2024). Male investors who work in masculine environments choose DE because their social environment values success and competition (Lawal & Sakariyaku, 2024; Rieger *et al*., 2022). Frino *et al*. (2015) demonstrated that male investors show higher DE levels than female investors, probably due to their assertive nature and performance-driven approach, making them more prone to biased trading decisions.

A higher-indulgence society tends to increase DE because people in such environments tend to react emotionally and seek immediate gratification (Rieger *et al*., 2022). People from indulgent cultures base their choices on immediate emotional cues, leading them to rush into selling their winning stocks but delay selling their losing ones (Hsee & Weber, 1999). Furthermore, Briley *et al*. (2000) state that DE in indulgent cultures becomes more likely as people choose based on emotional fulfillment and social peace.

Prior research has examined the relationship between the DE and cultural factors, emphasizing that cultural environments constitute key determinants of investor behavior (Breitmayer et al., 2019; Hsee & Weber, 1999; Rieger & Wang, 2012). Nevertheless, the existing literature remains limited in providing a comprehensive and in-depth analysis of this association. By investigating how cultural dimensions influence investor behavior across 64 global markets, this study seeks to deliver a more granular and systematic understanding of the mechanisms through which culture shapes the prevalence of DE.

6.3 Methodology and Data

To calculate DE, we collected daily trading volumes for all publicly traded firms in 64 countries from January 1, 2010, to June 30, 2024, through Capital IQ. The annual Gross National Income (GNI) per capita for these countries was collected from the World Bank database. Data on Hofstede's six cultural dimensions were obtained from Hofstede Insights (2024), which includes information for 119 countries.

Based on the methodology outlined by Chang and Lin (2015), DE is quantified and assessed using the following equations (Equation(10) to Equation (12)):

$$DE_{i,t} = \bar{V}^{u}_{i,t} / \bar{V}^{d}_{i,t} \tag{10}$$

$$\bar{V}^{u}_{i,t} = \sum_{j=1}^{q_{i,t}} V^{u}_{i,j,t} / q_{i,t} \tag{11}$$

$$\bar{V}^{d}_{i,t} = \sum_{k=1}^{r_{i,t}} V^{d}_{i,k,t} / r_{i,t} \tag{12}$$

where:

$V^{u}_{i,j,t}$ is a division of the trading volume of sector j with increased sector indexes in stock market *i* for day *t* by the average daily trading volume of sector *j* during the research interval

$V^{d}_{i,k,t}$ is a division of the trading volume of sector *k* with decreased sector indexes in stock market *i* for day *t* by the average daily trading volume of sector *k* during the research interval

$q_{i,t}$ -> number of sectors with increased sector indexes in the stock market *i* for day *t*

$r_{i,t}$ -> number of sectors with decreased sector indexes in the stock market *i* for day *t*

To relate cultural dimensions with DE, we employ a multinomial logistic approach, as displayed in the equation (13) to equation (15). Further, to account for institutional quality and the

development level of countries, we also include the Gross National Income per capita for each country.

$$DELv_{i,t} = \alpha_0 + \alpha_{1-6} Culture_i + \alpha_7 \ln(GNIPC_{i,t}) + \varepsilon_{i,t} \quad (13)$$

$$\sum_{h=1}^{3} \pi_h = 1 \quad (14)$$

$$\pi_1 = P(\text{DELv}_i = -1);\ \pi_2 = P(\text{DELv}_i = 0);\ \pi_3 = P(\text{DELv}_i = 1) \quad (15)$$

where:

$DELv_{i,t}$ is the level of the DE.

For the first quartile of DE (p25), $DELv_i$ = -1; for the fourth quartile of DE (p75), $DELv_i$ = 1; otherwise, $DELv_i$ = 0 and

π_1 , π_2, π_3 represents response probabilities (DELv = -1 for Low, DELv = 0 for Neutral, and DELv = 1 for High)

$Culture_i$ is IDV_i, UAVi, PDI_i, MAS_i, LTO_i, and IDG_i - Hofstede's Cultural Dimensions for each country.

$GNIPC_{i,t}$ is the Gross National Income per capita for each country.

α_0, $\alpha_{1\text{-}6}$, and α_7 are regression coefficients

ε_i refers to the error term

Table 13 presents the variable definitions, sources, and expected signs (based on references) for DE.

Table 13 - Variables Description – Disposition Effect

Variables	Definition	Source	Expected Signal	Reference
Dependent				
DE	Disposition Effect	Capital IQ (2025)		
Independent				
PDI	Power of Distance	Hofstede Insights (2024)	+	Breitmayer *et al*. (2019) Frino *et al*. (2015) Rieger *et al*. (2022)
IDV	Individualism	Hofstede Insights (2024)	-	Breitmayer *et al*. (2019) Briley *et al*. (2000) Rieger *et al*. (2022)

MAS	Masculinity	Hofstede Insights (2024)	+	Frino *et al*. (2015) Lawal & Sakariyaku (2024) Rieger *et al*. (2022)
UAV	Uncertainty Avoidance	Hofstede Insights (2024)	+	Breitmayer *et al*. (2019) Rieger *et al*. (2022)
LTO	Long-Term Orientation	Hofstede Insights (2024)	-	Briley *et al*. (2000) Rieger *et al*. (2022)
IDG	Indulgence	Hofstede Insights (2024)	+	Briley *et al*. (2000) Hsee & Weber (1999) Rieger *et al*. (2022)
GNIPC	Gross National Income per Capita	World Bank (2025)	-	Breitmayer *et al*. (2019)

Source: Elaborated by the authors

The generalized additive model (GAM) was used to assess robustness and control for nonlinearities in studying how Hofstede's cultural dimensions relate with DE. We also follow the method suggest by Liu (2019). The formalization of the approach appears in equation (16):

$$DE_{i,t} = \beta_0 + f_{1-6}(Culture_i) + f_7\big(\ln(GNIPC_{i,t})\big) + \varepsilon_{i,t} \quad (16)$$

where:

$\boldsymbol{f_{1-6}(\mathbf{Culture}_i)}$ is a smooth function calculated to fit each of the six Hofstede's dimensions: IDV_i, UAV_i, PDI_i, MAS_i, LTO_i, and IDG_i for each country *i*

$f_7(\mathrm{GNPC}_i)$ is a smooth function calculated to fit the Gross National Income for each country

β_0 is the regression coefficient

$\varepsilon_{i,t}$ refers to the error term

To reduce the distorting effects of outliers, the final dataset was winsorized at the lowest and highest DE percentiles, resulting in 198,618 daily observations. In addition, we perform diagnostic tests to assess multicollinearity through pairwise correlations and variance inflation factors (VIF) for all predictors. VIF reports an average of 2.27; all VIF values were below the conventional threshold of 5. The highest VIFs were observed for IDV (4.05) and GNIPC (3.04). VIF and correlation values indicate that multicollinearity is within acceptable limits; thus, multicollinearity is not a problem.

In addition, we conducted an intertemporal robustness analysis under exogenous shocks, detailed in Appendix K of this dissertation.

6.4 Results and Discussion

This study examines how Hofstede's cultural dimensions relate to DE across 64 markets from 2010 to 2024. The cross-country distribution of DE in Appendix H shows pronounced heterogeneity that aligns with variation in Hofstede cultural dimensions in Appendix I. The countries with the highest mean DE values (Venezuela, Serbia, Peru, Jamaica, Bulgaria) exhibit high power distance and uncertainty avoidance, but low long-term orientation or mixed scores, suggesting that their hierarchical social structures and fear of uncertainty could lead to loss-avoidant selling. The trading behavior of low-DE countries (such as the United States, Japan, China, Germany, and Finland) generally shows lower uncertainty avoidance and power distance, and higher long-term orientation, suggesting alignment with their deliberative trading style.

The descriptive statistics are presented in Table 14. The DE bias exhibits a mean value of 1.77 and a standard deviation of 2.897, indicating substantial dispersion across the 64 markets. Similarly, the cultural dimensions display relatively high variability, as reflected by their standard deviations in relation to their means. Among these, Uncertainty Avoidance (UAV) and Power Distance Index (PDI) report the highest mean (standard deviation) values, at 64.754 (22.482) and 29.919 (20.727), respectively. In contrast, Long-Term Orientation (LTO) shows the lowest mean value of 45.002, with a standard deviation of 20.478. Notably, these figures remain considerably below the theoretical maximum of 100 on the Hofstede scale.

Table 14 – Descriptive statistics - Disposition Effect

Variable	Obs	Mean	Std. Dev.	Min	Max
DE	198,618	1.77	2.897	0.056	22.637
GNIPC	231,804	26785	24005	780	105070
IDV	231,804	48.363	23.689	0	100
UAV	231,804	64.754	22.482	8	100
PDI	231,804	59.919	20.727	11	100
MAS	231,804	49.561	17.943	5	95
LTO	228,225	45.002	20.478	5	100
IDG	213,254	47.819	21.721	0	100

Source: Elaborated by the authors

Table 15 reports the results of the multinomial logistic regression. Columns 1 through 7 present partial specifications, while Column 8 displays the fully specified model, which forms the basis of our interpretation. The complete model reveals significant associations between cultural dimensions, economic development, and the likelihood of exhibiting a high disposition effect (DELv = 1) relative to a low disposition effect (DELv = –1).

Among the cultural variables, higher values of Individualism (IDV; $\alpha = -0.0020$), Masculinity (MAS; $\alpha = -0.0021$), Long-Term Orientation (LTO; $\alpha = -0.0021$), and Indulgence (IDG; $\alpha = -0.0026$)

are associated with lower odds of a high disposition effect. These coefficients are statistically significant, suggesting that investors embedded in cultures characterized by autonomy, future orientation, and emotional restraint are less susceptible to the behavioral bias of retaining losing assets. In other words, such investors appear more willing to liquidate underperforming positions, which may reflect cultural norms and institutional frameworks that promote rational decision-making and mitigate emotional biases. This interpretation is further supported by the coefficient for Gross National Income per Capita (GNIPC), which is negative and highly significant ($\alpha = -0.0455$, $p < 0.01$). This finding indicates that investors in wealthier countries are less prone to the disposition effect, potentially due to more sophisticated financial systems and higher levels of financial literacy (Breitmayer et al., 2019; Briley et al., 2000; Rieger et al., 2022). Collectively, these results underscore the interplay between cultural traits, economic development, and behavioral biases in shaping investor decision-making during periods of market stress.

However, the results show that MAS and IDG have a negative relationship with DE, which contradicts previous studies. For these, masculinity is associated with competitive risk-taking due to a higher risk tolerance that can lead investors to hold losing assets for longer, waiting for recovery, and to overconfidence in the belief that losing assets will recover due to the superiority of individual analysis (Frino et al., 2015; Lawal & Sakariyaku, 2024; Rieger et al., 2022). In the case of indulgent societies, DE would be associated with the emotional desire to realize the pride of a gain and the emotional impulse to avoid the pain and regret of formalizing a loss (Briley et al., 2000; Hsee & Weber, 1999; Rieger et al., 2022). The differences observed between societies may result from institutional discipline, social norms, and changes in masculine gender roles and cultural values.

Table 15 – Multinomial Logistic Estimations – Dependent variable: Disposition Effect

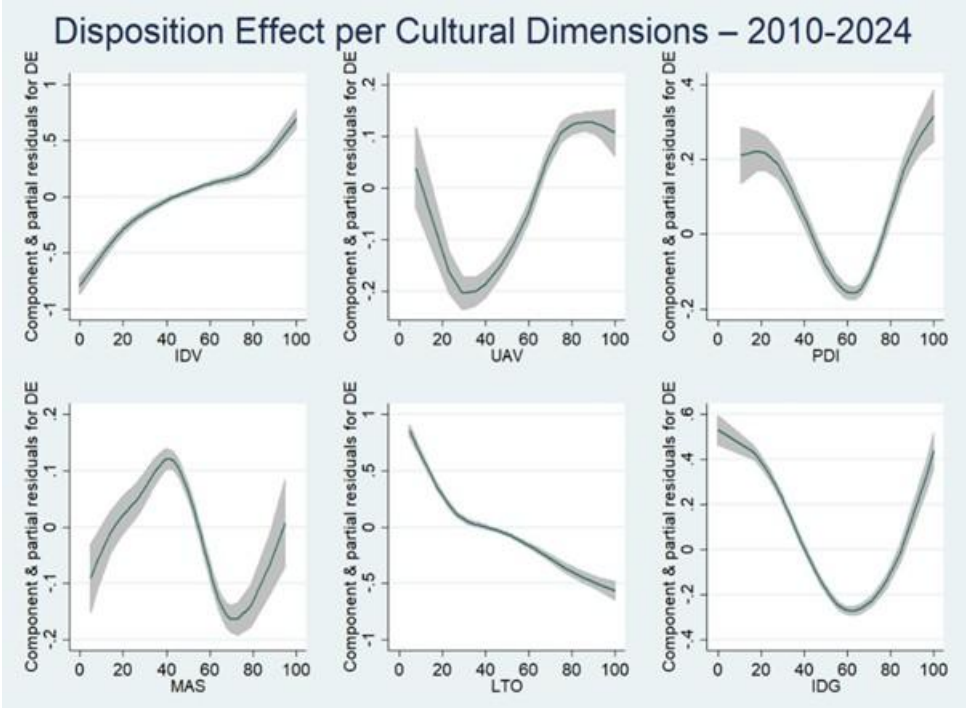
Period: 2010-2024

Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	0.2573***	-0.0942***	-0.3301***	0.1046***	0.2018***	0.1854***	1.0857***	0.779***
Individualism	-0.0054***							-0.0020***
Uncertainty Avoidance		0.0014***						0.0007*
Power Distance			0.0054***					0.0001
Masculinity				-0.0022***				-0.0021***
Long-term Orientation					-0.0048***			-0.0021***
Indulgence						-0.0043***		-0.0026***
GNI per capita							-0.1137***	-0.0455***
Pseudo R^2	0.003	0.004	0.004	0.002	0.016	0.001	0.006	0.024
χ^2	1193.17***	1446.45***	1451.57***	654.45***	6339.84***	326.27***	2604.94***	8979.86***

Notes: The dependent variable is the disposition effect (DE) *p<0.1; **p<0.05; ***p<0.01.
Source: Elaborated by the authors

To account for potential nonlinear relationships among variables, we employed a Generalized Additive Model (GAM). The results largely corroborate the findings from the multinomial logistic regression while uncovering several complex patterns that warrant further investigation. Figure 14 illustrates the estimated smooth functions depicting the relationship between each cultural dimension and the disposition effect (DE). For Uncertainty Avoidance (UAV) and Power Distance Index (PDI), the GAM curves indicate a positive association with DE, whereas Masculinity (MAS), Long-Term Orientation (LTO), and Indulgence (IDG) exhibit a negative relationship. Notably, the effect of Individualism (IDV) diverges from the coefficient obtained in the multinomial logistic model. Moreover, the relationships of MAS and IDG with DE display an inverted U-shaped pattern, suggesting that their influence varies with cultural intensity and interacts with other personality-related characteristics. This nonlinearity may highlight the complexity of cultural effects on behavioral biases.

Figure 14 – Disposition Effect per Cultural Dimensions – GAM Model



Source: Elaborated by the authors
Notes: The figure illustrates the nonlinear prediction of the disposition effect at the 95% confidence interval, based on assumed intervals of 0 to 100 for the six Hofstede cultural dimensions. The nonlinear prediction is based on the results of the GAM regression, as specified in Equation (16) and displayed in Appendix J.

Model complexity may explain why specific coefficients in the multinomial logistic regression model yielded unexpected results, either weak or counterintuitive. The analysis of cultural

dimensions reveals IDV as a dimension that shows the most interesting differences between modeling approaches. The multinomial logistic regression model shows a negative, statistically significant ($p < 0.01$) coefficient for IDV, indicating that people who score higher on individualism have lower odds of displaying DE. The study confirms previous research showing that people from individualistic cultures that value independence and self-reliance and focus on future goals tend to avoid DE behavioral biases (Briley *et al*., 2000; Rieger *et al*., 2022).

Nevertheless, the visualization of GAM shows different patterns in this relationship: DE probability directly relates to IDV values; higher IDV values result in greater DE. The study's results contradict theoretical predictions: individualism should protect people from these biases, yet the results show the opposite. The GAM plots demonstrate non-linear effects: IDV strength gains become more important as its values exceed certain thresholds.

The observed difference may be due to the standard regression model, which captures only average effects and might hide the complex behavioral patterns observed in the most extreme data points. The problem requires a visualization tool, such as GAM, to observe DE responses to IDV changes. The implementation of this model becomes necessary because it reveals threshold effects and nonlinear interactions that standard logit models cannot identify. The research suggests that individualistic behavior leads investors to develop excessive confidence and self-reliance, which makes them more vulnerable to DE when institutional safeguards are insufficient. The simple regression models indicate that individualism functions as a protective factor against DE, yet the GAM results demonstrate that individualism affects DE differently across specific contexts.

The influence of cultural dimensions may not be linear or uniform. The models' design structure provides a logical explanation for this phenomenon. The regression model provides standard effects for all variables, yet GAM models reveal non-linear patterns and threshold points that linear models fail to detect. The GAM results function as an extension of the regression analysis instead of showing conflicting results. Therefore, cultural elements influencing investment decisions do not affect investors equally throughout their entire range, and particular characteristics become influential only when circumstances exist. The research results indicate that researchers must use adaptable modeling approaches to study cultural behaviors that exist within specific social contexts.

6.5 Conclusions

This study investigates how Hofstede's cultural dimensions and economic development relate to the Disposition Effect (DE). Using daily stock market data from 64 countries between 2010 and

2024, we employ Multinomial logistic regression. Further, to capture potential nonlinearities, we also apply the Generalized Additive Models (GAMs). Regression analysis demonstrated statistically significant relationships between variables, notably the negative impact of IDV, MAS, LTO, and IDG.

The findings show that cultural elements shape how investors make their choices and reveal that these influences are often non-linear and require more sophisticated techniques, such as GAM models, to understand their relationships.

These results contribute to literature in several ways, primarily by introducing the Generalized Additive Model (GAM) as a necessary tool for capturing the nonlinear dynamics between cultural dimensions and the Disposition Effect, and by providing empirical evidence that challenges prior assumptions about the influence of Individualism, Masculinity, and Indulgence.

As practical implications, we understand that financial institutions, together with regulators, need to recognize that cultural elements affect investor decisions. Furthermore, the complexity revealed by the GAM models, particularly for Individualism, underscores the need for risk warnings and educational efforts to be culturally tailored rather than uniformly applied to mitigate investors' susceptibility to the disposition effect.

This study investigates how Hofstede’s cultural dimensions and economic development relate to the Disposition Effect (DE). Using daily stock market data from 64 countries between 2010 and 2024, we employ Multinomial logistic regression. Further to capture potential nonlinearities we also apply the Generalized Additive Models (GAMs). Regression analysis demonstrated statistically significant relationships between variables, notably the negative impact of IDV, MAS, LTO, and IDG.

The findings of this study have implications that go beyond the limits of regression analysis. The GAM plots revealed that IDV, UAV, and PDI are associated with higher probabilities of elevated DE levels, although their regression coefficients had low (UAV with $p<0.1$) or no statistical significance (PDI) or were negative (IDV, $p<0.01$). These observed discrepancies between the models are likely due to the linearity constraints of traditional modeling, rather than random statistical fluctuations. It is postulated that the fundamental dynamics of cultural influences do not adhere to a singular parametric pattern. The full expression of cultural characteristics that create threshold effects requires particular conditions, but their calculated impact on parametric results remains influenced by interactions with other factors. The inclusion of Gross National Income per capita (GNIPC)

reinforces the economic importance, since countries with higher levels of development have lower rates of prevalence of DE bias, indicating that economic wealth acts as a protective factor.

The results present surprising differences from current knowledge. A negative relationship was found between MAS and IDG (both in the logit and in the GAM) and IDV (in the GAM), contradicting previous studies that predicted that these cultural elements would drive DE bias. The exact reasons behind this change in investor behavior between historical and current observations remain uncertain, as they may stem from changes in cultural values, differences in research approaches, or natural changes in investor behavior. The results indicate that behavioral finance needs non-linear and flexible models to effectively analyze investor biases in markets with different cultures.

The findings demonstrate that cultural factors play a significant role in shaping how investors make investment decisions. These findings need to be evaluated in relation to multiple methodological constraints that affect their interpretation. The study examines 64 stock markets, but the selected sample may not capture all cultural elements, so studying more nations would enhance overall understanding. Hofstede's dimensions have gained wide acceptance, but may not capture all cultural factors influencing human behavior. The GLOBE project and Schwartz's value framework utilize different conceptual models that could produce distinct findings, thus expanding knowledge about operating systems. The decision-making process is influenced by external factors, including investors' economic position, religious beliefs, and financial knowledge. Future studies should examine these factors to understand the full extent of DE bias in different contexts.

This work shows that the use of logit regression and GAM provides advanced statistical methods for this area. Future research requires machine learning algorithms to work with advanced modeling techniques in order to study the intricate non-linear patterns that link cultural elements to expressions of DE bias.

The results of this research show that future studies require new methods to study investors' biases that exist between different cultural settings. Additionally, it is important to investigate the cultural-behavioral relationships in the context of major market events and global crises through an analysis of intertemporal bias stability, as shown in Appendix K of this dissertation. The growth of financial knowledge helps investors make investment choices and mitigate negative bias effects, including the Disposition Effect.

7 BEHAVIORAL SHIFTS IN TIMES OF CRISIS: HERDING AND CULTURE IN PANDEMIC-ERA MARKETS

7.1 Introduction

The COVID-19 pandemic exposed financial markets worldwide to extreme uncertainty, which showed how the social environment influences investor choices (Fang *et al*., 2021). The research investigates how cultural elements of nations, together with economic growth patterns, affected herding activities across three distinct time periods based on World Health Organization's (WHO) official declaration on the recognizing COVID-19 as a global pandemic: January 2015 to March, 10th 2020 as pre-pandemic period, March, 11th 2020 to December 2021 as pandemic period, and January 2022 to June 2024 as post-pandemic period (Padhan & Prabheesh, 2021). More specifically, the paper explores whether and how country-level cultural traits and economic development help explain cross-national differences in investor herding, and whether those relationships shifted as the COVID-19 crisis unfolded. The study explores the degree to which specific cultural dimensions and GNI per capita are associated with greater or lesser herding behavior, and whether the pandemic altered the direction, strength, or shape of those associations across the three periods.

Hofstede's cultural framework consists of six dimensions. Individualism versus Collectivism (IDV) captures whether people define themselves primarily as independent individuals or as members of tightly knit groups. Uncertainty Avoidance (UAV) measures the extent to which a society feels threatened by ambiguous or unknown situations and seeks to minimize that stress. Power Distance (PDI) reflects how a culture accepts and structures inequalities and hierarchical relationships. Masculinity versus Femininity (MAS) describes the distribution of emotional roles between genders and the value placed on competitiveness versus care. Long-term versus Short-term Orientation (LTO) indicates whether a society emphasizes future-oriented planning and persistence or favors past and present traditions and quick results. Indulgence versus Restraint (IDG) captures the degree to which a society permits gratification of human desires versus regulating and suppressing them. Each country receives a score on these dimensions, enabling cross-national comparisons (Hofstede, 2011).

Using a 64-country panel data set and employing multinomial logistic models and generalized additive models (GAMs), we document two central findings: economic development, measured by GNI per capita, is a stable and powerful protector against herding, and cultural traits matter in heterogeneous, time-varying ways that the pandemic amplified and, in some cases, reversed. Power distance emerged as a major driver of increased correlated trading in the recovery, long-term

orientation consistently suppressed herding and strengthened after the crisis, and uncertainty avoidance moved from neutral to a clear deterrent once uncertainty peaked. Individualism exerted a small but persistent positive effect. At the same time, indulgence's protective role largely vanished during the crisis, and masculinity flipped from weakly pro-herding to significantly anti-herding after the pandemic.

7.2 Literature Review

7.2.1 The Phenomenon of Investor Herding

In markets, herd behavior describes the tendency of investors to shadow the actions of a larger crowd, often sidelining their own private information or fundamental analysis (Chang & Lin, 2015). Herd behavior can destabilize markets, leading to market collapses and bubbles (Gabbori, 2024; Zhan, 2019).

Chang *et al*. (2000) suggest that herd behavior occurs when investors tend to ignore their own information, preferring to follow market trends and analyze how the dispersion of absolute returns across individual stocks relates to market volatility. Market conditions determine the level of herd behavior through investor sentiment (Schmeling, 2009) and socio-religious events, as observed in the Saudi Arabian market during Islamic holidays (Gabbori, 2024).

7.2.2 The Role of National Culture in Financial Decision-Making

National culture establishes permanent institutional systems that determine how investors generate value, establish rules, and process information (Tang & Koveos, 2008). Hofstede (2001) established six essential dimensions that explain how financial systems differ between nations: Individualism vs. Collectivism (IDV), Uncertainty Avoidance (UAV), Power Distance (PDI), Masculinity vs. Femininity (MAS), Long-Term vs. Short-Term Orientation (LTO), and Indulgence vs. Restraint (IDG).

Research studies in this field show that cultural dimensions affect all financial reporting elements of business operations (Boahen & Mamatzakis, 2025) and both personal investment choices and overall stock market performance (Zheng, 2015). The social nature of herding behavior, driven by group conformity, makes national culture the main factor shaping its prevalence and intensity across global financial markets.

7.2.3 The Nexus of National Culture and Herding Behavior

Research studies have established a direct relationship between cultural dimensions and herding behavior through their impact on stock price synchronization and market regime volatility.

The relationship between individualism (IDV) consistently emerges across studies. In societies with low individualism, stock price movements tend to be more tightly synchronized (Schmeling, 2009; Chang & Lin, 2015; Zhan, 2019; Zheng, 2015; Zhou, 2019).

The main reason for this phenomenon appears to be investors' tendency to follow the crowd, making them more likely to be influenced by their peers. In these contexts, collective opinion overrides individual opinion. The mentality is that "it is better to be wrong collectively than to be right alone." (Zheng, 2015). However, Danrimi *et al*. (2018) suggest that individualism is a significant factor in herd formation.

High-power-distance environments feature social inequality because investors obtain information through different conditions of access. High-power-distance investors in these environments follow herd behavior because they trust transaction-based implicit information, which emerges when others start a trend (Chang & Lin, 2015). The research by Lobão and Maio (2021) contradicts previous studies (Chang & Lin, 2015; Zheng, 2015; Zhou, 2019), which demonstrated that high PDI environments should lead to greater herding behavior due to obedience and information inequality.

The connection between masculinity and herding work produces both positive and negative effects. The competitive nature of high-MAS cultures leads to risk-taking behaviors, which Zheng (2015) links to market instability, but Lobão and Maio (2021) found that these investors disregard market trends. The combination of masculine workplace environments and market instability increases market volatility, but investors retain the ability to select their investments. Research evidence shows that Masculinity is the main reason investors follow one another in investment decisions (Chang & Lin, 2015; Danrimi *et al*., 2018; Zheng, 2015; Zhou, 2019).

High-UAV cultures maintain stability in their market approach, which prevents them from entering volatile market conditions (Zheng, 2015; Zhou, 2019). Schmeling (2009) shows that investors who avoid uncertainty create greater market unpredictability, leading to higher stock price volatility during unstable market periods.

People in nations with high LTO levels choose investment strategies that help them preserve financial stability for future development. The investment strategy that emphasizes stability and solidity helps to reduce market volatility. The investment strategies of nations with low LTO focus on short-term financial gains, which increases stock market volatility. Market bubbles and crashes

occur because of aggressive short-term investment methods, which, according to Zhou (2019), lead to herding behavior. The research by Saltik *et al*. (2024) suggests that people tend to avoid following others because they base their choices on personal goals.

Zhou (2019) identifies Indulgence (IDG) as a cultural element that, according to his research, reduces market volatility. The research shows how IDG affects market volatility by influencing investor decision-making. Research shows that high-IDG cultures experience lower market volatility because their investors make risky investments at high levels, which creates more herding behavior in societies with low IDG (high constraint).

7.2.4 Country Wealth and Herding Behavior

Research shows that investors from developed and less developed economies exhibit different herd behavior patterns because their economic development levels affect their willingness to follow others (Lobão & Maio, 2021; Zhan, 2019).

The strength of herd behavior grows stronger in markets that have not fully developed their market structure. The mutual fund industry operates in developing markets with inexperienced managers who make decisions by following other investors, leading to greater herding behavior (Lobão & Maio, 2021). The capital markets of developing countries face two primary problems: weak regulatory frameworks and unclear market information. The combination of these elements produces conditions that lead to herd behavior, according to Saltik *et al*. (2024).

The volatility of emerging markets exceeds that of developed markets. Stock price synchronization serves as a herding indicator, leading to greater market volatility in emerging markets than in developed markets when financial instability hits, as seen during the 2007-2008 crisis (Zhan, 2019).

7.3 Methodology and Data

Christie and Huang (1995) (hereafter, CH) examine how investor herding behavior affects stock market performance through an analysis of return data. The authors analyze market performance through cross-sectional standard deviation of returns (CSSD) to understand how herding behavior influences market behavior. The research investigates stock price clustering patterns under market stress to identify investor-led price adjustments, which show investors tend to follow market trends.

CH suggests that investors who follow market consensus will lose their individuality, abandoning their personal beliefs to conform to group behavior. In contrast, rational asset pricing

models assume that investors make decisions based on their analysis and the unique characteristics of individual securities. Therefore, the returns of individual securities can vary widely, depending on their sensitivity to market movements. Thus, there are conflicting predictions regarding security return dispersions between herd behavior and rational asset pricing models.

Based on CH, Chang *et al*. (2000) (hereafter referred to as CCK), using the cross-sectional absolute deviation of returns (CSAD) as a measure of dispersion, show that rational asset pricing models predict a linear, increasing relationship between equity return dispersion and the market return. However, suppose market participants tend to follow the market's aggregate behavior and disregard their own prior beliefs during periods of large average price movements. In that case, the linear, increasing relationship may not hold. Instead, the relationship could become non-linearly increasing or even decreasing. Therefore, in the absence of herding behavior and when stock returns follow the Capital Asset Pricing Model (CAPM), a positive linear relationship exists between the cross-sectional dispersion of stock returns and the absolute market return. However, if market-wide herding dominates, this linear relationship breaks down at higher absolute market returns.

This study follows the method of CCK to examine investors' herding behaviors in 64 stock markets, as shown in Equation (177)

$$CSAD_{i,t} = \gamma_{i,0} + \gamma_{i,1} R_{i,m,t} + \gamma_{i,2} |R_{i,m,t}| + \gamma_{i,3} R_{i,m,t}^2 + \varepsilon_{i,t} \qquad (17)$$

where:

$CSAD_{i,t}$ is the cross-sectional absolute deviation in the stock market *i* for day *t*

$R_{i,m,t}$ is the market return in the sector m to the stock market *i* for day *t*

$\gamma_{i,0}, \gamma_{i,1}, \gamma_{i,2}, \gamma_{i,3}$ are regression coefficients

$\varepsilon_{i,t}$ refers to the error term

CCK also evaluates the presence of herding behavior by examining the correlation between the CSAD and the volatility of the market return, as shown in Equation (18)

$$CSAD_{i,t} = \sum_{j=1}^{n_i} | R_{i,j,t} - R_{i,m,t}| / n_i \qquad (18)$$

where

$R_{i,j,t}$ is the sector index returns on industry *j* in the stock market *i* for day *t*

$R_{i,m,t}$ is the market return in the sector m to the stock market *i* for day *t*

n_i is the number of industries in the stock market *i*

As previously stated, CCK suggests that the variation in returns among stocks, as predicted by the CAPM, increases with the absolute value of market returns, exhibiting a linear relationship. However, if a market displays herding behavior, this linear relationship between variation in stock returns and market returns does not incrementally increase. Consequently, if the coefficient $\gamma_{i,3}$ in equation (17) has a significantly negative value; it signifies the presence of herding behavior among investors, and CSAD indicates herding tendencies.

7.3.1 Model Specification for Empirical Analysis

Following the methodology outlined by Chang and Lin (2015) and using DHerd as the dependent variable (DHerd = 1 if $\gamma_{i,3}< 0$ with significant value; DHerd = -1 if $\gamma_{i,3} > 0$ with significant value; otherwise DHerd = 0 (base category)) to a multinomial logistic regression analysis (Equations (19), (20), and (21)).

$$\text{DHerd}_i = \alpha_0 + \alpha_1 IDV_i + \alpha_2 UAV_i + \alpha_3 PDI_i + \alpha_4 MAS_i + \alpha_5 LTO_i + \alpha_6 IDG_i + \alpha_7(\ln(GNIPC_i)) + \varepsilon_i \quad (19)$$

$$\sum_{h=1}^{3} \pi_h = 1 \quad (20)$$

$$\pi_1 = P(DHerd_i = 1); \pi_2 = P(\text{DHerd}_i = 0), \pi_3 = P(\text{DHerd}_i = -1) \quad (21)$$

where:
PDI_i, IDV_i, MAS_i, UAV_i, LTO_i, and IDG_i are Hofstede's Cultural Dimensions for each country *i*
GNIPC is Gross National Income per capita
π_1, π_2, π_3 represents response probabilities
α_0, α_1, α_2, α_3, α_4, α_5, α_6, and α_7 are regression coefficients
ε_i refers to the error term

Table 16 presents the definitions and data sources for the variables employed in the study, along with the anticipated signs for herding behavior as suggested by prior literature. To compute the Cross-Sectional Absolute Deviation (CSAD), we gathered daily market and sectoral returns for all publicly listed firms across 64 countries between January 1, 2010, and June 30, 2024, using Capital IQ. Annual Gross National Income (GNI) per capita figures were sourced from the World Bank database. The research obtained Hofstede's six cultural dimensions data from Hofstede Insights (2025), which included 119 countries. The analysis follows the WHO's official pandemic declaration on March 11th, 2020 (Padhan & Prabheesh, 2021) through three distinct time periods, which examine behavioral changes from January 2015 to March 10th, 2020 (pre-pandemic), and from March 11th, 2020, to December 2021(pandemic), and then from January 2022 to June 2024 (post-pandemic)

Table 16 – Variables Description - Herding

Variables	Definition	Source	Expected Signal	Reference
Dependent				
CSAD	Cross-Sectional Absolute Deviation	Capital IQ (2025)		
DHerd	Herding			
Independent				
PDI	Power of Distance	Hofstede Insights (2024)	+	Chang & Lin (2015) Zheng (2015) Zhou *et al*. (2019)
			-	Lobão & Maio (2019)
IDV	Individualism	Hofstede Insights (2024)	-	Schmeling (2009) Chang & Lin (2015) Zengh (2015) Zhan (2019) Zhou *et al*. (2019)
			+	Danrimi *et al*. (2018)
MAS	Masculinity	Hofstede Insights (2024)	+	Chang & Lin (2015) Zheng (2015) Danrimi *et al*. (2018) Zhou *et al* (2019)
			-	Lobão & Maio (2019)
UAV	Uncertainty Avoidance	Hofstede Insights (2024)	-	Zheng (2015) Zhou *et al*. (2019)
			+	Schmeling (2009)
LTO	Long-Term Orientation	Hofstede Insights (2024)	+	Damrimi *et al*. (2018) Zhou *et al*. (2019)
			-	Saltik *et al*. (2024)
IDG	Indulgence	Hofstede Insights (2024)	+	Zhou *et al*. (2019)
GNIPC	Gross National Income per Capita	World Bank (2025)	-	Lobão & Maio (2019) Zhan (2019) Zhou *et al*. (2019) Saltik *et al*. (2024)

Source: Elaborated by the authors

Research on the relationships between national culture and financial capacity often includes economic development as a controlling factor (Nolan, 2020). The research community uses GNI per capita as an economic development control variable in studies of cultural and behavioral factors (Bialowolski *et al*., 2023). According to Nolan (2020), GNI per capita offers superior economic data

than GDP per capita because it includes all income sources, including corporate profits and international financial activities.

7.3.2 Robustness Test

For robustness, we use the Generalized Additive Model (GAM) of Hastie and Tibshirani (1986). GAM combines the strengths of generalized linear models (GLMs) with those of additive models, as described by Liu (2019). The research benefits from this flexible approach as it addresses problems related to the distribution of Hofstede's cultural dimensions residuals. The method eliminates the need for linear model constraints by allowing users to fit nonparametric or semiparametric additive models within an additive framework (Hastie & Tibshirani, 1986).

Equation (22) shows the GAM regression with CSAD as a proxy for herding:

$$CSAD_{i,t} = \beta_0 + f_1(PDI_i) + f_2(IDV_i) + f_3(MAS_i) + f_4(UAV_i) + f_5(LTO_i) + f_6(IDG_i) + f_7(\ln(GNIPC_i)) + \varepsilon_{i,t} \quad (22)$$

where:
$\boldsymbol{f_n}(\mathbf{Hof}\boldsymbol{stede'Dimension_i})$ is a smooth function calculated to fit each of the six Hofstede's dimensions
PDI_i, IDV_i, MAS_i, UAV_i, LTO_i, and IDG_i are Hofstede's Cultural Dimensions for each country *i*
$GNIPC_i$ is the gross national income per capita
β_0 is the regression coefficient
$\varepsilon_{i,t}$ refers to the error term

Using pairwise correlation coefficients and variance inflation factors (VIFs), we assess multicollinearity among the independent variables. The VIF average is 2.31, and all individual values remain below the standard threshold of 5. Moderate multicollinearity, which does not affect model stability, is detected in the IDV and GNIPC (VIF values of 4.74 and 3.00, respectively). These results supported the inclusion of GNIPC and Hofstede's six cultural dimensions in both linear and nonlinear model specifications.

The final dataset included 152,521 daily observations after winsorizing the Cross-Sectional Absolute Deviation (CSAD) distribution at its lowest and highest percentiles to minimize extreme-value effects.

7.4 Results and Discussion

The research examines how Hofstede's cultural dimensions influence herding behavior across 64 financial markets over a ten-year period, including three distinct time periods: pre-pandemic period, COVID-19 period, and post-pandemic period. Appendix L depicts the results of CSAD

regression (Equation (17)) following CCK, and Table 17 presents a summary of $\Upsilon_{i,3}$ (a coefficient to determine if investors' countries show herding (if $\Upsilon_{i,3} > 0$ and statistically significant) or anti-herding behavior (if $\Upsilon_{i,3} < 0$ and statistically significant) for the three time periods.

Table 17 - Coefficient $\Upsilon_{i,3}$

Country	N	Pre-Pandemic	N	Pandemic	N	Post-Pandemic
Argentina	1354	-0.323** (-2.46)	473	1.611 (0.99)	650	4.225*** (7.73)
Australia	1313	3.649*** (5.86)	461	1.29** (2.36)	627	1.217 (0.52)
Austria	1313	-2.258 (-1.62)	461	0.366 (0.29)	627	2.602 (0.6)
Bangladesh	1354	-6.454*** (-9.52)	473	-7.492*** (-10)	650	1.763** (2.09)
Belgium	1327	2.462*** (4.57)	466	-0.301 (-0.75)	638	6.937*** (5.05)
Brazil	1281	1.365*** (4.32)	449	-0.947 (-1.14)	622	1.739 (1.11)
Bulgaria	1278	0.048 (0.04)	444	0.222 (0.1)	614	-0.293 (-0.2)
Canada	1302	-0.271 (-0.63)	455	-1.404*** (-4.19)	626	8*** (3.15)
Chile	1291	0.041 (0.11)	450	0.678*** (2.64)	622	-1.408 (-0.85)
China	1262	-1.233*** (-3.67)	443	1.798 (1.14)	601	0.955 (0.78)
Colombia	1354	0.044 (0.31)	473	0.035 (0.25)	650	1.037* (1.78)
Croatia	1291	-1.192* (-1.95)	452	-0.848 (-0.85)	620	-0.429 (-0.42)
Czech Republic	1299	-3.228 (-0.2)	452	-0.61 (-0.62)	627	2.838 (0.83)
Denmark	1297	2.429*** (2.67)	452	1.134 (0.61)	624	8.613*** (8.05)
Egypt	1097	1.657*** (3.94)	439	0.401 (0.62)	600	-109.757 (-0.9)
Finland	1303	1.132* (1.84)	456	0.033 (0.06)	628	0.968 (0.65)
France	1327	1.873*** (6.72)	466	0.189 (0.62)	638	0.408 (0.36)
Germany	1311	0.606 (1.45)	460	0.008 (0.03)	638	0.877 (1.28)
Greece	1266	0.493* (1.82)	451	0.491* (1.67)	617	2.882*** (3.93)
Hong Kong	1277	2.834*** (3.98)	448	-0.618 (-0.57)	610	2.054*** (4.72)
Hungary	1291	0.419 (0.42)	454	-0.708 (-0.78)	627	-0.062 (-0.07)
India	1283	2.531*** (3.79)	451	-0.152 (-0.48)	617	7.687*** (6.48)
Indonesia	1262	5.531 (0.48)	440	1.552 (1.59)	595	2.355 (0.83)
Ireland	1318	1.849** (2.31)	462	-2.065* (-1.9)	633	5.453** (2.53)
Israel	1025	0.798 (0.67)	444	0.352 (0.47)	613	5.58*** (6.66)
Italy	1349	1.185*** (3.26)	472	-0.186 (-0.86)	649	1.356** (2.58)
Jamaica	1289	0.852 (0.4)	457	1.234 (0.25)	627	13.184 (1)
Japan	1266	-0.284 (-1.03)	444	1.807*** (3.14)	610	4.082*** (2.86)
Jordan	1585	1.077** (2.42)	567	0.778*** (10.72)	781	-1.011 (-0.79)
Kuwait	1585	-1.384*** (-3.7)	567	6.479*** (3.98)	781	-4.454** (-2.01)
Luxembourg	1324	5.828*** (3.23)	465	1.9* (1.92)	638	3.482* (1.89)
Malaysia	1270	12.233*** (7.8)	445	-2.739*** (-2.72)	604	1.069 (0.55)
Mexico	1303	-0.042 (-0.02)	457	3.753*** (2.72)	627	8.198 (1.34)
Morocco	1288	-4.333*** (-4.16)	450	-1.886*** (-3.78)	625	-5.507*** (-3.41)
Netherlands	1327	-0.303 (-0.31)	466	0.039 (0.07)	638	6.498*** (4.03)
New Zealand	1301	5.039** (2.51)	456	1.394 (1.23)	619	5.056* (1.71)
Nigeria	1285	-4.085*** (-5.11)	448	-4.73 (-0.5)	614	-11.066*** (-4.63)
Norway	1302	1.642*** (3.02)	455	0.051 (0.06)	626	4.013* (1.77)
Pakistan	1288	0.826 (0.86)	447	0.795 (1.05)	613	-1.397 (-1.54)

Peru	1354	-1.317*** (-2.75)	473	-0.09 (-0.13)	650	1.096 (1.54)
Philippines	1262	0.368 (0.38)	442	-1.283* (-1.7)	608	1.587 (1.03)
Poland	1295	-2.099*** (-4)	455	-0.687* (-1.73)	625	0.459 (0.82)
Portugal	1327	1.319** (2.06)	466	-1.789* (-1.9)	638	10.627** (2.48)
Romania	1328	0.922 (1.63)	466	-1.603** (-2.06)	636	-1.639* (-1.84)
Russia	1354	1.219*** (3.56)	473	0.336 (0.76)	47	-1.93*** (-3.95)
Saudi Arabia	1041	-0.767*** (-3.89)	451	-1.042 (-1.23)	616	-0.299 (-0.19)
Serbia	1328	1.624*** (3.19)	466	-1.052 (-1.37)	636	0.091*** (4.29)
Singapore	1354	-0.208 (-0.27)	473	-1.142 (-1.28)	650	6.262*** (3.35)
Slovenia	1328	-4.669*** (-2.58)	466	-2.176** (-2.29)	636	-0.691 (-0.95)
South Africa	1297	10.703 (1.61)	453	2.039** (2.56)	622	3.37 (1.51)
South Korea	1274	-1.198 (-0.97)	449	0.471 (0.96)	612	5.145*** (4.53)
Spain	1327	1.747*** (7.36)	464	-0.104 (-0.34)	638	4.04*** (3.4)
Sri Lanka	1245	3.474* (1.65)	405	-1.519 (-1.13)	588	1.639** (2.08)
Sweden	1303	1.026* (1.82)	457	-0.461 (-0.82)	628	3.091 (1.34)
Switzerland	1302	3.114*** (5.4)	458	-0.823 (-0.78)	628	5.283** (2.31)
Taiwan	1258	-0.753 (-1.59)	448	0.019 (0.02)	602	7.605*** (5.67)
Thailand	1268	4.11*** (6.05)	436	-0.532 (-1.05)	604	-1.362 (-0.46)
Tunisia	1585	1.055 (1.11)	567	-10.957*** (-6.29)	781	-3.632* (-1.75)
Turkey	1306	1.008 (0.75)	452	-1.987** (-2.27)	622	-0.13 (-0.65)
U.Arab Emirates	1118	0.68 (0.08)	451	6.918 (1.09)	625	-1.54 (-0.49)
United Kingdom	1313	1.636*** (2.92)	458	-0.603 (-1.47)	627	3.022 (1.57)
United States	1305	0.313 (0.28)	458	-0.62* (-1.93)	625	0.98 (0.97)
Venezuela	1242	0.291 (1.04)	435	3.237 (0.53)	599	-1.096 (-0.44)
Vietnam	1291	-2.423*** (-3.07)	458	-2.47*** (-2.74)	618	-2.358*** (-2.7)

Source: Elaborated by the authors

The majority of countries maintained neutral pre-pandemic behavior, but middle-income and emerging markets such as Argentina, Bangladesh, Nigeria, and Vietnam showed statistically significant herding behavior. The pandemic served as a shock that unified markets through herding behavior, which intensified across markets but was most pronounced in those with limited depth and weak information sharing (e.g., Bangladesh, Morocco, Tunisia, Vietnam). The post-pandemic period brought different reactions from countries, which included Malaysia, Philippines, Poland, Slovenia, and Turkey returning to neutral positions, while emerging markets like Morocco, Tunisia, Romania, and Vietnam maintained herding behavior, and high-income nations Canada, Ireland, Portugal, and the United States adopted anti-herding strategies when markets regained stability, and contrarian forces regained power.

Economic status strongly conditions these patterns. High-income countries tended toward neutral or anti-herding behavior overall, reflecting deeper liquidity, more active institutional participation, and greater capacity for independent price discovery (e.g., Australia, Belgium, France, Norway). Middle- and low-income countries experienced the most significant pandemic spikes in

herding and showed slower normalization afterward (e.g., Bangladesh, Malaysia, Philippines), consistent with higher retail participation, information asymmetries, and vulnerability to macro and liquidity shocks.

The distribution of economic activities and cultural elements follows patterns similar to those of geographic locations. Western Europe and North America showed greater resilience and a faster move toward anti-herding post-pandemic. The pandemic herding effect was most noticeable in Latin America and parts of Asia, Africa, and the Middle East, where the Philippines, Tunisia, and Turkey showed strong effects, and several countries, including Kuwait, Morocco, Nigeria, and Vietnam, experienced enduring post-pandemic impacts. Asia displayed diverse market behavior: its advanced economies operated similarly to other high-income nations (South Korea and Hong Kong), while its emerging markets exhibited herd behavior typical of developing regions (Kuwait, Saudi Arabia, and Vietnam).

Hofstede's cultural dimensions provide explanations for the existing cultural variations between different countries. The combination of high Power Distance and high Uncertainty Avoidance leads to increased herding behavior in Bangladesh, Morocco, and Poland during the pandemic, as these cultural traits make people more likely to follow established hierarchies and avoid ambiguity under stress. The countries with High Individualism show neutral or anti-herding results (e.g., Australia, Denmark, Finland, Germany, Netherlands) because their cultures support independent choices and oppose market trends. The long-term perspective, along with strong institutional systems in Hong Kong, Japan, and South Korea, prevented excessive short-term compliance. Masculinity (e.g., Austria, Hungary, Japan) and Indulgence (e.g., Australia, Mexico, New Zealand) showed no clear relationship between them. The research demonstrates that investor herding behavior during major market shocks results from a combination of economic development, market structure, geographic location, and deep cultural characteristics.

The multinomial logistic regression results are presented in Table 18. The pre-pandemic time appears in Panel A, followed by the pandemic period in Panel B, and the post-pandemic period in Panel C. The partial model estimates appear in columns 1-7, while the full specification results are presented in column 8. The complete model serves as our main focus for this discussion. The GAM regression models used to generate **Error! Reference source not found.** show that cultural d imensions relate to CSAD levels in non-linear ways. The time period before the pandemic appears in Panel A, the pandemic era in Panel B, and the time after the pandemic in Panel C. The Hofstede

cultural dimensions yield nonlinear CSAD (herding) predictions in each panel, with standardized values ranging from 0 to 100, indicating 95% confidence intervals.

The multinomial logistic regression results demonstrate that cultural elements and national income levels explain herd behavior across all three time periods, and the models achieve better predictive power after the pandemic. The pseudo R^2 rises from 0.128 in the pre-pandemic period to 0.139 during the pandemic and to 0.236 in the post-pandemic period, showing that the set of predictors explains substantially more variation in herd behavior during the recovery period. The GNI per capita indicator shows negative effects at all time points, whereas cultural dimensions exhibit substantial variation in strength and direction across time periods.

Higher GNI per capita serves as a strong protective factor against herd behavior, as the multinomial logit model shows a significant negative relationship across all time periods, with the strongest protection occurring during the pandemic. The research indicates that wealthier nations tend to show reduced patterns of correlated investment behavior. This result is corroborated by research from Lobão and Maio (2021), Saltik *et al*. (2024), Zhan (2019), and Zhou *et al*. (2019). The GAM results show that the relationship between herding (proxy for CSAD) and GNI per capita is non-linear: most of the reduction in herding behavior occurs as countries transition from low to middle-income levels, and the additional benefits become smaller and sometimes remain constant at higher GNI per capita values. The pandemic amplified the drop in income across the low-to-middle range, while the post-pandemic period shows a more substantial, more concentrated average negative effect around this threshold.

The research shows that Individualism (IDV) has a small positive effect on herd behavior, leading to increased herd behavior across all stages of the pandemic. The GAM component plots align with the results and show that the positive relationship holds mainly for low and moderate IDV scores before stabilizing at high IDV levels. This non-linearity may help explain the contrast with the results of most studies (Chang & Lin, 2015; Schmeling, 2009; Zhan, 2019; Zheng, 2015; Zhou *et al*., 2019), although it aligns with the findings of Danrimi *et al*. (2018).

Uncertainty aversion (UAV) shifts from no significant relationship before the pandemic to a significant negative association during and, especially, after the pandemic. The inverse relationship between herd behavior and UAVs is consistent with the results of Zheng (2015) and Zhou *et al*. (2019), but contrary to those of Schmeling (2009). The multinomial logit coefficients demonstrate this emerging deterrent effect, which the GAM plots specifically identify as occurring at higher UAV

values. Risk-averse cultures ceased their behavioral imitation as pandemic uncertainty grew, and this anti-herd behavior became stronger after the pandemic ended.

Table 18 – Multinomial logistic estimations. Dependent variable: Herding

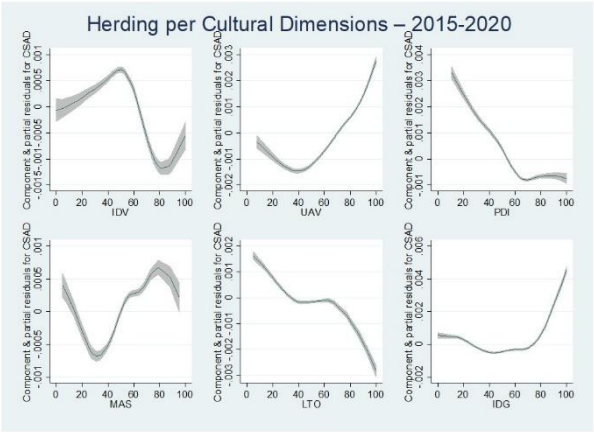
Panel A - Pre-Pandemic period: 2015-2020								
Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	1.1261***	-1.7380***	-3.8459***	-1.2712***	0.9924***	0.3078***	7.9969***	4.5818***
Individualism	-0.0420***							0.0115***
Uncertainty Avoidance		0.0148***						-0.0007
Power Distance			0.0473***					0.0296***
Masculinity				0.0108***				0.0030***
Long-term Orientation					-0.0411***			-0.0312***
Indulgence						-0.0242***		-0.1208***
GNI per capita							-0.9294***	-0.6401***
Pseudo R^2	0.053	0.008	0.046	0.036	0.039	0.022	0.061	0.128
χ^2	9284.57***	1321.22***	8084.17***	6196.98***	6784.48***	3570.91***	10659.93***	21108.82***
Panel B - Pandemic period: 03:2020-2021								
Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	-0.3352***	-1.2576***	-2.770***	-1.0929***	-0.4747***	-0.7585***	2.9801***	0.9492***
Individualism	-0.0152***							0.0275***
Uncertainty Avoidance		0.0033***						-0.0132***
Power Distance			0.0275***					0.0309***
Masculinity				0.0009				0.0072***
Long-term Orientation					-0.0124***			-0.0092***
Indulgence						-0.0069***		-0.0013
GNI per capita							-0.4237***	-0.4497***
Pseudo R^2	0.014	0.021	0.026	0.012	0.010	0.010	0.024	0.139
χ^2	752.24***	1107.94***	1389.31***	621.98***	503.88***	489.70***	1271.72***	6999.58***
Panel C - Post-Pandemic period 2022-06:2024								
Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	-0.0343	-2.1106***	-6.7722***	-0.6099***	-0.2114***	-0.3366***	4.2116***	5.4228***
Individualism	-0.0435***							0.0301***
Uncertainty Avoidance		0.0080***						-0.0440***
Power Distance			0.0730***					0.1431***
Masculinity				-0.0201***				-0.0959***
Long-term Orientation					-0.0396***			-0.1134***
Indulgence						-0.0364***		-0.0345***
GNI per capita							-0.6295***	-0.7218***
Pseudo R^2	0.0816	0.0014	0.0901	0.0081	0.1109	0.0292	0.0894	0.2361

χ^2	6156.18***	102.75***	6792.41***	608.22***	8263.65***	2001.78***	6738.74***	16213.98***

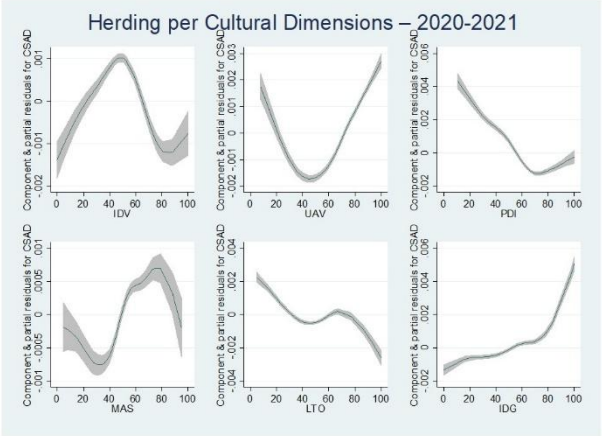
Source: Elaborated by the authors

Figure 15 - Predictive relationships of Herding by cultural dimensions

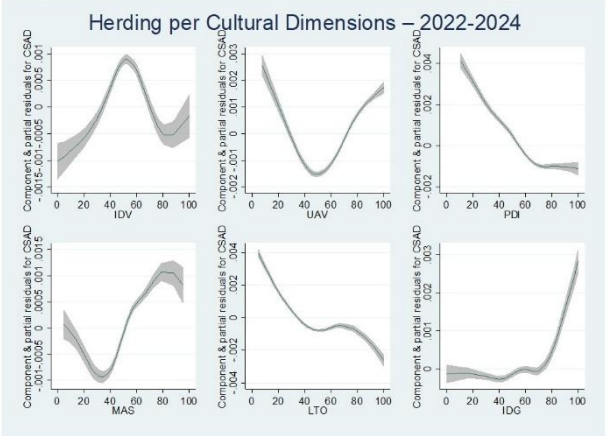
Panel A: Pre-Pandemic Period: 2015-2020



Panel B: Pandemic: 03:2020-2021



Panel C: Post-Pandemic: 2022-06:2024



Notes: The figure illustrates the nonlinear prediction of CSAD (a proxy for herding) at the 95% confidence interval, based on assumed intervals of 0 to 100 for the six Hofstede cultural dimensions. The nonlinear prediction is based on the results of the GAM regression, as specified in Equation **(22)** and displayed in Appendix M.
Source: Elaborated by the authors

Power distance (PDI) shows a clear upward direction after the pandemic. Multinomial logit coefficients are modestly positive before and during the pandemic but increase substantially afterward. GAMs' results indicate a positive time trend and show that herd behavior (CSAD) grows more rapidly at PDI levels that fall between medium and high after the pandemic. The positive relationship between PDI and herd behavior was also found by Chang and Lin (2015), Zheng (2015), and Zhou *et al*. (2019), although contradictory to the study by Lobão and Maio (2021). The research shows that societies with hierarchical systems tended to exhibit collective market behavior during recovery from crises, but the most severe impacts occurred in nations with the highest hierarchical values.

Long-term orientation (LTO) is consistently negative in the multinomial logistic model results, and GAMs show an increasingly pronounced decline in herd behavior at higher LTO levels, particularly in the post-pandemic period. This relationship aligns with the research of Saltik *et al*. (2024), but studies by Damrimi *et al*. (2018) and Zhou *et al*. (2019) show an inverse relationship.

The pre-pandemic period shows a clear negative relationship between indulgence (IDG), but this connection disappears during the pandemic and returns partially in the post-pandemic period. The GAM graphs demonstrate that at high levels of indulgence, the herd behavior becomes more pronounced in all time periods, as Zhou (2019) found. However, there were changes to intermediate IDG values during the crisis. The pandemic eliminated the protective effect of indulgence, as emergency situations cause people to lose their preference for spontaneous enjoyment.

Masculinity (MAS) is nearly neutral or slightly positive (in line with the results of Chang and Lin, 2015; Danrimi *et al*., 2018; Zheng, 2015; and Zhou *et al*., 2019) in the pre- and during-pandemic period, in the multinomial logistic model, but reverses to a significant negative effect in the post-pandemic period (aligned with Lobão and Maio, 2021). GAM models show that the reversal is concentrated in higher MAS scores. The study demonstrates that market stability does not guarantee the effectiveness of interventions because what works well in peaceful markets may not perform as well during times of crisis. The unstable nature of indulgence and masculinity shows that cultural effects do not remain constant. Taken together, these findings demonstrate that the pandemic altered not only the magnitude but also the direction of some cultural predictors.

The research shows that the pandemic changed investors' herd behavior, both in intensity and operational characteristics. The combined results of these two modeling approaches demonstrate this point: multinomial logistic models reveal the direction and statistical significance of average effects,

while GAMs show nonlinear patterns, threshold points, and score ranges where cultural traits have the most significant influence on herd behavior.

7.5 Conclusions

The study examines how cultural aspects and economic expansion influence financial market herding behavior yet reveals that the pandemic introduced new dynamics between these factors. Higher GNI per capita shows a consistent relationship with reduced herding behavior across all time periods, with the most considerable protective effect observed during the post-pandemic period. Generalized additive models (GAMs) show that most of the herding reduction occurs when countries move from low to middle-income status, but the additional benefits become less important after reaching middle-income status.

Our analysis reveals that cultural dimensions produce diverse and shifting patterns of influence on human behavior: Power Distance emerged as a key factor intensifying herd behavior post-pandemic, while Long-Term Orientation maintained its protective effects on recovery and reduced herding throughout the crisis. The results demonstrate that Uncertainty Avoidance had no impact on behavior before the pandemic but became a significant factor in blocking herding during and after the crisis. Individualism maintained a small positive relationship with herding, Indulgence lost much of its pre-pandemic protective effect during the crisis and only partially recovered afterward, and Masculinity reverted from marginally positive or neutral to significantly negative in the post-pandemic period. The conclusions from the multinomial logistic regression and GAM analyses indicate that these effects manifest as non-linear patterns that concentrate within specific score ranges; thus, their practical impact varies depending on a country's position on the cultural and income dimensions.

This study expands current knowledge by demonstrating that cultural dimensions and national income levels influence trend-following behavior across all nations and by revealing changes in these relationships between pre- and post-pandemic periods. The methodology employs multinomial logistic models along with generalized additive models to generate both accurate average forecasts and detailed functional patterns that reveal thresholds, plateaus, and sign reversals that linear models fail to identify. The analysis identifies Power Distance and Long-Term Orientation as the main cultural elements post-pandemic, but shows that per capita GNI negatively affects herding, which intensifies as countries move from low- to middle-income status. The observed changes over time in traits such as Indulgence and Masculinity suggest that cultural factors influencing investor behavior

can reverse direction under extreme stress. Our work builds on previous static literature, adding a time-sensitive and shock-based analysis.

The findings show that risk management strategies need to be specifically designed for different cultural environments, economic systems, and unpredictable patterns of risk occurrence. Governments should focus on low- and lower-middle-income markets because income growth in these ranges produces the most significant reduction in herding risk. Societies with high Power Distance require immediate attention during recovery because their hierarchical structures lead people to follow rules and practice reciprocal conduct. Crisis communication strategies that reduce ambiguity will achieve their best results in cultures with moderate to high Uncertainty Aversion, as these societies follow established rules and guidelines that can leverage their risk-averse nature to reduce copycat behavior. Combining investor education programs with market design strategies that emphasize long-term investment periods reduces short-term market constraints, particularly in regions with low Long-Term Orientation (LTO) and low Gross National Income (GNI). Surveillance systems require cultural and income thresholds for GAM diagnoses to detect concentrated systemic risk rather than conventional linear average indicators.

The results establish relationships between variables, but do not prove cause-and-effect relationships. Countries' cultural scores and GNI values serve as non-random treatment variables because they share common factors with unobserved confounders, policy interventions, and changes in market structure that affect herding. The cultural aggregation process at the national level fails to capture the diverse cultural patterns between regions within a country, while also neglecting the specific investment choices made by individual investors and the distinct behaviors of retail and institutional investors. The CSAD metric serves as a standard proxy for herding but only captures cross-sectional dispersion and does not distinguish between the reasons behind imitative behavior. The pandemic spread differently across nations due to distinct time patterns, policy approaches, and exposure levels, which this work does not fully capture. The GAM model produces descriptive nonlinear fits but cannot identify the fundamental connections and causal relationships between variables.

Future research needs to develop methods for causal identification using quasi-experimental policy timing variations and instrumental variables to measure the effects of cultural and income dimensions on herding. Research efforts should utilize country-level cultural indicators, along with trading data and investor survey responses, to examine how individual choices shape market trends and how market segments respond differently to disruptions. An analysis of cultural factors and GNI

levels in relation to pandemic severity and policy rigidity, using modeling, would help identify specific conditions in which economic protection and cultural elements strengthen or weaken the herding effect. Time-varying coefficient models, rolling windows, and state-space approaches should be used to analyze cultural effects over time, as they allow for tracking cultural development after multiple shocks and more accurate predictions of market fragility. Experimental or field intervention programs testing communication and market design methods based on cultural profiles would yield valuable insights for building more effective crisis prevention systems for correlated trading in the future.

8 FINAL CONSIDERATIONS

Our research investigated the influence of national culture on overconfidence biases, disposition effects, and herd behavior among investors from 64 countries between 2010 and 2024, including the period of the COVID-19 pandemic. The studies showed that cultural dimensions shape investor behavior. Furthermore, our research showed that cultural influences change in both magnitude and direction as societies adapt to external shocks.

The studies made important contributions to the literature, both in confirming previous studies and, mainly, in interesting findings:

a) Confirmations

Power Distance

• Investors from societies with high power distance, because they accept inequality and submit more to norms than to their own convictions, are less prone to overconfidence bias.

• On the other hand, they increase the possibility of incurring disposition effect bias, since, in these societies, admitting losses can be seen as a personal failing, and investors tend to hold onto falling assets to avoid appearing incompetent. Furthermore, investors in hierarchical cultures tend to delegate decisions or seek external validation, reducing autonomy and increasing inertia in decision-making.

• Also in herd behavior, investors in these cultures tend to follow dominant leaders and trends, especially during recovery periods.

Collectivism

• In societies with collectivist cultures, where the group is valued more than the individual, collective behavior is socially reinforced. In this way, they are more prone to herding.

Long-Term Orientation

• Cultures with a strong future orientation exhibit less disposition effect and herd behavior, as investors in these cultures tend to be more patient and less influenced by immediate gains or losses, valuing planning and resisting impulsive decisions.

Uncertainty Aversion

• Cultures averse to uncertainty are more prone to the disposition effect. Investors in these cultures prefer to achieve gains quickly and avoid risky decisions, holding assets at a loss out of fear of the unknown.

• They also reduce herd behavior and overconfidence, as they prefer to follow rigid norms to avoid imitation and suppress excessive self-confidence.

Indulgence

• Investors in indulgent societies, which value pleasure, gratification, and personal freedom, tend to overestimate their abilities, having a propensity for overconfidence bias, and, as they are more emotionally open cultures, tend to make more spontaneous, but not necessarily imitative, decisions, having a lower probability of herd behavior.

Masculinity

• Masculine cultures value assertiveness and competition, which reinforces self-confidence, leading to a propensity for overconfidence. In crises, this trait loses strength, giving way to more cooperative and cautious behaviors.

• Competitiveness can lead investors to avoid following the group to preserve reputation and performance, reducing herd behavior.

b) Findings

Individualism with a propensity for disposition effect.

The study on the disposition effect showed that more individualistic societies presented lower DE, as individualistic cultures tend to rely more on autonomous, rational decisions, reducing emotional attachment to losing assets. However, the protection of individualism against DE is not linear. At extreme levels, it can generate overconfidence and increased bias. The study revealed that the impact of individualism on DE is non-linear: it protects at moderate levels but can amplify bias at extreme levels. This reinforces the importance of using flexible models, such as GAM, to understand complex behaviors.

Critical attenuation of individualism in the overconfidence bias during the pandemic.

In individualistic societies, the overconfidence bias, which in normal times is driven by investors' belief in their own ability and competence, was likely attenuated by the recognition that the crisis's risks are systemic and uncontrollable. In contrast, in collectivist societies, an increase in overconfidence was observed, which can be explained by adherence to a collective optimism bias, in which responsibility for risk is socialized and diluted in the face of uncertainty.

Long-term orientation is related to overconfidence.

The focus on the future and caution, traditional elements of long-term oriented societies, should lead to more rational and less impulsive assessments, as demonstrated in our studies of disposition effects and herd behavior. As in previous studies, it was also expected that these cultures would show less tendency towards overconfidence, but our research did not confirm this.

Overconfidence likely stems from investors' confidence in long-term models and projections, which create an illusion of control and reinforce the belief that they are prepared for any scenario.

Indulgent societies are less prone to the disposition effect.

Investors in indulgent cultures focused on immediate gratification would tend to sell winning assets more quickly and, to avoid the pain of loss, hold onto losing stocks, thus demonstrating a disposition toward DE. But our study showed that indulgence, through greater tolerance for errors and losses, can lead to rational decisions with a lower propensity to hold declining assets, thus reducing DE.

Masculine cultures are less prone to the disposition effect.

Contrary to previous studies, investors from masculine societies are less vulnerable to the disposition effect. A likely explanation is that masculine cultures value self-control and rationality, leading investors to sell losing assets more quickly.

In addition to contributing to the field of behavioral finance in relation to culture, we believe that the research can assist financial educators and companies. The findings emphasize the importance of investor education to mitigate biases and that this education must be adapted to the culture in which the investor is embedded. For companies, the study provides guidance on considering culture in negotiation practices, market entry, and product launch strategies.

Our study has several limitations. Among them, we highlight: (i) Hofstede's indices are given at the national level and assume a homogeneous culture within a country, not capturing regional or subcultural variations or the impact of immigrant populations on financial decisions; (ii) Biases are measured by proxies (trading volume, probability of realizing gains/losses, deviation of sectoral returns from average market return) and not by questionnaires or direct psychological tests. These proxies can be affected by non-behavioral factors, such as market liquidity; (iii) being an observational study, there is no proposal to establish a causal relationship between culture and investor behavior. Furthermore, hidden variables at the macro level (e.g., regulation, financial literacy rate) can influence both culture and financial behavior; (iv) no differentiation is made between the trading volume performed by robots. We understand, based on several studies (e.g., Oyeniyi *et al.*, 2024; Yang *et al.*, 2025), that technology only systematizes behavioral biases, acting as a mechanism for transmitting human impulses. Therefore, transaction volume remains a valid proxy for the psychological inclinations of the country's investor culture; (v) the behavioral response observed during periods of crisis is specifically linked to an event of extreme uncertainty and was a global health shock (COVID-19). The results may not be generalizable to other types of crises, such as

financial shocks (e.g., 2008), wars, or political crises; (vi) the delimitation of the periods (Pre-, During, and Post-Pandemic) is determined by arbitrary dates. Behavioral changes may have occurred more gradually or at different times for each market.

Despite the limitations, we believe that by working with behavioral biases through the lens of culture, our work provides a basis for research into how culture influences investors. In short, culture is a powerful driving force that guides investor behavior.

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APPENDIX A - MOST PRODUCTIVE COUNTRIES

Table A-1 - Most productive countries

Country	Authors
United States	13
China	10
United Kingdom	10
Australia	8
Canada	5
Japan	5
Taiwan	5
U Arab Emirates	5
France	4
Germany	4
New Zealand	4
Indonesia	2
Malaysia	2
Poland	2
Saudi Arabia	2
Greece	1
India	1
Ireland	1
Israel	1
Netherlands	1
Pakistan	1
Portugal	1
Romania	1
Russia	1
Slovenia	1
South Korea	1
Spain	1
Tunisia	1

Source: Elaborated by the authors

APPENDIX B - Thematic Cluster and Geographical Coverage

Table B-1 – Thematic Cluster – Geography Coverage

Authors	**Geography**
Investors' Decision-Making	
Breuer *et al.* (2020); Choy *et al.* (2023); Docherty & Hurst (2018); Kim & Byun (2021); Nguyen & Truong (2013); Todea & Buglea (2017); Wang & Duxbury (2021); Wijayana & Gray (2018); Zaremba *et al.* (2020)	Global
Mbarki *et al.* (2022)	Asia-Pacific
Lee *et al.* (2019); Persakis & Koutoupis (2024)	Europe
Zouaoi *et al.* (2011)	Europe and the United States
Shah (2024)	Middle East and North Africa
Switzer *et al.* (2017)	G7 countries
Speidell (2009)	Frontier economy countries
Azzi & Suchard (2019); Cheema & Nartea (2014); Duxbury & Yao (2017); Ng & Wu (2010); Zeng *et al*. (2023)	China
Nguyen *et al.* (2018)	India and Pakistan
Adachi *et al.* (2017); Aman *et al.* (2023); Ikeda (2023)	Japan
Cronqvist *et al*. (2015)	Sweden
Chang *et al.* (2018); Chang (2020)	Taiwan
Huang (2015); Qadan & Zoua'bi (2019)	United States
Lee *et al.* (2023)	United States and China
Market and Investors' Cultural Preference	
Bergsma & Jiang (2016); Chourou *et al.* (2018); Zhang *et al.* (2023)	Global
Durand *et al.* (2013)	Asia-Pacific
Berk *et al*. (2017)	Frontier economy
Bley & Saad (2010)	Arabian Gulf countries
Chen *et al.* (2020); Huang, Chiu, *et al.* (2022)	China
Mahendra *et al.* (2021)	India
Huang, Rhee, *et al.* (2022)	Japan
Halari *et al*. (2015)	Pakistan
Wasiuzzaman (2018)	Saudi Arabia
Chen & Chien (2011)	Taiwan
Culture and Behavioral Biases	
Anderson *et al.* (2011); Beugelsdijk & Frijns (2010); Bhatta *et al.* (2018); Breitmayer *et al.* (2019);Cakici & Zaremba (2023); Chang & Lin (2015); Drobetz *et al*. (2023); El Hajjar *et al.* (2024); Schmeling (2009); Zhan (2019)	Global
Roque & Cortez (2014)	OECD countries
Liu *et al*. (2023); Wang & Zhang (2024)	China
Kosaka *et al.* (2017)	Japan
Gabbori *et al*. (2024)	Saudi Arabia
Balkanska (2018)	United States

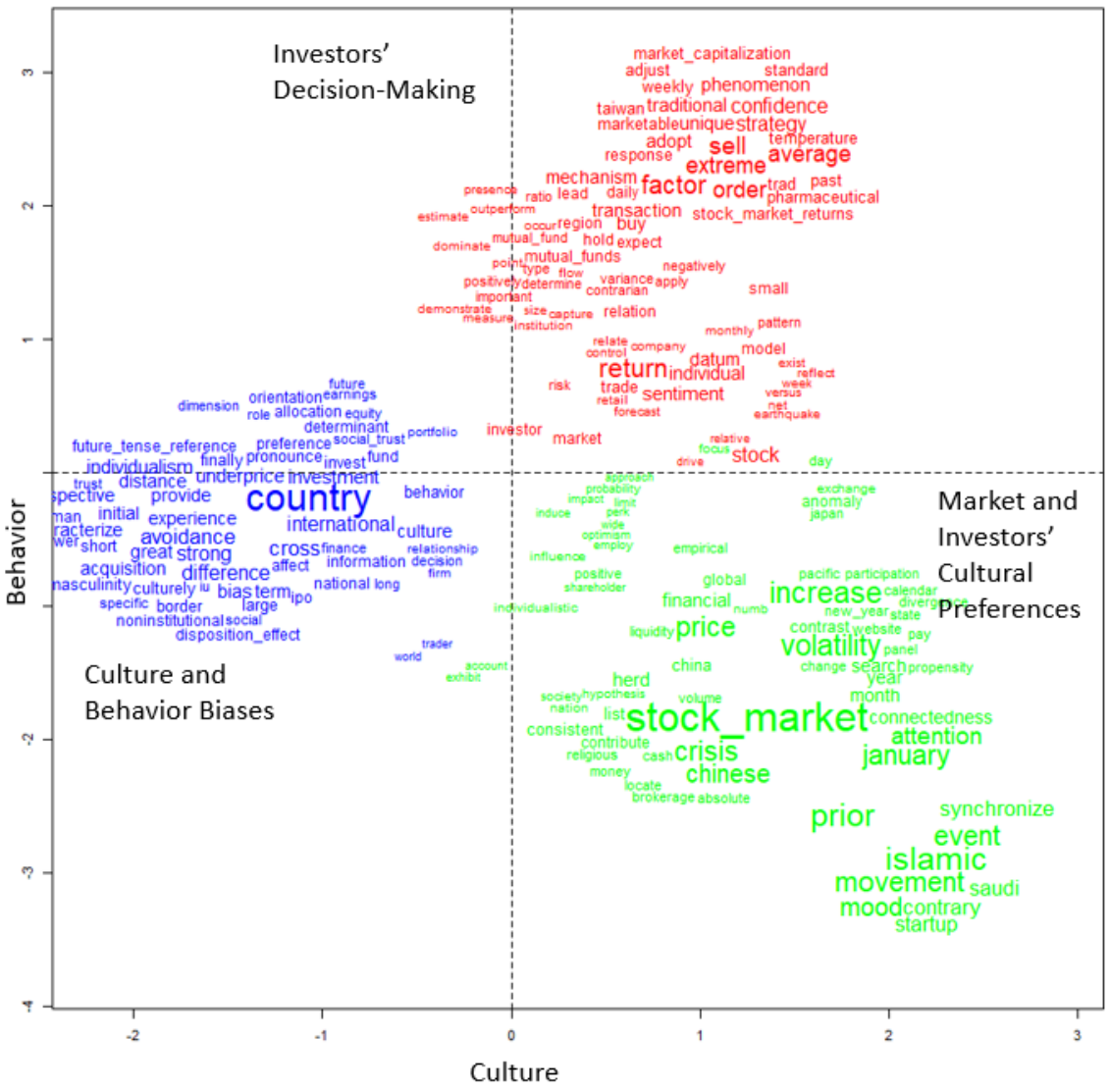
APPENDIX C - Studies by Thematic Cluster and Hofstede's Cultural Dimensions

Table C-1 – Thematic Cluster by Cultural Dimensions

Authors	Cultural Dimensions
Investors' Decision-Making	
Persakis & Koutoupis (2024)	power distance, individualism/collectivism, uncertainty avoidance, short/long-term orientation, and indulgence/restraint
Wijayana & Gray (2018)	power distance, individualism/collectivism, masculinity/femininity, and uncertainty avoidance
Choy *et al.* (2023)	power distance, individualism/collectivism, and masculinity/femininity
Shah (2024)	individualism/collectivism, masculinity/femininity, and uncertainty avoidance
Nguyen & Truong (2013); Zouaoi *et al.* (2011)	individualism/collectivism, and uncertainty avoidance
Docherty & Hurst (2018)	individualism/collectivism, and short/long-term orientation
Kim & Byun (2021)	uncertainty avoidance, and short/long-term orientation
Wang & Duxbury (2021)	individualism/collectivism, and uncertainty avoidance
Cheema & Nartea (2014); Lee *et al.* (2019); Lee *et al*.(2023); Switzer *et al*. (2017); Todea & Buglea (2017); Zaremba *et al*.(2020)	individualism/collectivism
Breuer *et al.* (2020)	short/long-term orientation
Zeng *et al.* (2023)	uncertainty avoidance
Market and Investors' Cultural Preference	
Chourou *et al.* (2018)	power distance, individualism/collectivism, and uncertainty avoidance
Berk *et al*. (2017); Durand *et al.* (2013)	individualism/collectivism
Culture and Behavioral Biases	
Breitmayer *et al.* (2019); Cakici & Zaremba (2023); Drobetz *et al.* (2023); Zhan (2019)	power distance, individualism/collectivism, masculinity/femininity, uncertainty avoidance, short/long-term orientation, and indulgence/restraint
Chang & Lin (2015)	power distance, individualism/collectivism, masculinity/femininity, uncertainty avoidance, and short/long-term orientation
Anderson *et al.* (2011)	individualism/collectivism, masculinity/femininity, uncertainty avoidance, and short/long-term orientation
Roque & Cortez (2014)	power distance, individualism/collectivism, masculinity/femininity, and uncertainty avoidance
Beugelsdijk & Frijns (2010); Schmeling (2009)	individualism/collectivism and uncertainty avoidance

APPENDIX D – Correspondence Analysis Map

Figure D-1 – Correspondence Analysis Map



Source: Elaborated by the authors

APPENDIX E - Mean and Median of Trade Volume per Country (in millions)

Table E-1 – Mean and Median of Trade Volume (in millions)

Country	#	Mean	Std. Dev.	Median	Min	Max
Argentina	2442	25.81	15.38	22.91	0.00	151.20
Australia	2401	1718.06	969.64	1361.52	229.77	7893.08
Austria	2395	6.43	3.55	5.52	0.01	76.85
Bangladesh	1822	198.88	138.03	156.16	0.59	932.27
Belgium	2431	12.44	6.92	10.97	2.01	180.23
Brazil	2352	962.57	491.97	825.04	104.86	3124.77
Bulgaria	2336	0.61	1.24	0.28	0.01	26.33
Canada	2383	695.10	719.30	494.79	62.99	8492.08
Chile	2363	941.59	2060.57	689.86	86.73	86184.76
China	2306	71842.43	25158.23	71568.67	25830.41	164982.17
Colombia	2450	24.75	25.26	19.96	0.00	562.33
Croatia	2363	0.07	0.07	0.05	0.01	0.91
Czech Republic	2378	1.53	1.57	1.15	0.16	35.29
Denmark	2373	28.36	9.73	26.23	10.64	141.93
Egypt	2136	620.03	575.28	496.18	57.75	18219.80
Finland	2387	70.80	36.49	63.27	16.80	648.87
France	2431	144.70	65.70	129.15	19.35	905.24
Germany	2409	149.36	50.64	139.62	45.11	598.35
Greece	2334	30.89	25.29	26.50	3.01	462.27
Hong Kong	2335	6051.65	2651.48	5475.68	1539.13	23407.93
Hungary	2372	7.01	5.39	5.63	0.99	129.53
India	2351	2117.17	1268.95	1730.23	0.00	11502.50
Indonesia	2297	13048.22	10001.99	7803.60	2119.47	110148.71
Ireland	2413	156.71	194.28	113.07	2.39	3870.62
Israel	2082	115.10	64.25	98.62	12.99	557.96
Italy	2451	496.22	187.34	466.25	0.02	2595.12
Jamaica	2373	25.12	44.02	15.90	0.17	938.49
Japan	2320	2565.13	546.19	2483.86	0.01	6146.27
Jordan	2655	4.98	3.88	4.71	0.00	86.50
Kuwait	2296	177.78	121.74	149.38	0.01	2182.95
Luxembourg	2427	43.20	21.38	38.92	2.33	253.62
Malaysia	2319	3104.98	1939.31	2523.74	673.98	22871.26
Mexico	2387	215.82	136.68	195.27	14.15	2944.90
Morocco	2363	1.65	30.93	0.67	0.03	1503.25
Netherlands	2431	109.59	39.24	101.72	17.98	419.68
New Zealand	2376	35.77	22.08	31.38	4.02	467.88
Nigeria	2347	366.99	308.82	290.57	55.65	5492.71
Norway	2383	98.74	50.60	88.68	19.82	876.77
Pakistan	2348	288.65	199.31	232.65	38.98	2329.17
Peru	2454	12.50	55.69	7.27	0.30	1757.92
Philippines	2312	1197.75	1562.68	835.61	172.94	28036.64

Poland	2375	54.21	24.94	47.34	15.70	298.86
Portugal	2431	83.40	46.95	73.24	6.97	396.59
Romania	2383	19.13	17.27	15.21	0.00	345.97
Russia	1867	9939.85	13935.72	7116.84	0.00	309973.70
Saudi Arabia	2108	390.53	241.73	332.28	102.50	3719.66
Serbia	2346	0.01	0.03	0.01	0.00	0.82
Singapore	2470	906.18	464.02	805.43	0.00	5303.33
Slovenia	2359	0.04	0.12	0.03	0.00	5.85
South Africa	2372	219.46	103.21	200.92	8.76	2246.69
South Korea	2335	1357.12	683.63	1181.08	433.24	5265.14
Spain	2429	374.56	166.45	339.26	94.20	2249.35
Sri Lanka	2238	91.94	153.71	45.10	3.72	2642.02
Sweden	2388	321.29	258.27	253.80	22.11	5584.58
Switzerland	2388	192.63	72.04	179.25	25.38	678.88
Taiwan	2308	4404.52	2318.40	3657.94	1223.15	21818.35
Thailand	2308	7595.63	4612.09	6118.80	1886.53	46363.36
Tunisia	2380	0.75	0.60	0.59	0.06	8.14
Turkey	2380	4470.55	2866.41	3829.84	267.07	15124.15
U Arab Emirates	2194	413.62	364.64	344.26	50.74	10297.66
United Kingdom	2398	2429.47	1312.50	2111.15	139.11	17553.22
United States	2388	12494.49	10092.40	9873.89	2120.75	118096.27
Venezuela	2271	1187.91	15244.42	0.72	0.00	388103.34
Vietnam	2367	649.11	420.55	462.92	127.40	2534.34

Source: Elaborated by the authors

APPENDIX F - Hofstede's Cultural Dimensions by Country

Table F-1 - Hofstede's Cultural Dimensions by Country

Country	IDV	UAV	PDI	MAS	LTO	IDG
Argentina	51	86	49	56	29	62
Australia	73	51	38	61	56	71
Austria	77	70	11	79	47	63
Bangladesh	5	60	80	55	38	20
Belgium	81	94	65	54	61	57
Brazil	36	76	69	49	28	59
Bulgaria	50	85	70	40	51	16
Canada	72	48	39	52	54	68
Chile	49	86	63	28	12	68
China	43	30	80	66	77	24
Colombia	29	80	67	64	6	83
Croatia	42	80	73	40	40	33
Czech Rep.	70	74	57	57	51	29
Denmark	89	23	18	16	59	70
Egypt	13	55	80	55	22	0
Finland	75	59	33	26	63	57
France	74	86	68	43	60	48
Germany	79	65	35	66	57	40
Greece	59	100	60	57	51	50
Hong Kong	50	29	68	57	93	17
Hungary	71	82	46	88	45	31
India	24	40	77	56	51	26
Indonesia	5	48	78	46	29	38
Ireland	58	35	28	68	51	65
Israel	56	81	13	47	47	
Italy	53	75	50	70	39	30
Jamaica	39	13	45	68		
Japan	62	92	54	95	100	42
Jordan	20	65	70	45	20	43
Kuwait	28	70	73	45	31	29
Luxembourg	60	70	40	50	64	56
Malaysia	27	36	100	50	47	57

Country	IDV	UAV	PDI	MAS	LTO	IDG
Mexico	34	82	81	69	23	97
Morocco	24	68	70	53	25	25
Netherlands	100	53	38	14	67	68
New Zealand	69	49	22	58	55	75
Nigeria	0	55	80	60	8	84
Norway	81	50	31	8	55	55
Pakistan	5	70	55	50	19	0
Peru	20	87	64	42	5	46
Philippines	17	44	94	64	46	42
Poland	47	93	68	64	49	29
Portugal	59	99	63	31	42	33
Romania	46	90	90	42	32	20
Russia	46	95	93	36	58	20
Saudi Arabia	48	64	72	43	27	14
Serbia	42	92	86	43	37	28
Singapore	43	8	74	48	67	46
Slovenia	81	88	71	19	50	48
S. Africa	23	49	49	63	18	63
S.Korea	58	85	60	39	86	29
Spain	67	86	57	42	47	44
Sri Lanka	35	45	80	10	45	
Sweden	87	29	31	5	52	78
Switzerland	79	58	34	70	42	66
Taiwan	40	69	58	45	87	49
Thailand	19	64	64	34	67	45
Tunisia	27	75	70	40	24	
Turkey	46	85	66	45	35	49
U A Emir	36	66	74	52	22	22
U.K.	76	35	35	66	60	69
U.S.A.	60	46	40	62	50	68
Venezuela	26	76	81	73	0	100
Vietnam	30	30	70	40	47	35

Source: Hofstede Insights (2024)

APPENDIX G - GAM Estimations. Dependent Variable: Overconfidence

Table G-1 – GAM Estimations – Dependent Variable: Overconfidence

Variables	Model (1) Pre-Pandemic Period (2015-03:2020)	Model (2) Pandemic Period (03:2020-2022)	Model (3) Pos-Pandemic Period (2022-06:2024)
constant	0.8434*** (345.18)	1.432*** (236.02)	1.3487*** (262.35)
Individualism	0.0011*** (6.16) 285.71***	-0.0065*** (-14.11) 103.94***	0.0011*** (2.78) 162.50***
Uncertainty Avoidance	0.0011*** (8.45) 213.07***	-0.0046*** (-14.24) 136.08***	-0.0047*** (-17.32) 655.65***
Power Distance	-0.0004** (2.24) 71.84***	0.0001 -0.312 292.19***	0.0034*** (8.51) 574.51***
Masculinity	0.0017*** (11.24) 342.30***	-0.0042*** (-11.35) 130.54***	-0.0028*** (-8.96) 363.59***
Long-Term Orientation	0.0014*** (8.89) 202.58***	0.0099*** (25.04) 876.52***	0.0003 (1.01) 816.34
Indulgence	0.0006*** (4..27) 125.42***	0.0050*** (15.39) 876.52***	0.0007*** (2.73) 128.89***
non-linearity χ^2	1240.92***	1771.52***	2701.49***

Notes: Generalized additive model estimations where the dependent variable is overconfidence. Z-value (in parentheses) is the ratio of the linear coefficient to its standard error to assess how strongly a variable is associated with overconfidence. The third line represents the gain value. *p<0.1; **p<0.05; ***p<0.01.
Source: Elaborated by the authors

APPENDIX H – Description Statistics of Disposition Effect by Country

Table H-1 – Description Statistics of Disposition Effect by Country

Country/Region	Obs	Mean	Std. Dev.	Median	Country/Region	Obs	Mean	Std. Dev.	Median
Africa					**Asia**				
Egypt	2571	2.188	3.792	1.104	Bangladesh	2266	1.589	1.995	1.203
Morocco	3451	2.822	4.759	1.085	China	2453	1.114	0.291	1.074
Nigeria	3430	1.671	2.302	1.088	Hong Kong	2840	1.119	0.496	1.052
South Africa	3327	1.363	1.917	1.035	India	3156	1.172	0.602	1.079
Tunisia	3477	1.999	2.560	1.257	Indonesia	3155	1.311	0.898	1.096
Europe					Japan	2520	1.113	0.497	1.033
Austria	3322	1.351	1.867	0.999	Malaysia	3023	1.253	1.148	1.083
Belgium	3438	1.216	1.179	1.028	Pakistan	3139	1.707	2.004	1.242
Bulgaria	3412	3.092	5.325	0.984	Philippines	3235	2.046	3.169	1.123
Croatia	3233	2.591	4.088	1.148	Singapore	3471	1.452	1.895	1.063
Czech Republic	2590	2.625	4.358	0.951	South Korea	3185	1.321	1.235	1.086
Denmark	3456	1.666	2.709	1.069	Sri Lanka	3047	2.575	3.920	1.347
Finland	3140	1.127	0.777	1.006	Taiwan	2985	1.224	0.659	1.094
France	2904	1.219	1.027	1.006	Thailand	2897	1.210	0.694	1.076
Germany	3074	1.133	0.854	1.002	Vietnam	3114	1.278	1.208	1.099
Greece	3325	1.577	2.163	1.069	**Middle East**				
Hungary	3466	2.207	3.659	1.111	Israel	2733	1.361	1.906	1.038
Ireland	3457	1.944	3.071	1.071	Jordan	3195	1.900	2.520	1.227
Italy	3240	1.339	1.471	1.084	Kuwait	3099	2.063	3.339	1.154
Luxembourg	3493	1.661	2.719	1.048	Saudi Arabia	2456	1.361	1.250	1.140
Netherlands	3383	1.964	3.777	1.029	Turkey	3038	1.433	1.693	1.114
Norway	3342	1.611	2.400	1.071	U Arab Emirates	3053	2.285	3.650	1.195
Poland	3306	1.200	0.988	1.017	**North America**				
Portugal	3383	1.718	2.896	1.034	Canada	3280	1.244	0.968	1.016
Romania	3449	2.197	3.417	1.132	United States	2827	1.089	0.496	1.002
Russia	2827	2.349	4.033	1.088	**Latin America and the Caribbean**				
Serbia	2909	4.391	6.503	1.411	Argentina	3066	1.804	2.560	1.158
Slovenia	3063	2.311	3.653	1.161	Brazil	3046	1.130	0.606	1.017
Spain	3202	1.570	2.277	1.035	Chile	3253	1.880	2.830	1.107
Sweden	3112	1.283	1.364	1.012	Colombia	3296	2.415	4.049	1.093
Switzerland	3284	1.219	1.155	1.007	Jamaica	3283	3.001	5.190	1.018
United Kingdom	3069	1.300	1.341	1.005	Mexico	3360	1.648	2.835	1.013
Oceania					Peru	3155	3.155	5.268	1.179
Australia	3166	1.229	1.259	1.025	Venezuela	1184	5.171	7.843	0.988
New Zealand	3507	1.455	1.625	1.066					

Source: Elaborated by the authors

APPENDIX I - Hofstede's Cultural Dimensions by Region and Country

Table I-1 – Hofstede's Cultural Dimensions score per Region and Country

Region/Country	IDV	UAV	PDI	MAS	LTO	IDG
Africa						
Egypt	13	55	80	55	22	0
Morocco	24	68	70	53	25	25
Nigeria	0	55	80	60	8	84
South Africa	23	49	49	63	18	63
Tunisia	27	75	70	40	24	
Asia						
Bangladesh	5	60	80	55	38	20
China	43	30	80	66	77	24
Hong Kong	50	29	68	57	93	17
India	24	40	77	56	51	26
Indonesia	5	48	78	46	29	38
Japan	62	92	54	95	100	42
Malaysia	27	36	100	50	47	57
Pakistan	5	70	55	50	19	0
Philippines	17	44	94	64	46	42
Singapore	43	8	74	48	67	46
South Korea	58	85	60	39	86	29
Sri Lanka	35	45	80	10	45	
Taiwan	40	69	58	45	87	49
Thailand	19	64	64	34	67	45
Vietnam	30	30	70	40	47	35
Europe						
Austria	77	70	11	79	47	63
Belgium	81	94	65	54	61	57
Bulgaria	50	85	70	40	51	16
Croatia	42	80	73	40	40	33
Czech Republic	70	74	57	57	51	29
Denmark	89	23	18	16	59	70
Finland	75	59	33	26	63	57
France	74	86	68	43	60	48
Germany	79	65	35	66	57	40
Greece	59	100	60	57	51	50
Hungary	71	82	46	88	45	31
Ireland	58	35	28	68	51	65
Italy	53	75	50	70	39	30
Luxembourg	60	70	40	50	64	56
Netherlands	100	53	38	14	67	68

Norway	81	50	31	8	55	55
Poland	47	93	68	64	49	29
Portugal	59	99	63	31	42	33
Romania	46	90	90	42	32	20
Russia	46	95	93	36	58	20
Serbia	42	92	86	43	37	28
Slovenia	81	88	71	19	50	48
Spain	67	86	57	42	47	44
Sweden	87	29	31	5	52	78
Switzerland	79	58	34	70	42	66
United Kingdom	76	35	35	66	60	69
Latin America and the Caribbean						
Argentina	51	86	49	56	29	62
Brazil	36	76	69	49	28	59
Chile	49	86	63	28	12	68
Colombia	29	80	67	64	6	83
Jamaica	39	13	45	68		
Mexico	34	82	81	69	23	97
Peru	20	87	64	42	5	46
Venezuela	26	76	81	73	0	100
Middle East						
Israel	56	81	13	47	47	
Jordan	20	65	70	45	20	43
Kuwait	28	70	73	45	31	29
Saudi Arabia	48	64	72	43	27	14
Turkey	46	85	66	45	35	49
U Arab Emirates	36	66	74	52	22	22
North America						
Canada	72	48	39	52	54	68
United States	60	46	40	62	50	68
Oceania						
Australia	73	51	38	61	56	71
New Zealand	69	49	22	58	55	75

Source: Elaborated by the authors using the Hofstede Insights database (2024)

APPENDIX J – GAM Estimation (2010-2024)– Disposition Effect

Table J-1 – GAM Estimations (2010-2024) – Dependent Variable: Disposition Effect

Variables	Model Period (2010-06:2024)
Constant	1.7371***
	(269.15)
Individualism	0.0114***
	(19.77)
	321.53***
Uncertainty Avoidance	0.0047***
	(13.53)
	190.76***
Power Distance	-0.0014***
	(-2.77)
	589.43***
Masculinity	-0.0026***
	(-6.49)
	299.65***
Long-Term Orientation	-0.0125***
	(-28.35)
	416.65***
Indulgence	-0.0083***
	(23.21)
	1036.27***
GNI per capita	-0.1691***
	(-17.77)
	692.10***
non-linearity χ^2	3546.39

Notes: Generalized additive model estimations where the dependent variable is disposition effect. Z-value (in parentheses) is the ratio of the linear coefficient to its standard error to assess how strongly a variable is associated with overconfidence. The third line represents the gain value. *p<0.1; **p<0.05; ***p<0.01.
Source: Elaborated by the authors

APPENDIX K - Intertemporal Analysis of the Disposition Effect under COVID-19

K.1 Definition and justification of the periods

For the intertemporal robustness analysis of the Disposition Effect (DE), we divided the data sample into three distinct periods in order to test the stability and modulation of the influence of cultural dimensions under a global external shock that generated extreme volatility and widespread uncertainty in capital markets.

The temporal division is crucial to contextualize changes in investor behavior, as exogenous shocks intensify emotions and risk-taking that underlie behavioral biases. The analysis follows the official statement of the World Health Organization (WHO) of March 11, 2020 (Padhan & Prabheesh, 2021) and structures the periods according to the crisis and recovery cycle: (i) January 2015 to March 10, 2020 (pre-pandemic), (ii) March 11, 2020 to December 2021 (pandemic) and (iii) January 2022 to June 2024 (post-pandemic).

K.2. Subsample Analysis Methodology

To test the stability and modulation of the results presented in Chapter 6, the statistical analysis methodology was replicated in its entirety but applied to three independent temporal subsamples. The objective is to isolate and compare the impact of market shocks on regression coefficients and nonlinear patterns of cultural dimensions on the Disposition Effect (DE).

K.3. Results and Specific Discussion of Temporal Modulation

The results in Table K-1 (Multinomial logistic estimations), Table K-2 (GAM estimations), and Figure K-1 (Predictive relationships of DE levels) reveal the modulation of the influence of cultural dimensions on the Disposition Effect (DE) under the shock of COVID-19.

The GNI per capita coefficient showed negative values in all time periods while maintaining high statistical significance, which proved that economic wealth protects against DE. The UAV cultural dimension coefficient displayed positive values throughout all periods at a high level of statistical significance. The GAM curve showed an upward trend throughout all time periods (Panels A, B, and C), with UAV cultural dimension standing as the most stable factor in the sample.

Similarly, the MAS and IDG dimensions remained negative or non-significant in all subsamples. The GAM plots (Panels A, B, and C) reinforce that the unexpected finding that these dimensions do not drive DE is a robust result in the sample studied.

The IDV variable shows negative coefficients throughout all time periods, yet the GAM curve indicates DE intensity increased during the pandemic because of pandemic-related stress (Panel B), maintaining the contradiction of the negative linear relationship as observed in the main study.

The PDI dimension culture failed to show any statistical relationship during all assessment time points. The GAM plots (Panels A, B, and C) of PDI show a steady non-linear pattern, which indicates that social hierarchy and acceptance of authority do not follow a linear relationship with DE, and their small effects remained unchanged throughout the crisis.

The LTO coefficients lost their statistical significance during the pandemic and post-pandemic periods, yet GAM curves (Panels A, B, and C) maintained their negative patterns, which proved that DE resistance in long-term oriented cultures endures through external disruptions.

Table K-1 – Multinomial logistic estimations. Dependent variable – Disposition Effect

Panel A - Pre-Pandemic period: 2015-2020

Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	0.2402***	-0.1297***	-0.3342***	0.0010	0.21778***	0.1598***	1.1032***	0.6512***
Individualism	-0.0054***							-0.0002
Uncertainty Avoidance		0.0018***						0.0007
Power Distance			0.0052***					0.0012
Masculinity				-0.0002				0.0001
Long-term Orientation					-0.0056***			-0.0034***
Indulgence						-0.0040***		-0.0025***
GNI per capita							-0.1176***	-0.0563***
Pseudo R^2	0.005	0.004	0.007	0.001	0.022	0.002	0.010	0.032
χ^2	804.95***	651.56***	1033.09***	207.78***	3104.93***	242.87***	1508.82***	4246.72

Panel B - Pandemic period: 03:2020-2021

Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	0.3494***	-0.0077	-0.3213***	0.1881***	0.2340***	0.2369***	0.9976***	0.1947
Individualism	-0.0064***							-0.0060***
Uncertainty Avoidance		0.0008						0.0003
Power Distance			0.0060***					0.0005
Masculinity				-0.0029***				-0.0019
Long-term Orientation					-0.0043***			-0.0019
Indulgence						-0.0045***		-0.0031***
GNI per capita							-0.0992***	0.0405
Pseudo R^2	0.004	0.002	0.005	0.002	0.014	0.001	0.007	0.020
χ^2	200.41***	123.48***	239.97***	94.73***	719.51***	29.01***	370.82***	950.98***

Panel C - Post-Pandemic period 2022-06:2024								
Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Constant	0.2363***	-0.2431***	-0.4509***	0.0721*	0.1342***	0.1676***	1.2695***	0.8350***
Individualism	-0.0064***							-0.0051***
Uncertainty Avoidance		0.0026***						0.0036***
Power Distance			0.0063***					-0.0014
Masculinity				-0.0029***				-0.0039***
Long-term Orientation					-0.0050***			0.0003
Indulgence						-0.0053***		-0.0029***
GNI per capita							-0.1379***	-0.0523**
Pseudo R^2	0.002	0.005	0.004	0.002	0.015	0.001	0.007	0.026
χ^2	175.35***	389.73***	262.96***	120.84***	1073.56***	53.11***	493.19***	1709.03***

Notes: Multinomial logistic estimations where the dependent variable is the disposition effect. *p<0.1; **p<0.05; ***p<0.01.

Source: Elaborated by the authors

Table K.2 - GAM Estimations – Dependent Variable: Disposition Effect

Variables	Model (1) Pre-Pandemic Period (2015-03:2020)	Model (2) Pandemic Period (03:2020-2022)	Model (3) Pos-Pandemic Period (2022-06:2024)
Constant	1.6749*** (164.38)	1.6499*** (100.58)	1.6802*** (113.47)
Individualism	0.0114*** (12.51) 125.03***	0.0063*** (4.22) 61.09***	0.0155*** (11.40) 92.65***
Uncertainty Avoidance	0.0042*** (7.57) 44.96***	0.0070*** (7.89) 22.75***	0.0098*** (12.09) 90.43***
Power Distance	-0.0006 (-0.79) 200.89***	-0.0037*** (-2.91) 105.77***	0.0026*** (2.24) 111.94***
Masculinity	-0.0010* (-1.62) 178.23***	-0.0036*** (-3.54) 42.89***	-0.0046*** (-5.11) 61.06***
Long-Term Orientation	-0.0148*** (-21.20) 201.77***	-0.0118*** (-10.45) 100.13***	-0.0086*** (-8.50) 128.15***
Indulgence	-0.0089*** (-15.83) 479.23***	-0.0061*** (-6.93) 217.96***	0.0022*** (-2.82) 430.47***
GNI per capita	-0.1363*** (-8.74)	-0.1365*** (-5.34)	-0.3189*** (-13.77)

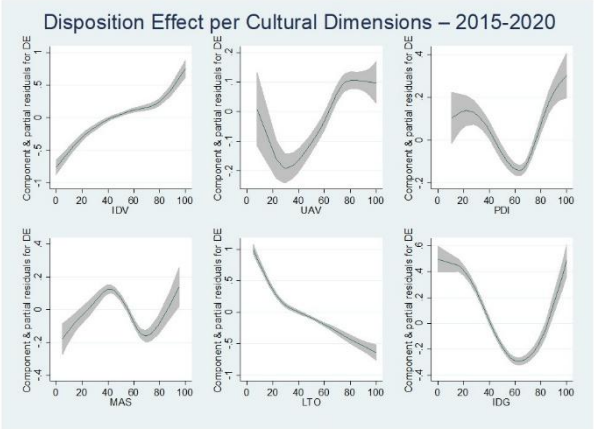
	357.79***	73.26***	206.96***
non-linearity χ^2	1587.91***	623.84***	1121.66***

Notes: Generalized additive model estimations where the dependent variable is overconfidence. Z-value (in parentheses) is the ratio of the linear coefficient to its standard error to assess how strongly a variable is associated with overconfidence. The third line represents the gain value. *p<0.1; **p<0.05; ***p<0.01.

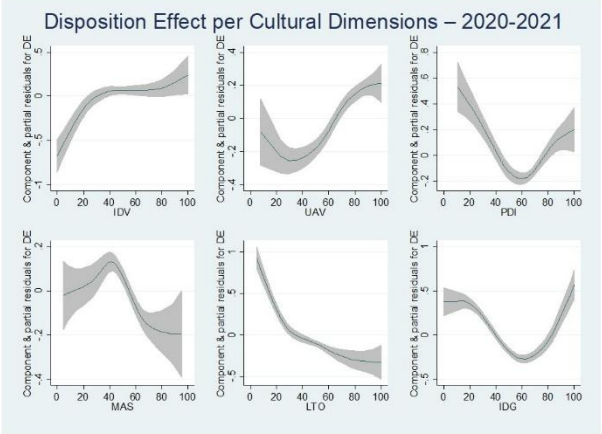
Source: Elaborated by the authors

Figure K-1 - Predictive relationships of disposition effect levels by cultural dimensions

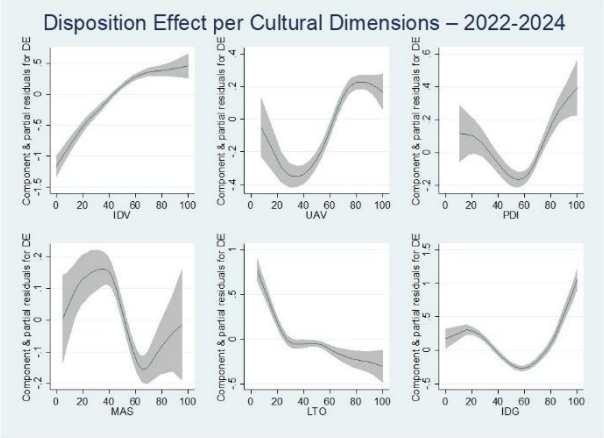
Panel A: Pre-Pandemic Period: 2015-2020



Panel B: Pandemic: 03:2020-2021



Panel C: Post-Pandemic: 2022-06:2024



Notes: The figure illustrates the nonlinear prediction of the disposition effect at the 95% confidence interval, based on assumed intervals of 0 to 100 for the six Hofstede cultural dimensions. The nonlinear prediction is based on the results of the GAM regression, as specified in the Equation (16) and displayed in Figure K.1.
Source: Elaborated by the authors

K.4. Conclusion

The analysis shows that cultural relationships with DE are sensitive to the market regime. It shows that the protective effect of LTO lost significance, and that DE was higher for higher IDV during the crisis. Therefore, this complementary analysis shows that bias stability depends on stress.

APPENDIX L – INVESTORS' HERDING

This study uses the methods by Chang *et al*. (2000) to investigate herding behavior. If statistically significant and $\Upsilon_{i,3} < 0$, herding; $\Upsilon_{i,3} > 0$, anti-herding

Table L-1 – Regression coefficients - Herding

Panel A - Pre-Pandemic (Jan 2015 - Mar 2020)

Country	N	$\Upsilon_{i,0}$	$\Upsilon_{i,1}$	$\Upsilon_{i,2}$	$\Upsilon_{i,3}$	Adj-R^2	F-statistic
Argentina	1354	0.009*** (16.99)	-0.043** (-2.51)	0.476*** (16.32)	-0.323*** (5.86)	0.265	164***
Australia	1313	0.006*** (47.62)	0.019** (2.23)	0.039* (1.91)	3.649*** (4.57)	0.113	56.51***
Austria	1313	0.009*** (32.75)	0.018 (0.95)	0.485*** (10.51)	-2.258*** (4.32)	0.162	85.52***
Bangladesh	1354	0.004*** (21.92)	0.056*** (3.95)	0.729*** (24.71)	-6.454 (0.04)	0.403	304.99***
Belgium	1327	0.006*** (48.49)	0.021*** (2.64)	0.147*** (7.24)	2.462 (0.11)	0.255	152.58***
Brazil	1281	0.007*** (40)	0.024*** (3.29)	0.156*** (8.34)	1.365 (0.31)	0.263	152.98***
Bulgaria	1278	0.009*** (26.22)	0.021 (0.58)	0.5*** (7.46)	0.048*** (2.67)	0.093	44.76***
Canada	1302	0.006*** (42.39)	-0.024* (-1.92)	0.271*** (11.9)	-0.271*** (3.94)	0.186	100.4***
Chile	1291	0.005*** (49.61)	0.03*** (4.33)	0.224*** (13.74)	0.041* (1.84)	0.327	209.79***
China	1262	0.004*** (30.57)	0 (-0.05)	0.21*** (11.46)	-1.233*** (6.72)	0.234	129.14***
Colombia	1354	0.004*** (25.65)	0.013** (2.01)	0.698*** (56.91)	0.044 (1.45)	0.839	2358.35***
Croatia	1291	0.004*** (31.33)	0.017 (0.98)	0.511*** (16.57)	-1.192* (1.82)	0.287	174.02***
Czech Rep.	1299	0.006*** (2.79)	0.111 (0.68)	0.512 (1.1)	-3.228*** (3.98)	0.001	1.24
Denmark	1297	0.007*** (33.86)	0.009 (0.9)	0.297*** (9.36)	2.429 (0.42)	0.36	243.98***
Egypt	1097	0.008*** (33.64)	0.04*** (4.17)	0.194*** (7.65)	1.657*** (3.79)	0.292	151.65***
Finland	1303	0.007*** (37.53)	0.003 (0.31)	0.109*** (4.33)	1.132 (0.48)	0.08	38.67***
France	1327	0.004*** (53.37)	-0.002 (-0.49)	0.071*** (5.91)	1.873** (2.31)	0.29	181.75***
Germany	1311	0.005*** (45.69)	0.004 (0.62)	0.12*** (7.43)	0.606 (0.67)	0.183	99.13***
Greece	1266	0.008*** (26.94)	0.042*** (4.19)	0.403*** (16.17)	0.493*** (3.26)	0.486	399.15***
Hong Kong	1277	0.006*** (39.95)	0.041*** (5.92)	0.083*** (3.51)	2.834 (0.4)	0.194	103.64***
Hungary	1291	0.008*** (29.07)	0.044*** (3.1)	0.379*** (9.76)	0.419** (2.42)	0.226	126.41***
India	1283	0.005*** (45.14)	0.007 (0.89)	0.133*** (6.19)	2.531*** (3.23)	0.207	112.54***
Indonesia	1262	0.009*** (4.36)	-0.011 (-0.08)	-0.092 (-0.25)	5.531*** (7.8)	-0.002	0.13
Ireland	1318	0.009*** (31.44)	0.006 (0.35)	0.291*** (7.37)	1.849** (2.51)	0.168	89.43***

Israel	1025	0.007*** (15.25)	0.021 (0.55)	0.152** (2.09)	0.798*** (3.02)	0.015	6.04***
Italy	1349	0.007*** (26.76)	0.006 (0.49)	0.255*** (10.15)	1.185 (0.86)	0.243	145***
Jamaica	1289	0.018*** (31.04)	-0.01 (-0.23)	0.549*** (5.92)	0.852 (0.38)	0.078	37.28***
Japan	1266	0.004*** (45.3)	0.015*** (3.24)	0.106*** (8.13)	-0.284** (2.06)	0.151	75.82***
Jordan	1585	0.003*** (29.87)	0.041*** (3.83)	0.773*** (36.93)	1.077 (1.63)	0.73	1429.99***
Kuwait	1585	0.004*** (26.09)	-0.008 (-0.58)	0.649*** (25.15)	-1.384*** (3.56)	0.463	456.46***
Luxembourg	1324	0.011*** (18.88)	0.069*** (3.13)	0.247*** (3.34)	5.828*** (3.19)	0.136	70.28***
Malaysia	1270	0.004*** (36.73)	0.021* (1.89)	0.046 (1.48)	12.233 (1.61)	0.211	114.44***
Mexico	1303	0.007*** (18.17)	0.007 (0.28)	0.304*** (4.5)	-0.042*** (7.36)	0.051	24.35***
Morocco	1288	0.006*** (42.62)	-0.006 (-0.44)	0.481*** (14.24)	-4.333* (1.65)	0.219	121.59***
Netherlands	1327	0.008*** (36.26)	0.007 (0.51)	0.216*** (5.99)	-0.303* (1.82)	0.087	43.07***
New Zealand	1301	0.006*** (44.77)	-0.005 (-0.38)	0.149*** (3.72)	5.039*** (5.4)	0.114	56.52***
Nigeria	1285	0.007*** (36.13)	0.028** (2.4)	0.727*** (23.22)	-4.085*** (6.05)	0.539	500.71***
Norway	1302	0.007*** (45.29)	0.035*** (4.24)	0.21*** (9.52)	1.642 (1.11)	0.287	175.76***
Pakistan	1288	0.007*** (44.53)	0.04*** (5.06)	0.124*** (4.31)	0.826 (0.75)	0.127	63.28***
Peru	1354	0.003*** (19.14)	0.012* (1.75)	0.829*** (39.64)	-1.317 (0.08)	0.827	2159.08***
Philippines	1262	0.007*** (30.34)	-0.002 (-0.19)	0.277*** (8.23)	0.368*** (2.92)	0.169	86.79***
Poland	1295	0.006*** (46.91)	0.026*** (3.79)	0.313*** (16.15)	-2.099 (0.28)	0.328	211.93***
Portugal	1327	0.007*** (36.69)	0.013 (1.2)	0.276*** (10.21)	1.319 (1.04)	0.271	165.56***
Romania	1328	0.005*** (14.97)	0.035 (1.47)	0.46*** (10.23)	0.922 (0.99)	0.199	110.69***
Russia	1354	0.006*** (26.21)	0.027*** (3.25)	0.534*** (24.25)	1.219** (2.36)	0.663	886.72***
Saudi Arabia	1041	0.006*** (36.89)	0.017* (1.94)	0.234*** (13.86)	-0.767 (0.29)	0.235	107.4***
Serbia	1328	0.004*** (21.75)	-0.024** (-1.98)	0.727*** (27.52)	1.624 (0.1)	0.666	881.58***
Singapore	1354	0.006*** (43.08)	0.041*** (4.98)	0.314*** (13.23)	-0.208*** (2.64)	0.341	234.18***
Slovenia	1328	0.005*** (14.62)	0.038 (1.09)	0.719*** (10.13)	-4.669 (1.14)	0.115	58.42***
South Africa	1297	0.012*** (9.08)	0.012 (0.18)	-0.205 (-0.95)	10.703 (0.25)	0.001	1.23
South Korea	1274	0.007*** (47.27)	0.01 (0.96)	0.157*** (5.03)	-1.198 (0.61)	0.06	27.98***
Spain	1327	0.006*** (48.79)	0.013** (2)	0.14*** (10.12)	1.747 (0.62)	0.338	226.21***
Sri Lanka	1245	0.007*** (41.47)	0.058*** (3.01)	0.196*** (4.05)	3.474 (0.06)	0.082	38.25***

Sweden	1303	0.006*** (42.74)	-0.006 (-0.72)	0.093*** (4.23)	1.026 (0.62)	0.085	41.49***
Switzerland	1302	0.006*** (40.47)	-0.027*** (-2.67)	0.179*** (7.56)	3.114 (0.03)	0.289	177.41***
Taiwan	1258	0.003*** (39.33)	0.015** (2.28)	0.267*** (16.63)	-0.753* (1.67)	0.39	269.33***
Thailand	1268	0.005*** (31.95)	0.023* (1.86)	0.051* (1.92)	4.11 (1.59)	0.113	54.98***
Tunisia	1585	0.004*** (31.52)	0.012 (1.12)	0.769*** (27.97)	1.055 (0.47)	0.67	1075.01***
Turkey	1306	0.007*** (19.28)	0.003 (0.22)	0.086 (1.63)	1.008 (0.25)	0.019	9.58***
U.A. Emir.	1118	0.013*** (4.16)	-0.019 (-0.11)	0.308 (0.71)	0.68*** (3.14)	0	0.84
UK	1313	0.005*** (38.94)	-0.002 (-0.31)	0.124*** (6.24)	1.636*** (10.72)	0.171	90.94***
USA	1305	0.005*** (22.04)	-0.018 (-1.09)	0.15*** (3.63)	0.313*** (3.98)	0.041	19.8***
Venezuela	1242	0.027*** (4.36)	-0.084 (-0.85)	0.602*** (3.28)	0.291* (1.92)	0.081	37.66***
Vietnam	1291	0.006*** (40.1)	0.03*** (3.3)	0.31*** (11.57)	-2.423*** (2.72)	0.222	123.69***

Panel B - Pandemic (Mar 2020 - Dez 2021)

Country	**N**	**$\Upsilon_{i,0}$**	**$\Upsilon_{i,1}$**	**$\Upsilon_{i,2}$**	**$\Upsilon_{i,3}$**	**Adj-R^2**	**F-statistic**
Argentina	473	0.016*** (7.75)	0.018 (0.43)	0.339** (2.38)	1.611 (0.07)	0.108	20.14***
Australia	461	0.006*** (23.72)	0.031** (2.53)	0.151*** (4.45)	1.29 (1.23)	0.32	73.14***
Austria	461	0.009*** (14.94)	0.012 (0.42)	0.592*** (7.67)	0.366 (0.06)	0.405	105.55***
Bangladesh	473	0.004*** (12.16)	0.048** (2.39)	0.772*** (17.29)	-7.492 (1.05)	0.468	139.55***
Belgium	466	0.007*** (21.07)	0.052*** (3.32)	0.334*** (9.84)	-0.301 (0.76)	0.371	92.31***
Brazil	449	0.008*** (6.34)	-0.007 (-0.2)	0.365*** (3.92)	-0.947** (2.56)	0.079	13.81***
Bulgaria	444	0.006*** (10.94)	0.001 (0.03)	0.686*** (7.14)	0.222 (0.96)	0.304	65.42***
Canada	455	0.006*** (22.83)	0.027** (2.3)	0.348*** (10.86)	-1.404 (0.02)	0.383	95.12***
Chile	450	0.007*** (26.14)	0.046*** (4.97)	0.223*** (9.42)	0.678 (1.09)	0.56	191.66***
China	443	0.008*** (23.44)	-0.014 (-0.8)	0.073 (1.35)	1.798 (0.53)	0.048	8.5***
Colombia	473	0.004*** (14.39)	0.054*** (6.81)	0.668*** (32.37)	0.035*** (7.73)	0.91	1590.67***
Croatia	452	0.004*** (9.79)	0.053* (1.72)	0.571*** (8.56)	-0.848 (0.52)	0.374	90.99***
Czech Rep.	452	0.005*** (12.73)	0.011 (0.49)	0.568*** (9.92)	-0.61 (0.6)	0.463	130.77***
Denmark	452	0.011*** (18.3)	-0.022 (-0.79)	0.235*** (2.9)	1.134** (2.09)	0.089	15.69***
Egypt	439	0.008*** (25.45)	0.003 (0.18)	0.229*** (5.93)	0.401*** (5.05)	0.275	56.26***
Finland	456	0.007*** (22.12)	0.046*** (3.36)	0.254*** (7.14)	0.033 (1.11)	0.317	71.4***
France	466	0.004*** (20.3)	0.035*** (3.73)	0.273*** (11.96)	0.189*** (3.15)	0.559	197.78***
Germany	460	0.006*** (28.58)	0.017* (1.83)	0.204*** (9.2)	0.008 (0.78)	0.395	100.79***

Greece	451	0.008*** (29.45)	0.021** (2.22)	0.197*** (7.62)	0.491* (1.78)	0.471	134.4***
Hong Kong	448	0.007*** (24.33)	0.034*** (3.02)	0.177*** (4.18)	-0.618 (0.83)	0.162	29.8***
Hungary	454	0.007*** (12.62)	0.03 (1.17)	0.478*** (7.67)	-0.708*** (8.05)	0.276	58.44***
India	451	0.006*** (24.68)	0.008 (0.78)	0.291*** (11.18)	-0.152 (0.65)	0.517	161.65***
Indonesia	440	0.008*** (16.66)	0.072*** (3.02)	0.146** (2.48)	1.552 (0.36)	0.133	23.49***
Ireland	462	0.009*** (14.68)	0.099*** (4.06)	0.52*** (7.64)	-2.065 (1.28)	0.292	64.23***
Israel	444	0.007*** (24.06)	-0.003 (-0.24)	0.224*** (5.89)	0.352*** (3.93)	0.305	65.67***
Italy	472	0.007*** (23.86)	0.017 (1.37)	0.296*** (11.83)	-0.186*** (4.72)	0.429	119.12***
Jamaica	457	0.017*** (23.53)	-0.017 (-0.35)	0.639*** (4.31)	1.234*** (6.48)	0.187	35.98***
Japan	444	0.005*** (21.01)	0.002 (0.16)	0.105*** (3.58)	1.807 (0.83)	0.29	61.34***
Jordan	567	0.003*** (20.23)	-0.004 (-0.26)	0.719*** (38.32)	0.778** (2.53)	0.965	5178.73***
Kuwait	567	0.004*** (15.29)	0.008 (0.36)	0.457*** (7.86)	6.479*** (6.66)	0.543	225.04***
Luxembourg	465	0.015*** (26.26)	0.026 (1.43)	0.25*** (4.25)	1.9** (2.58)	0.284	62.47***
Malaysia	445	0.005*** (18.37)	0.036** (2.2)	0.382*** (8.8)	-2.739 (1)	0.269	55.5***
Mexico	457	0.007*** (20.06)	0.013 (0.89)	0.34*** (6.64)	3.753*** (2.86)	0.517	163.67***
Morocco	450	0.005*** (37.01)	-0.036*** (-2.66)	0.296*** (10.72)	-1.886* (1.89)	0.361	85.65***
Netherlands	466	0.009*** (23.81)	0.046*** (2.68)	0.295*** (7.14)	0.039 (0.55)	0.27	58.32***
New Zealand	456	0.009*** (22.17)	-0.027 (-1.23)	0.291*** (4.8)	1.394 (1.34)	0.246	50.51***
Nigeria	448	0.01*** (5.6)	-0.021 (-0.14)	0.637* (1.82)	-4.73*** (4.03)	0.011	2.65**
Norway	455	0.008*** (22.26)	0.037** (2.22)	0.246*** (5.44)	0.051* (1.71)	0.217	42.9***
Pakistan	447	0.007*** (29.96)	0.052*** (4.12)	0.173*** (4.81)	0.795* (1.77)	0.282	59.26***
Peru	473	0.003*** (3.53)	0.024 (1.16)	0.875*** (13.8)	-0.09 (1.54)	0.685	343.24***
Philippines	442	0.008*** (12.18)	0.018 (0.7)	0.395*** (6.4)	-1.283 (1.03)	0.186	34.69***
Poland	455	0.008*** (25.87)	0.044*** (3.26)	0.273*** (8.91)	-0.687 (0.82)	0.306	67.81***
Portugal	466	0.007*** (13.26)	0.013 (0.57)	0.552*** (9.38)	-1.789** (2.48)	0.312	71.22***
Romania	466	0.005*** (21.74)	-0.007 (-0.5)	0.478*** (13.17)	-1.603*** (4.29)	0.594	227.4***
Russia	473	0.006*** (14.1)	-0.006 (-0.45)	0.544*** (15.55)	0.336*** (3.35)	0.717	400.2***
Saudi Arabia	451	0.006*** (27.47)	0.009 (0.57)	0.25*** (6.86)	-1.042 (1.51)	0.237	47.71***
Serbia	466	0.002*** (8.45)	-0.008 (-0.63)	0.917*** (24.76)	-1.052*** (4.53)	0.841	822.98***
Singapore	473	0.01*** (26.73)	0.022 (1.33)	0.311*** (6.58)	-1.142*** (3.4)	0.226	46.82***

Slovenia	466	0.005*** (13.67)	0.026 (1.01)	0.61*** (10.93)	-2.176** (2.08)	0.358	87.32***
South Africa	453	0.011*** (22.36)	0.069*** (3.66)	0.259*** (4.94)	2.039 (1.34)	0.345	80.27***
South Korea	449	0.006*** (24.31)	0.035*** (3.41)	0.158*** (5.31)	0.471** (2.31)	0.294	63.23***
Spain	464	0.006*** (21.08)	0.078*** (7.01)	0.331*** (12.47)	-0.104*** (5.67)	0.54	182.38***
Sri Lanka	405	0.008*** (17.09)	0.062*** (2.86)	0.409*** (6.46)	-1.519 (1.57)	0.288	55.5***
Sweden	457	0.007*** (21.75)	0.032** (2.29)	0.253*** (7.32)	-0.461 (0.97)	0.258	53.87***
Switzerland	458	0.009*** (18.07)	-0.006 (-0.2)	0.314*** (4.79)	-0.823** (-2.46)	0.113	20.49***
Taiwan	448	0.005*** (17.2)	-0.002 (-0.16)	0.233*** (5.21)	0.019 (-1.62)	0.235	46.82***
Thailand	436	0.006*** (19.92)	0.086*** (5.88)	0.232*** (6.15)	-0.532*** (-9.52)	0.264	53.07***
Tunisia	567	0.003*** (14.46)	0.053*** (2.76)	1.002*** (19.18)	-10.957 (-0.63)	0.625	315.16***
Turkey	452	0.01*** (19.18)	-0.01 (-0.46)	0.229*** (4.14)	-1.987*** (-3.67)	0.052	9.2***
U.A. Emir.	451	0.015*** (6.21)	0.07 (0.63)	0.007 (0.02)	6.918* (-1.95)	0.008	2.23*
UK	458	0.005*** (20.68)	0.044*** (3.74)	0.287*** (10.2)	-0.603 (-0.2)	0.38	94.46***
USA	458	0.006*** (23.75)	0.018* (1.8)	0.195*** (7.28)	-0.62 (-1.03)	0.25	51.86***
Venezuela	435	0.137 (1.47)	0.468 (0.19)	-1.832 (-0.47)	3.237*** (-3.7)	-0.005	0.25
Vietnam	458	0.006*** (22.7)	-0.011 (-0.87)	0.243*** (6.17)	-2.47 (-0.02)	0.174	33.04***

Panel C - Post-Pandemic (Jan 2022 - Jun 2024)

Country	N	$\Upsilon_{i,0}$	$\Upsilon_{i,1}$	$\Upsilon_{i,2}$	$\Upsilon_{i,3}$	Adj-R^2	F-statistic
Argentina	650	0.016*** (17.6)	-0.021 (-0.93)	0.122** (2.33)	4.225*** (-4.16)	0.279	84.65***
Australia	627	0.006*** (25.26)	-0.01 (-0.63)	0.106* (1.92)	1.217 (-0.31)	0.048	11.54***
Austria	627	0.01*** (21.6)	-0.018 (-0.66)	0.4*** (3.89)	2.602*** (-5.11)	0.152	38.33***
Bangladesh	650	0.003*** (15.04)	0.081** (2.46)	0.758*** (15.74)	1.763*** (-2.75)	0.543	257.71***
Belgium	638	0.008*** (31.43)	0.006 (0.39)	0.05 (1.15)	6.937*** (-4)	0.203	54.94***
Brazil	622	0.008*** (25.91)	0.019 (1.43)	0.114** (2.39)	1.739*** (-3.89)	0.093	22.32***
Bulgaria	614	0.004*** (16.41)	0.044* (1.95)	0.658*** (12.08)	-0.293 (-0.27)	0.432	156.16***
Canada	626	0.007*** (26.7)	-0.025* (-1.73)	0.021 (0.36)	8*** (-2.58)	0.115	27.95***
Chile	622	0.007*** (27.39)	0.031*** (2.75)	0.308*** (6.74)	-1.408 (-0.97)	0.292	86.4***
China	601	0.007*** (31.24)	0.063*** (5.12)	0.163*** (4.34)	0.955 (-1.59)	0.173	42.75***
Colombia	650	0.005*** (13.96)	0.006 (0.63)	0.681*** (19.01)	1.037*** (-3.07)	0.791	820.72***
Croatia	620	0.003*** (20.6)	0.054*** (2.96)	0.567*** (15.11)	-0.429*** (-10)	0.498	205.41***
Czech Rep.	627	0.007*** (11.1)	-0.029 (-0.69)	0.439*** (3.64)	2.838 (-0.75)	0.121	29.6***
Denmark	624	0.013*** (27.09)	0.016 (0.78)	0.045 (0.86)	8.613 (-1.14)	0.304	91.5***
Egypt	600	-0.041 (-0.84)	-3.469** (-2.25)	9.513 (1.61)	-109.757*** (-4.19)	0.01	2.99**
Finland	628	0.008*** (26.81)	-0.025* (-1.87)	0.165*** (3.46)	0.968 (-0.85)	0.138	34.47***
France	638	0.005*** (17.69)	0.008 (0.53)	0.181*** (3.98)	0.408 (-0.62)	0.103	25.29***
Germany	638	0.007*** (33.43)	-0.004 (-0.35)	0.134*** (4.63)	0.877 (-0.57)	0.157	40.5***

Greece	617	0.007*** (34.5)	0.016 (1.62)	0.082*** (2.68)	2.882 (-0.78)	0.24	65.86***
Hong Kong	610	0.007*** (30.04)	0.041*** (5.27)	0.167*** (7.19)	2.054 (-0.48)	0.543	241.79***
Hungary	627	0.009*** (15.47)	0.039 (1.48)	0.506*** (8.17)	-0.062* (-1.9)	0.261	74.64***
India	617	0.006*** (27.32)	-0.021 (-1.59)	0.055 (1.42)	7.687 (-0.86)	0.301	89.5***
Indonesia	595	0.007*** (23.46)	0.012 (0.51)	0.254*** (3.76)	2.355*** (-2.72)	0.117	27.26***
Ireland	633	0.014*** (21.91)	0.012 (0.48)	0.089 (1.03)	5.453*** (-3.78)	0.086	20.7***
Israel	613	0.008*** (39.89)	0.041*** (4.12)	-0.049* (-1.68)	5.58 (-0.5)	0.149	36.6***
Italy	649	0.006*** (27.83)	0.01 (1.17)	0.262*** (9.82)	1.356 (-0.13)	0.507	223.4***
Jamaica	627	0.019*** (20.93)	0.189*** (2.81)	0.145 (0.58)	13.184* (-1.7)	0.029	7.14***
Japan	610	0.005*** (23.78)	-0.012 (-1.44)	0.088** (2.28)	4.082* (-1.73)	0.256	70.92***
Jordan	781	0.003*** (18.08)	0.01 (0.8)	0.833*** (23.59)	-1.011* (-1.9)	0.771	876.29***
Kuwait	781	0.004*** (15.46)	0.005 (0.2)	0.694*** (11.26)	-4.454** (-2.06)	0.337	132.94***
Luxembourg	638	0.014*** (32.75)	0.033* (1.9)	0.16** (2.48)	3.482 (-1.23)	0.149	38.31***
Malaysia	604	0.004*** (28)	0.016 (1.08)	0.198*** (4.89)	1.069 (-1.37)	0.146	35.48***
Mexico	627	0.008*** (6.92)	-0.004 (-0.07)	0.124 (0.64)	8.198 (-1.28)	0.017	4.65***
Morocco	625	0.007*** (31.94)	-0.017 (-0.91)	0.427*** (8.61)	-5.507** (-2.29)	0.187	49***
Netherlands	638	0.008*** (29.71)	0.024 (1.6)	0.094* (1.86)	6.498 (-0.34)	0.218	60.16***
New Zealand	619	0.007*** (33.92)	0.022 (1.38)	0.132** (2.29)	5.056 (-1.13)	0.12	29.07***
Nigeria	614	0.008*** (24.88)	-0.013 (-0.5)	1.11*** (15.9)	-11.066 (-0.82)	0.529	230.7***
Norway	626	0.008*** (24.37)	0.026 (1.64)	0.171*** (2.67)	4.013 (-0.78)	0.173	44.7***
Pakistan	613	0.006*** (30.9)	0.055*** (5.39)	0.231*** (7.24)	-1.397 (-1.05)	0.248	68.13***
Peru	650	0.003*** (10.3)	-0.005 (-0.49)	0.806*** (23.32)	1.096*** (-6.29)	0.853	1253.88***
Philippines	608	0.007*** (24.49)	0.042*** (3.5)	0.24*** (5.15)	1.587** (-2.27)	0.274	77.26***
Poland	625	0.007*** (19.5)	0.031** (2.42)	0.26*** (7.59)	0.459 (-1.47)	0.265	76.09***
Portugal	638	0.008*** (15.25)	-0.009 (-0.36)	0.15 (1.38)	10.627* (-1.93)	0.134	33.87***
Romania	636	0.005*** (18.94)	-0.012 (-0.73)	0.494*** (12.44)	-1.639*** (-2.74)	0.376	128.46***
Russia	47	-0.008 (-1.28)	-0.035 (-0.72)	1.137*** (7.36)	-1.93 (-0.2)	0.749	46.76***
Saudi Arabia	616	0.006*** (30.19)	0.022* (1.77)	0.209*** (5)	-0.299 (-0.85)	0.156	38.95***
Serbia	636	0.002*** (15.01)	-0.011* (-1.81)	0.925*** (108.03)	0.091 (-0.42)	0.989	18830.83***
Singapore	650	0.008*** (28.04)	-0.007 (-0.46)	0.226*** (4.22)	6.262 (-0.9)	0.304	95.37***
Slovenia	636	0.005*** (20.8)	-0.026 (-1.59)	0.577*** (16.31)	-0.691 (-0.07)	0.562	272.44***
South Africa	622	0.008*** (18.48)	0.005 (0.27)	0.257*** (3.47)	3.37 (-0.79)	0.199	52.51***
South Korea	612	0.007*** (32.6)	0.049*** (4.82)	0.002 (0.06)	5.145** (-2.01)	0.166	41.47***
Spain	638	0.006*** (29.22)	-0.016 (-1.52)	0.144*** (3.96)	4.04*** (-3.41)	0.295	90.03***
Sri Lanka	588	0.008*** (23.01)	0.074*** (5.38)	0.269*** (6.32)	1.639*** (-4.63)	0.394	128.36***
Sweden	628	0.008*** (20.4)	-0.035** (-1.99)	0.131* (1.92)	3.091 (-1.54)	0.094	22.57***
Switzerland	628	0.008*** (27.25)	-0.021 (-1.2)	0.103* (1.7)	5.283* (-1.84)	0.11	26.96***
Taiwan	602	0.004*** (21.82)	0.044*** (4.41)	0.15*** (3.92)	7.605*** (-3.95)	0.491	194.55***
Thailand	604	0.005*** (23.4)	0.004 (0.28)	0.148** (2.49)	-1.362 (-0.19)	0.037	8.7***
Tunisia	781	0.003*** (19.17)	-0.004 (-0.23)	0.874*** (18.61)	-3.632 (-0.95)	0.669	527.65***
Turkey	622	0.01*** (34.04)	0.025*** (2.82)	0.081*** (4.01)	-0.13 (-0.46)	0.067	15.98***
U.A. Emir.	625	0.007*** (15.56)	0.133*** (3.55)	0.523*** (5.64)	-1.54* (-1.75)	0.135	33.36***
UK	627	0.006*** (27.7)	-0.022 (-1.35)	0.046 (0.89)	3.022 (-0.65)	0.05	12.08***
USA	625	0.005*** (27.47)	0.008 (0.83)	0.136*** (4.16)	0.98 (-0.49)	0.183	47.71***

Venezuela	599	0.017*** (5.84)	-0.053 (-0.44)	0.666*** (2.86)	-1.096 (-0.44)	0.03	7.27***
Vietnam	618	0.006*** (27.42)	-0.017* (-1.86)	0.254*** (7.74)	-2.358*** (-2.7)	0.255	71.23***

Note: Figures in parentheses are the t-statistic. *** p<0.01 **p<0.05 *p<0.1

Source: Elaborated by the authors

APPENDIX M - GAM Estimations. Dependent Variable: Herding

Table M-1 – GAM Estimations – Dependent Variable: Herding

Variables	Model (1) Pre-Pandemic Period (2015-03:2020)	Model (2) Pandemic Period (03:2020-2022)	Model (3) Pos-Pandemic Period (2022-06:2024)
Constant	0.008644*** (443.85)	0.010415*** (268.54)	0.009525*** (320.90)
Individualism	0.000016*** (-9.28) 1487.85***	-0.000008*** (-2.35) 588.85***	0.000006** (2.19) 614.03***
Uncertainty Avoidance	0.000047*** (45.10) 1013.08***	0.000047*** (22.55) 672.47***	0.000028*** -17.37 1219.13***
Power Distance	-0.000047*** (-30.99) 767.30****	-0.000063*** (-20.79) 460.13****	-0.000061*** (-26.64) 350.45***
Masculinity	0.000013*** (11.03) 544.92***	0.000019*** (8.14) 194.24***	0.000025*** -13.97 360.78***
Long-Term Orientation	-0.000032*** (-23.83) 471.48***	-0.000032*** (-12.04) 330.61***	-0.000054*** (-27.19) 707.21***
Indulgence	0.000011*** (10.66) 2946.27***	0.000038*** (18.17) 499.62***	0.000013*** (8.52) 407.50***
GNI per capita	-0.000022*** (-7.32) 1438.83***	-0.000646*** (-10.72) 581.11***	-0.000423*** (-9.20) 878.81***
non-linearity χ^2	8669.73***	3327.03***	4537.91***

Notes: Generalized additive model estimations where the dependent variable is herding. Z-value (in parentheses) is the ratio of the linear coefficient to its standard error to assess how strongly a variable is associated with overconfidence. The third line represents the gain value. *p<0.1; **p<0.05; ***p<0.01.
Source: Elaborated by the authors